

# Canonical Minimal Polynomial Invariants from Noisy Data via Adaptive Sparse Nullspace Recovery and Gröbner Basis

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## Abstract

Recovering polynomial conservation laws from data faces two distinct failures. The first, which Kaiser, Kutz, and Brunton (Kaiser et al., 2018) identified and left open, is nullspace ambiguity: when several invariants hold at once, plain nullspace recovery returns an arbitrary linear mixture of them, and sparsity alone cannot separate that mixture. Moreover, fixed-step symplectic integration crystallizes this failure, because the integrator conserves a nearby modified invariant  $H_{dt} = H + O(dt^p)$  in place of the true Hamiltonian  $H$ , so naive recovery risks returning a numerically real but physically spurious quantity. The second failure is algebraic redundancy: the same invariant can be discovered as, for instance,  $x^2 + y^2 - 1$ , its square, or a rescaling of either, and that absence of a canonical form breaks downstream sum-of-squares (SOS) and satisfiability-modulo-theories (SMT) verification. We present SR-GB+CSNP, a pipeline that uses combinatorial nullspace pursuit to treat rationality, not sparsity, as the primary discriminating prior. Candidate supports are scored by an information-theoretic bit-cost over rational coefficients and minimised by a Branch-and-Bound search that certifies global optimality of the recovered support, resolving ambiguities that a sparsity-only criterion cannot. For quadratic Hamiltonians we prove in closed form that this rationality criterion prefers  $H$  over  $H_{dt}$ , unconditionally for first-order integrators such as symplectic Euler, where the modified invariant carries a monomial  $H$  does not. The result is weaker for symmetric integrators such as Störmer–Verlet, whose modified invariant keeps  $H$ ’s support and perturbs only a coefficient: there it holds outside a small set of resonant step sizes, measure-zero in the idealized zero-tolerance limit but of positive measure at any finite working tolerance. Redundancy is eliminated separately, by reduced Gröbner basis canonicalisation over  $\mathbb{Q}$ , connected to noisy data through noise-adaptive snap-rounding and a Normal-Form Pruning Lemma. An adaptive minimum-description-length (MDL) loop discovers the polynomial degree when it is not known in advance. Both the rationality-based ambiguity resolution and the Gröbner canonicalisation rest on one shared assumption: that the true invariant has simple rational coefficients in the variables supplied. That assumption is not unit-free, and recovery breaks down under an arbitrary rescaling of the physical quantities. Experiments cover eight benchmark classes. On the fixed- $dt$  harmonic oscillator, sampled from the exact rotation flow, SR-GB+CSNP separates the true invariant from a  $dt$ -specific irrational-coefficient artifact at **100%** where a sparsity-only criterion scores **0%**; on genuine symplectic-Euler trajectories, however, a third relation, a pure algebraic identity of the update rule itself, ties  $H$  at equal cost, and no method tested, SR-GB+CSNP included, recovers  $H$  from that wider dictionary. On the 26 single-invariant Feynman polynomial equations, every method agrees at zero noise, but under noise SR-GB+CSNP pulls ahead, holding **73.1%** at  $\sigma = 0.02$  and **59.7%** at  $\sigma = 0.05$  against KRONIC’s 28.2%/20.4% and SINDy-ST’s 42.2%/29.7%; within that suite the degree-2 subset alone stays near-perfect (**100%/96.7%**) while the noise-sensitive degree-3 subset degrades sharply.

**Keywords:** conservation law discovery; polynomial invariants; combinatorial nullspace pursuit; minimum description length; Gröbner bases; symbolic regression; rationality prior; symplectic integrators.

# 1 Introduction

For systems where more than one polynomial conservation law holds at once, plain nullspace recovery has a known failure. Kaiser, Kutz, and Brunton (Kaiser et al., 2018) observed that when several invariants coexist, singular value decomposition (SVD) returns an arbitrary linear mixture of them in place of the individual physical quantities, and they left open how that mixture is to be separated into its constituents; Section 2 discusses the closest attempt since, which resolves this for simultaneous *linear* laws but leaves the nonlinear case addressed here untouched. This *nullspace ambiguity* is the first of two problems this paper addresses.

Moreover, data from a fixed-step symplectic integrator crystallizes this ambiguity instead of avoiding it. Backward error analysis for symplectic integrators (Hairer et al., 2006) shows that such an integrator does not approximately conserve the true Hamiltonian  $H$ ; it exactly conserves a nearby modified invariant  $H_{dt} = H + O(dt^p)$ . Naive recovery from this data therefore risks returning a numerically real but physically spurious quantity, and it does so with full confidence, because  $H_{dt}$  is genuinely and exactly conserved on the sampled trajectory. Section 4.6 proves that the selection criterion introduced below prefers  $H$  over  $H_{dt}$  for quadratic Hamiltonians. This holds unconditionally for first-order integrators, and for symmetric ones it holds outside a set of *resonant* step sizes, values of  $dt$  where the modified invariant’s perturbed coefficient happens to sit as close to a low-denominator rational as  $H$ ’s own coefficient, so the two look equally simple to the rationality tie-break. That set is measure-zero only as the tolerance vanishes, with positive measure at any fixed finite working tolerance. Section 4.6 states the finite-tolerance version, and documents one case the argument does not resolve, where several exactly-vanishing relations tie on the wider, unrestricted transition dictionary.

The second failure is *algebraic redundancy*. The smallest instance is the unit-circle relation  $x^2 + y^2 = 1$ , which in mechanical terms is the energy level set of a unit harmonic oscillator and the rod constraint of a planar pendulum. A method that correctly identifies this relation may, depending on how it happens to be parameterised, output any of  $\{x^2 + y^2 - 1\}$ ,  $\{x^2 + y^2 - 1, (x^2 + y^2 - 1)^2\}$ , or  $\{x^2 + y^2 - 1, x^2 - 1 + y^2\}$ . All three are mathematically correct, but none is canonical. A downstream sum-of-squares (SOS) solver (Prajna et al., 2002), a satisfiability-modulo-theories (SMT) verifier, or a physics-informed regulariser that treats these as independent constraints will therefore produce degenerate results; in the regulariser’s case it doubly- or triply-penalises what is in fact a single algebraic relation. Both failures arise wherever polynomial invariants are extracted from data, whether in classical mechanics (Kaiser et al., 2018), robotic kinematic constraints, or the Feynman symbolic-regression benchmark (Udrescu and Tegmark, 2020). State trajectories from trained control agents are a further setting where such constraints may exist, though whether they do is itself an open question that Section 6 takes up as future work. The goal in each setting is not merely *a* polynomial that fits the data, but *the* canonical minimal generating set of the ideal the data satisfies.

The central insight behind the method proposed here is that

*nullspace ambiguity can be resolved by casting invariant recovery as globally optimised sparse-support selection under an information-theoretic rationality cost; algebraic canonicalisation then yields the unique ideal representation.*

We propose **SR-GB+CSNP** (Sparse Recovery, Gröbner Basis, and Combinatorial Sparse Nullspace Pursuit), a *pipeline* that resolves nullspace ambiguity through **CSNP** (Combinatorial Sparse Nullspace Pursuit), a lexicographic Branch-and-Bound (BB) search that enforces the rationality cost as its primary criterion, sparsity as secondary, and geometric stability as a tertiary tie-break. It eliminates redundancy separately, by computing the **reduced Gröbner basis** (Cox et al., 2015) of the recovered ideal. These two mechanisms are in turn supported by adaptive monomial expansion, QR-based degree sweeping, algebra-guided pruning, minimum-description-length (MDL)-based stopping, and noise-adaptive snap-rounding, all detailed in Section 3.

For **single-invariant** systems ( $d = 1$ ), a sparsity-enforced nullspace method such as **SINDy-null (KRONIC)** (Kaiser et al., 2018) already recovers the unique generator reliably, and our experiments confirm it (100% on the circle, sphere, cubic, and holonomic constraints). The focus of this work is instead the  $d > 1$  regime, where such methods return arbitrary mixtures of the coexisting invariants. That is the gap CSNP with lexicographic BB fills: a global combinatorial search over monomial supports with an information-theoretic priority favouring rational, sparse, and geometrically stable invariants.

The pipeline carries an exact recovery condition (Proposition 4.1), a global optimality certificate for the BB search (Theorem 4.4), a Normal-Form Pruning Lemma justifying algebra-guided monomial elimination

(Lemma 4.3), and a conditional adaptive exact-recovery result that extends the recovery guarantee to the full adaptive pipeline exactly when its stated assumptions hold (Proposition 4.5). It also carries the pairwise integrator-separation result above (Proposition 4.6).

Left open are exact algebraic ties among competing relations on unrestricted transition dictionaries (Section 4.6), a sharp noise ceiling for degree-3 invariants (Section 5.5), and recovery of invariants with inherently irrational coefficients (Section 6.9). One restriction applies from the start: the rationality prior is not invariant to a change of units. Rescaling each physical quantity by an arbitrary factor destroys recovery on three of the four benchmarks stress-tested in Section 6.4, and per-column standardization does not repair it. The method therefore assumes data supplied in units where the true invariant already has simple rational coefficients. Every curated benchmark in this paper satisfies that assumption by construction, though arbitrary field data need not.

### Contributions.

#### *Algorithmic.*

1. *Ambiguity resolved by rationality, not sparsity alone.* The Lexicographic Rationality Cost, a bit-cost over rational coefficients (Section 3.4), is the primary CSNP selection criterion, without tuning of the criterion itself; Section 5.8 shows it succeeds (100%) exactly where a sparsity-only criterion fails (0%) on the fixed- $dt$  oscillator, and the same section’s multi-invariant deflation benchmark recovers all four entangled 2D-oscillator generators in every seed where SINDy-null on the identical difference dictionary never returns the full set.
2. *Certified search in place of greedy selection.* Branch-and-Bound with dominance pruning and the admissible bound  $R \geq |S|$  (Section 3.5) delivers a global optimality certificate over  $(R, k, \text{gap})$  (Theorem 4.4), not merely a feasible support.
3. *Automatic degree discovery.* Progressive monomial expansion, algebra-guided pruning (Lemma 4.3), and global MDL selection (Section 3.10) remove the requirement to specify the maximum degree  $D$  in advance.

#### *Theoretical.*

4. An exact recovery condition for CSNP (Proposition 4.1), global optimality of the BB search (Theorem 4.4), the Normal-Form Pruning Lemma with the counterexample showing why it must be stated on normal forms (Lemma 4.3), and a conditional adaptive exact-recovery proposition (Proposition 4.5) that extends the recovery guarantee to the full adaptive pipeline exactly when its assumptions hold. For the fixed- $dt$  symplectic sub-case (Section 4.6) the cost-ordering part of assumption (4) is proved rather than assumed; the remaining assumptions are checked empirically, not derived.
5. A closed-form, partial resolution of the integrator trap: CSNP’s selection criterion prefers the true Hamiltonian  $H$  over a symplectic integrator’s exactly-conserved modified invariant  $H_{dt}$  (Proposition 4.6), unconditionally for first-order integrators and, for symmetric ones, outside a set of resonant step sizes that is measure-zero only in the zero-tolerance limit (positive measure at any fixed finite working tolerance, Section 4.6); exact ties among other candidate relations on the unrestricted transition dictionary are documented as open.

#### *Empirical.*

6. Validation across eight benchmark classes, synthetic varieties, degree-lifting robustness, transition invariants, noise sensitivity, the Feynman polynomial subset, program loop invariants, holonomic robotic equality constraints, and a point-vortex system, using exact recovery of the ideal, not merely a fitting expression, as the metric. Noise-tolerant snap-rounding lifts exact recovery on the circle to  $\sigma = 0.14$  with 100% success and retains 96.7% at  $\sigma = 0.16$ , extending the noise tolerance far beyond naïve rounding, which fails already at  $\sigma = 0.04$ . The evaluation also covers the failure cases: the Lemma 4.3 counterexample, the wide-dictionary tie of Section 6.5, and the failed tolerance sweep.

In the following, Section 2 reviews related work and positions SR-GB+CSNP within the broader literature, and Section 3 presents the SR-GB+CSNP pipeline in detail, including the new adaptive algorithm. Section 4 contains the theoretical analysis, and Section 5 reports experimental results. Section 6 addresses limitations and directions for future work, before Section 7 concludes.

## 2 Related Work

**Symbolic regression.** Classical genetic programming (Koza, 1992) and modern frameworks such as PySR (Cranmer, 2023) discover analytic expressions from data, but what they return is only one of many syntactically equivalent forms: algebraic equivalence between outputs is neither checked nor enforced. PySR is, moreover, limited to explicit functions  $y = f(x)$ , so it cannot recover implicit, multi-valued varieties such as the unit circle. The AI Feynman framework (Udrescu and Tegmark, 2020) instead applies dimensional analysis and neural symmetry detection, and reports strong results on the Feynman database from which the polynomial benchmark of Section 5.5 is drawn, though at a computational cost that can run to hours per equation under its original evaluation’s time cap. Both PySR and AI Feynman solve the *explicit* regression problem  $y = f(x)$ , and neither one targets implicit varieties  $p(x) = 0$ , canonicalises its output, or handles multi-dimensional nullspaces. On the implicit, possibly-multi-invariant task addressed here they are therefore inapplicable by construction.

**Sparse identification (SINDy) and conservation-law discovery.** Sparse Identification of Nonlinear Dynamics (SINDy) (Brunton et al., 2016) regresses an explicit dynamical model  $\dot{X} = \Theta(X)\Xi$  from data via sequential thresholded least squares (STLSQ), with  $X$  the  $N \times n$  matrix of stacked state observations. The nullspace-based approach for discovering conservation laws was first formalised by Kaiser, Kutz & Brunton (Kaiser et al., 2018) as **KRONIC**. That work already observed that plain SVD returns arbitrary mixtures when multiple invariants coexist. Later variants, including SINDy with automatic differentiation (SINDy-AD) (Kaheman et al., 2022), improve sparsity promotion and noise handling, but both still require a time dimension to differentiate, which restricts them to trajectory data. A more direct descendant tackles the  $d > 1$  case within this same SVD-nullspace framework. Oellerich and Emelianenko (Oellerich and Emelianenko, 2024) extend KRONIC to systems with several simultaneous conservation laws, and disambiguate the resulting basis by reducing it to row-echelon form, not by any sparsity or rationality criterion. On the worked example in that paper, two coexisting *linear* laws, that reduction is a genuine canonical form: for a linear ideal, reduced row-echelon form and a Gröbner basis coincide, so no tie-break over competing supports is needed in the first place. Whether the same reduction disentangles the *nonlinear* invariants this paper targets, where several generators can share monomial support in a way that is not triangular under any fixed pivot order, turns out to depend on which nonlinear case: on our point-vortex benchmark (Section 5.9), row-echelon reduction isolates the two linear impulses cleanly, reproducing the regime Oellerich and Emelianenko tested, but never once returns the one dense quadratic generator as an individual row, not even at the vortex count where CSNP recovers the full ideal in every trial. A closer response to the nullspace degeneracy itself is SINDy-PI (Kaheman et al., 2020), which avoids forming a nullspace at all: it cycles through the library terms, treats each in turn as an explicit left-hand side, and fits it in parallel against the remaining terms, keeping the regression that is most self-consistent. That works well when a single implicit relation can be pivoted around one term, but it does not resolve the situation this paper targets, where several independent invariants span a multi-dimensional nullspace and no lone left-hand term picks one out. CSNP selects among those directions directly instead of testing them one term at a time. SINDy-RL (Zolman et al., 2025) extends the framework to reinforcement learning, using sparse dictionary surrogates for the environment model, reward, and controller. It constructs explicit dynamical models  $\dot{x} = \Xi \Theta(x, u)$ , written here in that paper’s per-sample convention with  $x$  a column vector, describing how the state evolves under a policy, whereas SR-GB+CSNP discovers *algebraic constraints* the trajectories implicitly satisfy; the two solve different problems and complement each other.

**Weak-form and ensemble reformulations.** WSINDy (Messenger and Bortz, 2021) and ensemble-SINDy variants attack noise amplification in the design matrix  $\Phi$  itself, by integrating against test functions or aggregating across bootstrap resamples before the sparse-regression step runs. This is a front-end concern, orthogonal to the one this paper addresses: none of these reformulations selects *among* directions of a multi-dimensional nullspace once  $\Phi$  is built, which is CSNP’s target. A weak-form or ensemble front end could in principle feed a cleaner  $\Phi$  into CSNP without substituting for its selection step.

**Neural variance-minimisation approaches.** A separate line of work learns conservation laws by training a network to output a scalar that stays nearly constant along a trajectory, minimising a variance-based loss instead of solving a linear nullspace problem at all. NGCG (Ray, 2026) is a recent instance: a multi-restart variance minimiser produces a candidate near-constant quantity, system-specific symbolic extraction (polynomial or log-basis Lasso, or a genetic-programming fallback) turns it into a closed form, and a strict

constancy gate paired with a diversity filter rejects spurious laws, reporting zero false positives on systems with no true invariant. Its learning objective is defined directly over trajectories  $\{x^{(i)}(t)\}$ , so, like SINDy-FD and SINDy-AD above, it needs a time dimension, and its variance-minimisation mechanism forms no nullspace to select within. The evaluation reported there, moreover, targets one conserved quantity per system, not the disambiguation among several simultaneously-valid candidates competing under sparsity, the problem CSNP’s rationality tie-break is built for. No result in that evaluation speaks to the multi-dimensional-nullspace case this paper addresses.

**Approximate vanishing ideals and variety learning.** The Approximate Vanishing Ideal (AVI) framework (Heldt et al., 2009; Kreuzer et al., 2009) extends the Buchberger–Möller algorithm to numerical data by using an  $\varepsilon$ -truncated ideal membership test, while Livni et al. (Livni et al., 2013) introduced vanishing component analysis, a polynomial threshold function approach that recovers sparse generators from point-cloud data. Breiding et al. (Breiding et al., 2018) developed a sampling-based approach for learning algebraic varieties from noisy point clouds.

AVI computes the entire border basis of the vanishing ideal, not a single isolated sparse invariant, and it offers no mechanism analogous to BB for selecting among the directions of a multi-dimensional nullspace. For  $d = 1$ , the border basis reduces to one polynomial and a practical AVI implementation coincides with the *Dense SVD+GB* ablation evaluated in Section 5.1. For  $d > 1$ , however, AVI returns the full border basis in place of the individual conserved quantity. Closing that gap is what CSNP and BB are designed to do, and Appendix G extends this comparison to the broader AVI literature.

**Koopman operator and HAVOK methods.** Koopman operator theory (Koopman, 1931) and its data-driven realisation HAVOK (Brunton et al., 2017) discover linear representations of nonlinear dynamics by lifting observations into a space of observables and identifying zero-eigenvalue eigenfunctions as conserved quantities. KRONIC (Kaiser et al., 2018) is the Koopman-eigenfunction route to conserved quantities, so the Koopman family and the nullspace family meet in that method. HAVOK differs more sharply: it is a delay-embedding decomposition for chaotic time series and produces a linear model with a forcing term, not an algebraic constraint  $p(x) = 0$  of the kind considered here.

**Conservation law and invariant discovery.** Hamiltonian neural networks (Greydanus et al., 2019) and Lagrangian networks (Lutter et al., 2019) learn conserved quantities by embedding physical structure directly into the architecture. SR-GB+CSNP, by contrast, makes no architectural assumptions and operates directly on raw state observations, at the cost of being restricted to polynomial invariants of bounded degree.

**Interpretable policy distillation.** A growing body of work distils trained neural RL policies into interpretable forms, such as symbolic expressions for robot locomotion (Acero and Li, 2024), but these methods target the *policy function*  $\pi(a_t | s_t)$  and need the full MDP tuple together with access to the agent’s architecture. SR-GB+CSNP targets a complementary object, algebraic invariants of state trajectories, not the policy function itself. What such invariants would be good for in that setting is a separate question from whether they exist; both are left open here (Section 6).

**Trace-based program invariant generation.** A separate line of work discovers polynomial invariants from program execution traces, not physical trajectories. Sankaranarayanan, Sipma and Manna (Sankaranarayanan et al., 2004) generate polynomial loop invariants symbolically, fixing a template with unknown coefficients and using Gröbner basis computation to solve for the coefficients that make it identically zero along every program path. Rodríguez-Carbonell and Kapur (Rodríguez-Carbonell and Kapur, 2007) then extend this to the entire ideal of invariants for simple loops. The Guess-and-Check approach of Sharma et al. (Sharma et al., 2013) and the DIG invariant generator (Nguyen et al., 2014) instead guess candidate relations from a matrix of concrete trace values by solving for its nullspace, then check and, if necessary, refute and re-guess against an SMT solver such as Z3 (de Moura and Bjørner, 2008). This is, in spirit, the closest prior work to the pipeline proposed here, since the guess step is essentially the same SVD-nullspace computation SR-GB+CSNP uses, applied to discrete program state instead of continuous trajectories. What differs is the failure mode each setting confronts. DIG’s traces are ordinarily exact integer or rational values, so its nullspace is genuinely low-rank and any vector inside it is an equally valid, exactly satisfied invariant. There is no noise to disambiguate among competing directions there, and the SMT check only rules out relations that hold on the sampled traces but not in general. SR-GB+CSNP instead targets floating-point trajectories corrupted by sensor noise, discretization error, or roundoff, where the empirical nullspace is only approximately low-rank and its raw SVD basis mixes the true invariant with sampling artefacts of comparable weight. The rationality cost and CSNP search resolve that ambiguity, a step DIG’s exact



counterexample-refinement loop neither needs nor, by construction, can perform. Section H reports a direct empirical comparison between DIG and SR-GB+CSNP.

**Conservation laws in chemical reaction networks.** An analogous nullspace-based notion of conservation arises in chemical reaction network theory, where the left null space of the stoichiometric matrix gives the linear conserved moieties of the network (Schuster and Höfer, 1991). That construction is the linear, degree-one special case of the problem considered here: the nullspace itself is already the answer, and no selection among competing directions is required. Selecting among such directions once the invariants are nonlinear is the step CSNP adds.

**Gröbner bases in machine learning.** The classical theory of Gröbner bases and their uniqueness properties is well established (Cox et al., 2015). This paper takes that theory as given and puts it to work as the canonicalisation stage; the contribution lies in the interface layer between noisy numerical data and exact algebraic computation. Within machine learning, Gröbner bases have already been applied to solving polynomial systems (Berthomieu et al., 2021), and recent work on learning Gröbner bases with Transformer architectures (Kera et al., 2024) illustrates a growing interest in bridging neural methods and exact algebraic computation.

**Where each line of prior work stops.** Table 1 distills the paragraphs above into a single comparison.

Table 1: Comparison with prior work. “Failure addressed here” means the cited method targets the same problem and the boundary listed was measured directly in this paper; “scope difference” means the cited method solves a different problem by design, and no comparison is implied.

| Prior work                                     | Achievement                                                          | Boundary                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SINDy-null / KRO-NIC (Kaiser et al., 2018)     | Sparse dynamics and single-invariant nullspace recovery              | <i>Failure addressed here:</i> for $d > 1$ returns an arbitrary mixture, not the individual generators; the authors identified this and left it open. Validated on the 2D-oscillator deflation task, where SINDy-null recovers neither the energy generators nor the angular momentum in any seed (a single seed returns only the cross term) and never the full generating set (Section 5.8). |
| SINDy-ST (Brunton et al., 2016)                | Sparse regression via sequential thresholding                        | <i>Failure addressed here:</i> redundant generating sets persist even at $\sigma = 0$ (4–8 polynomials per variety, Section 5.1), and without a rationality tie-break the all-targets ensemble still trails SR-GB+CSNP by nearly 30 points under noise on the 26 Feynman equations (42.2% vs. 73.1% at $\sigma = 0.02$ , Section 5.5).                                                         |
| Hamiltonian NNs (Greydanus et al., 2019)       | Conserved quantities via architectural inductive bias                | <i>Scope difference:</i> targets learned continuous dynamics, not an implicit algebraic certificate, and performs no canonicalisation; complementary to this work.                                                                                                                                                                                                                             |
| AVI (Heldt et al., 2009; Kreuzer et al., 2009) | Full border basis of the approximate vanishing ideal from noisy data | <i>Failure addressed here:</i> returns every relation in the nullspace instead of selecting the physically meaningful one, and border-basis cardinality inflates with noise even when the true ideal has not changed (Kepler angular momentum: mean cardinality 3.2 at $\sigma = 0$ to 20.3 at $\sigma = 0.02$ , Section 5.7).                                                                 |
| DIG (Nguyen et al., 2014)                      | Exact polynomial invariants from program traces via guess-and-check  | <i>Scope difference:</i> its exact-vanishing test is correct for symbolically exact traces; it has no analogue of noise-adaptive snap-rounding for relations that hold only approximately on floating-point data, confirmed by a positive control on an exactly-rational trace (Section H).                                                                                                    |
| PySR (Cranmer, 2023)                           | Explicit symbolic regression $y = f(x)$                              | <i>Scope difference:</i> implicit, multi-valued varieties $p(x) = 0$ fall outside its design by construction.                                                                                                                                                                                                                                                                                  |

## 2.1 Comparison with Adjacent Selection Criteria

Four literatures use ideas close enough to this paper’s that the distinction matters.

**Versus plain MDL/BIC model selection.** Classical MDL and BIC model selection (Rissanen, 1978; Schwarz, 1978) score candidate models by a description-length or penalised-likelihood trade-off between fit

and *model complexity*, ordinarily measured by parameter *count*. The rationality cost of Section 3.4 is sensitive to the *arithmetic* of the fitted coefficients themselves, so that two candidate supports of identical cardinality  $|S|$  can separate sharply on cost if one rounds cleanly to small rationals and the other does not. Ordinary MDL/BIC, scoring by count alone, cannot make this distinction.

**Versus MML and Bayesian model selection.** Minimum Message Length (Wallace and Boulton, 1968) and Bayesian model selection more broadly integrate a prior over model space into a posterior, and typically report a score without certifying that no better-scoring model exists. CSNP’s rationality cost is used as a hard lexicographic primary key inside a Branch-and-Bound search with an admissible bound ( $R \geq |S|$ , Section 3.5), not integrated against a likelihood. The output is a certified optimum over  $(R, k, \text{gap})$  (Theorem 4.4), not a posterior sample.

**Versus integer-relation algorithms.** PSLQ and LLL-based lattice reduction (Ferguson et al., 1998; Lenstra et al., 1982) solve a different problem: given a fixed tuple of real numbers, find an exact integer relation among them. That problem presupposes the tuple whose relation is sought. It has no notion of *selecting which tuple*, among many nullspace directions, is the physically meaningful one. CSNP’s search is over *supports* of a nullspace basis, with rationality cost as the tie-break criterion once a support is fixed.

**Versus sparse coding and compressed sensing.** The spark condition underlying exact sparse recovery (Donoho and Elad, 2003) has a direct nullspace analogue used in our recovery guarantee (Proposition 4.1). What compressed sensing does not supply is a criterion for choosing *among* equally sparse candidates once uniqueness fails, which is the  $d > 1$  regime Kaiser et al. (Kaiser et al., 2018) identified as unresolved and this paper’s rationality-cost tie-break targets.

### 3 Method: SR-GB+CSNP

**Problem formulation.** Given a dataset  $\mathcal{D} = \{x_i\}_{i=1}^N$  with  $x_i \in \mathbb{R}^n$ , the objective is to recover generators of the unknown polynomial ideal  $I \subset \mathbb{R}[x_1, \dots, x_n]$  such that  $p(x_i) \approx 0$  for all  $p \in I$  and all  $i$ . The ideal  $I$  is the natural algebraic object of interest because it is unique up to algebraic equivalence, whereas any particular generating set is not. The goal is hence not merely *some* generator of  $I$ , but a *canonical* one: the reduced Gröbner basis (Cox et al., 2015), unique for any fixed monomial ordering and minimal in the sense that no generator’s leading monomial divides another’s.

#### 3.1 Notation

Table 2 summarises the principal notation used throughout the paper. Throughout this paper we use the **graded reverse lexicographic (grevlex)** monomial ordering; the reduced Gröbner basis is unique for this fixed ordering (Cox et al., 2015). By “minimal” we mean minimal with respect to leading monomials: no generator’s leading monomial is divisible by another’s. This is the standard algebraic notion and is distinct from minimality of cardinality among all generating sets of the same ideal. The end-to-end pipeline is also illustrated diagrammatically in Figure 1.

Table 2: Notation summary.

| Symbol                                                     | Meaning                                                                                              |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $\mathcal{D} = \{x_i\}_{i=1}^N$                            | Input dataset of $N$ observations, each $x_i \in \mathbb{R}^n$                                       |
| $n$                                                        | Number of state variables                                                                            |
| $D$                                                        | Current monomial degree in the adaptive loop                                                         |
| $M$                                                        | Number of monomials in the current active library                                                    |
| $\varphi(x)$                                               | Monomial feature map $\mathbb{R}^n \rightarrow \mathbb{R}^M$                                         |
| $\Phi \in \mathbb{R}^{N \times M}$                         | Design (lifted data) matrix                                                                          |
| $U \in \mathbb{R}^{N \times N}$                            | Left singular vectors of $\Phi$ (from SVD (Golub and Loan, 2013))                                    |
| $\Sigma \in \mathbb{R}^{N \times M}$                       | Diagonal matrix of singular values of $\Phi$                                                         |
| $V \in \mathbb{R}^{M \times M}$                            | Right singular vectors of $\Phi$ ; last $d$ columns form $V_{\text{null}}$                           |
| $\sigma$                                                   | True noise standard deviation (additive Gaussian)                                                    |
| $\hat{\sigma}$                                             | Estimated noise magnitude via Marchenko–Pastur tail median                                           |
| $\rho_i = \sigma_i / \sigma_{i+1}$                         | Consecutive singular value ratio; largest $\rho_i$ locates the spectral gap                          |
| $r$                                                        | Numerical rank: $r = \arg \max_i \rho_i$ (gap heuristic, robust for unknown $\sigma$ )               |
| $d = M - r$                                                | Nullspace dimension; $d > 1$ triggers CSNP                                                           |
| $V_{\text{null}} \in \mathbb{R}^{M \times d}$              | Nullspace basis (last $d$ right singular vectors)                                                    |
| $\alpha \in \mathbb{R}^d$                                  | Nullspace coordinate vector; $c = V_{\text{null}} \alpha$                                            |
| $c \in \mathbb{R}^M$                                       | Coefficient vector of recovered invariant polynomial                                                 |
| $S \subseteq \{1 \dots M\}$                                | Support set (active monomials) of $c$                                                                |
| $k =  S $                                                  | Sparsity (number of non-zero monomials)                                                              |
| $k_{\text{max}}$                                           | Support-size cap for the BB search (default 6)                                                       |
| $R(c)$                                                     | Rationality cost (information-theoretic bit cost)                                                    |
| $\text{gap}(S) = \sigma_d(V_{\text{null}}[\text{not } S])$ | Stability gap (tertiary tie-breaker)                                                                 |
| $\text{UB} = (R, k, \text{gap})$                           | Upper bound cost tuple in BB                                                                         |
| $\text{LB} =  S_{\text{partial}} $                         | Lower bound (sparsity) for pruning                                                                   |
| $\rho_{\min}^A = 20$                                       | Minimum gap ratio for the submatrix rank test of Section 3.5 (feasibility classification)            |
| $Q_{\text{max}} = 16$                                      | Maximum denominator for snap-rounding, shared with the search oracle’s denominator ceiling           |
| $D_{\text{max}}$                                           | Safety bound for the adaptive loop (default 10)                                                      |
| $\mathcal{J}$                                              | Recorded sequence of MDL costs $\text{MDL}(D)$ , one entry per degree at which a candidate was found |
| $I \subset \mathbb{R}[x]$                                  | Unknown polynomial ideal satisfied by the data                                                       |
| $\text{GB}(I)$                                             | Reduced Gröbner basis of $I$ over $\mathbb{Q}$                                                       |

### 3.2 Adaptive Monomial Expansion and SVD Update

Instead of fixing a maximum degree upfront, the pipeline starts with degree  $D = 1$  and progressively adds monomials of exact degree  $D$  to the active library. At each step, the design matrix  $\Phi$  is augmented with new columns corresponding to all monomials of degree  $D$  not already present.

The pipeline maintains a QR factorization  $\Phi = QR$ , extended by Householder-based column insertion (Golub and Loan, 2013) as each new batch of monomials arrives, and takes a fresh SVD of  $R$  alone at every degree. Since  $Q$  has orthonormal columns,  $R$ ’s singular values and right singular vectors are exactly  $\Phi$ ’s; because  $R$  is only  $M \times M$ , independent of the sample count  $N$ , this fresh SVD costs  $O(M^3)$  and matches a direct SVD of  $\Phi$  to machine precision, with no drift accumulated across degrees. The left singular vectors of  $\Phi$  live in sample space and are never needed: CSNP and the MDL cost below consume only singular values and right singular vectors, both of which  $R$  already supplies exactly. After each expansion, CSNP is run on the current nullspace to search for sparse invariants.

This adaptive strategy removes the need to specify the maximum degree  $D$  when it is not known in advance; the algorithm discovers it via the MDL-based stopping criterion described in Section 3.10.



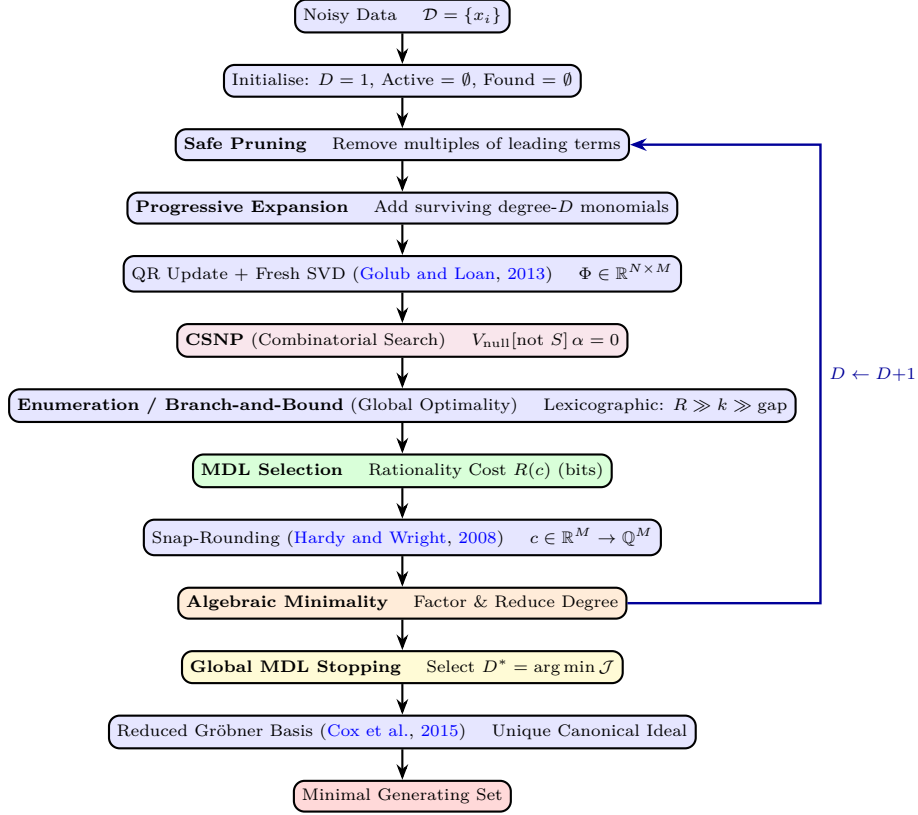


Figure 1: SR-GB+CSNP pipeline with adaptive loop. The CSNP step enforces the exact nullspace constraint  $V_{\text{null}}[\text{not } S]\alpha = 0$  as a hard filter and routes each instance by predicted cost to direct enumeration or the BB engine, with an anytime fallback beyond both budgets (Section 3.6). The exact engines perform a provably convergent global search over monomial supports with lexicographic priority: **rationality cost**  $R$  (first), then **sparsity**  $k$  (second), then the **geometric stability gap** (third; BB engine only, Section 3.5). The safe pruning step removes algebraically redundant monomials using a partial Gröbner basis. The global MDL stopping selects the optimal degree from the explored range. Citations indicate the established methods on which each stage builds; the contributions of this paper are the BB optimisation, the information-theoretic rationality cost, the adaptive expansion and pruning, and their integration into a coherent end-to-end pipeline.

**Remark 3.1** (Bias in Lifted Monomial Columns under Input Noise). *With additive noise on each observed variable,  $z = x + e$ ,  $e \sim \mathcal{N}(0, \sigma^2)$ , the lifted columns are biased estimates of the true monomials:  $\mathbb{E}[z^2] = x^2 + \sigma^2$ ,  $\mathbb{E}[z^3] = x^3 + 3\sigma^2 x$ , and so on up the degrees. The bias is invisible at low noise, but once  $\sigma$  grows the sampled moments genuinely sit on a slightly perturbed variety, and the search then converges to that perturbed variety, not the true one; on the unit circle, for instance,  $\mathbb{E}[(x + e_1)^2 + (y + e_2)^2] = 1 + 2\sigma^2$ , so the fitted constant drifts away from  $-1$  and eventually snaps to a wrong nearby fraction. The remedy is classical errors-in-variables estimation, applied to the columns and not to any gate or prior: when the caller supplies a noise level  $\sigma$ , each power  $z^a$  is replaced by the Hermite-style polynomial  $h_a(z)$  given by  $h_0 = 1$ ,  $h_1 = z$ , and the recurrence  $h_{a+1} = z h_a - a\sigma^2 h_{a-1}$  (so  $h_2 = z^2 - \sigma^2$  and  $h_3 = z^3 - 3\sigma^2 z$ ), which satisfies  $\mathbb{E}[h_a(x + e)] = x^a$  exactly; since the noise is independent across variables, the product of per-variable estimators is unbiased for every mixed monomial as well. At  $\sigma = 0$  the recurrence collapses to the plain powers, so noiseless behaviour is untouched. The correction removes the bias of the lifted columns but not their variance, which is what ultimately bounds the recoverable noise regime measured in Section 5.3.*

### 3.3 Nullspace Estimation via SVD

The nullspace of  $\Phi$  is estimated using Singular Value Decomposition (SVD) (Golub and Loan, 2013):  $\Phi = U\Sigma V^T$ , where  $U \in \mathbb{R}^{N \times N}$  contains the left singular vectors,  $\Sigma \in \mathbb{R}^{N \times M}$  is the diagonal matrix of singular values, and  $V \in \mathbb{R}^{M \times M}$  contains the right singular vectors. SVD is preferred over iterative nullspace solvers because it reveals the full nullspace basis in a single pass, in the columns of  $V$  that correspond to near-zero singular values. CSNP needs this full basis to search over sparse linear combinations.

The numerical rank  $r$  is estimated with a gap-based heuristic applied directly to the singular value spectrum. The consecutive ratios

$$\rho_i = \frac{\sigma_i}{\sigma_{i+1}}, \quad i = 1, \dots, M-1, \quad (1)$$

are computed, after which the rank is set to the index of the largest ratio:

$$r = \arg \max_i \rho_i. \quad (2)$$

If the largest ratio is  $\rho_i$ , the gap falls between  $\sigma_i$  and  $\sigma_{i+1}$ , so the signal subspace is exactly  $\sigma_1, \dots, \sigma_i$  and the rank is  $i$ . In effect, Equation (2) locates the spectral “elbow”, the point at which the signal subspace ends and the noise floor begins, without requiring an externally supplied threshold.

**Remark 3.2** (Rank Estimation on Flat or Gapless Spectra). *The pure ratio-gap rule of Equation (2) assumes a genuine spectral elbow exists, and can fail in either direction when it does not. In the first, a flat spectrum with no pronounced elbow, seen on the degree-3 Feynman equations where the largest ratio  $\rho_i$  is only  $\approx 1.5$ , makes  $r$  over-estimated, so the recovered nullity  $d = M - r$  is too small to contain the true invariant. In the second, a smoothly decaying spectrum with no genuine gap, characteristic of a large library lifted to high monomial degree, puts the largest consecutive ratio near the top of the spectrum rather than at the true nullspace boundary; a pure gap rule then returns a tiny rank and a nullity close to  $M$ , which blows up the downstream BB search. We therefore trust the gap only when it clears a minimum ratio  $\rho_{\min}$  (Table 20 gives the default); otherwise  $r$  is instead set by the standard absolute numerical-rank criterion (Golub and Loan, 2013),  $r = \#\{i : \sigma_i > \tau\}$  with  $\tau = \max(10^{-8}, 10^{-6}\sigma_1)$ , which does not depend on the spectrum having any particular shape. When a genuine elbow is present the condition  $\rho_i \geq \rho_{\min}$  always holds and the estimate is identical to Equation (2); when it is absent, the threshold fallback prevents a spurious rank estimate.*

The absolute noise magnitude  $\sigma$  is estimated separately using the Marchenko–Pastur tail heuristic:

$$\hat{\sigma} = \frac{\text{median}(\sigma_{r+1}, \dots, \sigma_M)}{\sqrt{N}}. \quad (3)$$

This estimate feeds only the downstream quantitative filters that require an absolute noise scale, namely the residual acceptance threshold  $\tau_{\text{resid}} = \max(10^{-4}, 3\hat{\sigma}) \cdot \sqrt{N} \cdot \Phi_{\text{rms}}$  in CSNP (Appendix A.3 gives the  $\Phi_{\text{rms}}$  scale-correction rationale), the snap-rounding tolerance  $\tau = \max(0.05, 3\hat{\sigma})$ , and the geometric condition  $\delta = \max(10^{-8}, 2\hat{\sigma}/\sqrt{N})$ .

Decoupling rank estimation from noise magnitude estimation avoids instability when the nullity is small (e.g.,  $d = 1$ ), because the tail would then contain too few singular values to estimate  $\sigma$  well. The pipeline finds the correct structural rank even when the absolute noise estimate is itself noisy.

### 3.4 Information-Theoretic Rationality Cost

The rationality cost  $R(c)$  is derived from the Minimum Description Length (MDL) principle (Rissanen, 1978). For each coefficient  $c_i$ , we find the simplest rational  $p_i/q_i$  with denominator  $q_i \leq Q_{\max}$  that lies within the noise-adaptive tolerance  $\varepsilon = \max(10^{-3}, 3\sigma)$ . The denominator ceiling  $Q_{\max}$  (Table 20) is a single constant shared with the snap-rounding step of Section 3.8, so the cost oracle that drives the search and the rounding stage that produces the exact answer draw from the same family of denominators,  $q \leq Q_{\max}$ , at every noise level; their acceptance windows differ, since  $\varepsilon$  scores a floating-point proxy during search while snap-rounding’s own, wider band (Section 3.8) applies once a fraction is actually committed to. If no such rational exists,  $R(c) = +\infty$ . The bit cost is:

$$R(c) = \sum_{i \in S} (\log_2(|p_i|) + \log_2(q_i) + 1), \quad (4)$$

where the  $+1$  accounts for the sign bit. This cost naturally enforces rational coefficients as the primary criterion without heuristic weights. Smaller  $R$  means coefficients closer to simple rationals; a  $\pm 1$  coefficient costs exactly 1 bit, so an all-unit support of size  $k$  costs  $R = k$ , the cheapest any support of that size can be. Sampling artefacts with irrational coefficients, by contrast, resist snap-rounding and receive high (or infinite) cost. The adaptive tolerance  $\varepsilon$  improves robustness to noise without breaking the theoretical guarantee  $R(c) \geq |S|$ : every declared support entry contributes at least 1 bit (entries whose estimate falls below  $\varepsilon$  are still charged this floor rather than dropped), and coefficients outside  $\varepsilon$  of every admissible rational yield  $+\infty$ .

The  $\varepsilon$ -window matters only while  $c$  is still a floating-point vector, where the intended fraction is unknown and the window expresses that uncertainty; there the cost is a proxy, charging each entry the cheapest admissible fraction it might be. Once snap-rounding (Section 3.8) has produced exact rationals, Equation (4) is evaluated on those fractions directly, with no window. Final candidate selection therefore always uses the exact post-snap cost (Remark 3.3).

**A worked example.** Consider the fixed- $dt$  harmonic oscillator of Section 5.8, sampled at  $dt = 0.1$  from the exact rotation flow. Two candidates sit in the nullspace. The energy  $x^2 + v^2$  has coefficients that a noisy raw SVD estimate might return as, say, 0.9998; snap-rounding sees this as within  $\varepsilon$  of the rational 1, so it costs 1 bit per coefficient and  $R = 4$  bits in total. The sampling artifact carries a coefficient equal to  $\cos(dt) = \cos(0.1) = 0.995004\dots$ , which is irrational and matches no fraction  $p/q$  with  $q \leq Q_{\max}$  within  $\varepsilon$ , so  $R = \infty$  for that candidate. Sparsity alone prefers the artifact, since its transition-invariant form has three terms against the energy’s four; rationality rejects it outright.

### 3.5 Branch-and-Bound Optimisation

When  $d > 1$ , plain SVD returns an arbitrary basis for the nullspace in place of the physically meaningful individual invariants. Not every such  $d$  calls for the same response, however: Table 3 separates the three structurally different reasons  $d > 1$  arises, only the first of which the search in this section is meant to resolve. We formulate the sparse nullspace recovery as a mixed-integer rank-constrained optimisation and solve it via **Branch-and-Bound (BB)** with dominance pruning (Nemhauser and Wolsey, 1988).

Table 3: Three sources of nullspace dimension  $d > 1$ , and how each is handled.

| Source                                                             | Example                                                                                                                               | Resolved by                                 | Status |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------|
| (a) Intrinsic: genuinely coexisting invariants                     | 2D oscillator, $d = 4$ : two per-axis energies, a cross term, angular momentum                                                        | CSNP with deflation (Section 3.13)          | solved |
| (b) Apparent, redundant: higher-degree multiples from over-lifting | circle at $D = 3$ : $p$ and $y \cdot p$ both vanish                                                                                   | minimality factorization (Section 3.7)      | solved |
| (c) Apparent, exact tie: dictionary-level algebraic dependencies   | unrestricted transition dictionary on genuine symplectic-Euler trajectories: update-rule ties, e.g. $(x_t - x_{t+1})^2$ (Section 6.5) | none yet; needs a structure-aware tie-break | open   |

### 3.5.1 Admissible Lower Bound

For any partial support  $S$ , the rationality cost of any completion satisfies  $R \geq |S|$  because each active monomial contributes at least one bit. Thus  $\text{LB} = |S|$  is admissible.

### 3.5.2 Dominance Pruning

If  $S$  is a subset of a previously detected hard-infeasible support, then by Lemma 4.2 all super-sets of  $S$  are also hard-infeasible. We cache such sets and prune any node whose support is dominated.

### 3.5.3 Symmetry Breaking

Monomials are ordered canonically; we only branch by adding indices greater than the current maximum, avoiding duplicate permutations.

### 3.5.4 Warm-Start with Beam Search

Before exact BB, a beam search of fixed width (Table 20) obtains an initial upper bound, accelerating pruning.

### 3.5.5 Direct Enumeration for Small $d$

When the nullspace dimension is small enough that the dual subset count  $\binom{M}{d-1}$  fits the enumeration budget of Section 3.6, we search in nullspace-coefficient space directly, not over primal supports. Every size- $(d-1)$  subset  $Z$  of coordinate indices pins a candidate direction  $\alpha$  up to scale:  $\alpha$  is the generalised cross product of the corresponding  $d-1$  rows of  $V_{\text{null}}$ , orthogonal to all of them by construction, so  $c = V_{\text{null}}\alpha$  vanishes exactly on  $Z$  whenever those rows are independent. Reading the pinned direction off a  $(d-1) \times d$  determinant is cheaper per candidate than the BB engine’s general feasibility classification, which instead tests the rank of the full  $(M - |S|) \times d$  complement of a primal support, and the whole batch of subsets is evaluated at once, not one at a time. Every surviving candidate is then scored by the primary sub-objective  $(R(c), |S|)$  of Equation (5) ( $S$  the candidate’s own thresholded support, not the pinning subset  $Z$  that produced it; the tertiary  $\text{gap}(S)$  term is not evaluated here, see the note after Equation (5) below), and only the globally best-scoring candidate under that sub-objective is returned. This enumeration costs  $O\left(\binom{M}{d-1}\right)$ , independent of both  $k_{\text{max}}$  and the answer’s own support size  $k = |S|$ , which is what lets it reach dense invariants ( $k$  close to  $M$ ) that the general BB path cannot afford (its cost is exponential in  $k$ , not in  $d$ ). It is guarded by its own combinatorial budget (Section 3.6), and beyond it the general BB search or the L1 fallback is used instead.

### 3.5.6 The Cost Tuple

For a candidate support  $S$  and coefficient vector  $c = V_{\text{null}}\alpha$ , we define the lexicographic cost tuple:

$$\text{cost}(S, c) = (R(c), |S|, \text{gap}(S)), \quad (5)$$

where  $R(c)$  is the rationality cost from Equation (4),  $|S|$  is the sparsity, and  $\text{gap}(S) = \sigma_d(V_{\text{null}}[\text{not } S])$  is the stability gap (tertiary tie-breaker; smaller gaps indicate a more isolated null direction). The optimisation problem is:

$$\min_{S: \text{rank}(V_{\text{null}}[\text{not } S])=d-1, |S| \leq k_{\max}} (R(c), |S|, \text{gap}(S)), \quad (6)$$

with minimisation over the *lexicographic ordering* of the tuple.

In the implementation, the tertiary  $\text{gap}(S)$  term is evaluated by the general BB engine (Section 3.5) as part of its feasibility oracle, since that oracle already computes  $A$ 's spectrum to classify  $S$  and  $\sigma_d$  falls out for free. The direct-enumeration engine above and the anytime branching corner of Section 3.6 instead rank candidates by the primary sub-objective  $(R(c), |S|)$  alone. This asymmetry was tested directly: an uncapped, gap-aware version of both was implemented and measured against the full 26-equation, 30-seed Feynman suite. It changed nothing at  $\sigma = 0$  or  $\sigma = 0.02$ , and cost one trial at  $\sigma = 0.05$  (I.39.22, one seed) relative to the  $(R, |S|)$ -only version, with no offsetting gain anywhere in the suite: on that trial a spurious two-term candidate happened to carry a smaller measured gap than the true four-term invariant, and the gap-aware ranking preferred it. This is the same hazard already noted for the feasibility oracle's own gap heuristic, a smaller gap is not always the more trustworthy one in a finite, noisy sample, so these two engines keep the  $(R, |S|)$  sub-objective and the gap term remains exclusive to the general BB engine.

### 3.5.7 Feasibility Classification

Given a candidate support  $S$ , we extract  $A = V_{\text{null}}[\text{not } S]$  and compute its singular values  $\sigma_1 \geq \dots \geq \sigma_d$  ( $A$  has  $d$  columns). Instead of comparing  $\sigma_d$  against an absolute, noise-derived tolerance, the implementation applies the same gap heuristic used for the top-level rank estimate (Equation (2)) directly to  $A$ 's own spectrum: it locates the largest consecutive ratio  $\sigma_i/\sigma_{i+1}$  and trusts it only if it clears a fixed minimum gap ratio  $\rho_{\min}^A$ , set well above the  $\rho_{\min}$  of Remark 3.2 (Table 20 gives both defaults) because a spurious gap is more likely to appear by chance in a small  $(M - |S|) \times d$  submatrix than in the full spectrum of  $\Phi$ . If the largest ratio clears  $\rho_{\min}^A$  at position  $i$ ,  $A$ 's numerical rank is taken to be  $i$ ; otherwise  $A$  is conservatively treated as full column rank ( $= d$ ), which is the Hard Infeasible outcome below. Writing  $\text{rank}(A)$  for this gap-based estimate, we classify:

1. **Hard Infeasible:**  $\text{rank}(A) = d$ . No null vector exists on this support (or no reliable gap was found).
2. **Soft Infeasible (Reducible):**  $\text{rank}(A) \leq d - 2$ . Multiple null vectors exist; the search instead looks for a smaller *primal* subset  $S^* \subset S$  at which  $\text{rank}(A_{S^*}) = d - 1$  exactly.
3. **Feasible:**  $\text{rank}(A) = d - 1$ . A unique null vector exists.

For feasible supports, the null vector is the right singular vector corresponding to the smallest singular value  $\sigma_d$ . The gap test above is noise-independent by construction, since  $A$ 's own spectrum already reflects the noise level indirectly through  $V_{\text{null}}$ . It is not, however, infallible: a genuine one-dimensional near-kernel can fail to clear  $\rho_{\min}^A$  purely because its neighbouring ratio is unlucky, without  $\sigma_d$  itself ever leaving the noise floor. A conservative guard covers this case: whenever the gap test alone would return Hard Infeasible ( $\text{rank}(A) = d$ ) and the relative smallest singular value  $\sigma_d/\sigma_1$  falls at or below a fixed multiple of the noise estimate ( $\kappa\sigma$ , with  $\kappa$  given in Table 20), the support is instead treated as feasible ( $\text{rank}(A) = d - 1$ ) and passed on to evaluation, where the existing rationality and residual gates remain free to reject it. This is a second, separately tuned absolute threshold layered on the gap test, deliberately so: it only ever pulls a support back from Hard Infeasible into evaluation and never removes one the gap test would otherwise admit, so it cannot introduce a new false Feasible verdict, only reduce false Hard Infeasible ones.



### 3.5.8 Branch-and-Bound Master

The BB master maintains:

- A priority queue of partial supports  $S_{\text{partial}}$ , ordered by lower bound  $\text{LB} = |S_{\text{partial}}|$  (since  $R \geq k$ ).
- An upper bound  $\text{UB} = (R^*, k^*, \text{gap}^*)$  from the best feasible solution found so far.
- A cache of hard-infeasible supports for dominance pruning.

#### Pruning Rules:

1. **Lower Bound Pruning:** If  $\text{LB} > \text{UB}[0]$  (rationality), prune the node. Ties on  $\text{LB} = \text{UB}[0]$  are not pruned: since  $\text{gap}(S)$  is part of the lexicographic objective of Equation (9), a partial support tied with the incumbent on  $(R, k)$  so far could still complete to a strictly better gap, and pruning it would make Theorem 4.4’s optimality claim false over the full  $(R, k, \text{gap})$  tuple, not merely over  $(R, k)$ . This costs some additional node exploration at equal  $(R, k)$  compared to a rule that also cut on the tie, but keeps the certificate exact.
2. **Dominance Pruning:** If  $S_{\text{partial}}$  is a subset of any cached hard-infeasible support, prune the node. This is justified by the monotonicity of rank (Lemma 4.2).

**Branching:** Add one monomial index at a time, enforcing canonical ordering ( $i > \max(S_{\text{partial}})$ ) to avoid duplicates.

## 3.6 CSNP: Combinatorial Sparse Nullspace Pursuit

Algorithm 1 states the full procedure. Its circuit search selects one of three engines, routed by predicted computational cost (Step 3 of the algorithm gives the exact budgets), all three ordering candidates by the primary sub-objective  $(R(c), |S|)$  of Equation (5) and the general BB engine (Section 3.5) alone additionally applying the tertiary  $\text{gap}(S)$  tie-break, for reasons the Cost Tuple discussion of that section explains: direct support enumeration when the dual cost is affordable, the general BB search at the largest affordable support cap  $k' \leq k_{\text{max}}$  otherwise, and, when neither exact engine fits its budget, an anytime branching search over per-coordinate zero/nonzero commitments whose incumbent is seeded by an L1 relaxation. Only that last engine is heuristic, and only when its node cap is exhausted before the tree is; exhausting the tree certifies the same optimum the two exact engines do.

*Note on the geometric threshold:* In Steps 3–4, the threshold is noise-adaptive:  $10^{-8}$  in the noiseless limit, relaxing to  $2\sigma/\sqrt{N}$  for larger noise levels.

**Remark 3.3** (Progressive Nullspace-Dimension Search). *Algorithm 1 presents CSNP for a single, already-determined nullspace dimension  $d$ . In practice we do not commit to one upfront  $d$  from the shape of the singular-value spectrum, since Remark 3.2 shows this can fail in both directions: under-estimating  $d$  discards the true invariant’s direction entirely, while over-estimating  $d$  folds noise-dominated directions into the nullspace estimate and breaks the rank-based feasibility classification of Section 3.5. The search instead considers candidate nullspace dimensions  $d = 1, 2, 3, \dots$  directly, up to a small cap: for each candidate  $d$ , the corresponding  $d$  smallest singular vectors are taken as a candidate nullspace basis, CSNP is run on exactly that basis (enumeration, BB, or the anytime fallback, routed by predicted cost as in Section 3.6), and the full post-processing pipeline of Section 3.7–3.12 is applied before anything is accepted or compared. In the enumeration regime the search hands over its few top-ranked candidates, not a single winner, since its in-search score is the floating-point proxy of Section 3.4 and the proxy’s ranking can disagree with the exact post-snap cost precisely in the noisy regime where the  $\varepsilon$ -window is wide. Each handed-over candidate is fully post-processed, and all survivors, within one  $d$  and across different  $d$ , are ranked by the same exact lexicographic cost  $(R(c), |S|)$  evaluated on their snapped rational coefficients; the scan terminates early once a candidate attains the minimum possible cost for its sparsity ( $R = |S|$ , i.e. all unit-integer coefficients). This early exit is a useful heuristic, not a cross- $d$  optimality certificate: a larger  $d$  could in principle hold a sparser candidate with  $R' = |S'| < |S|$ , which the exit forgoes checking; we have not observed this on any benchmark, but the certificate of Theorem 4.4 remains per- $d$ .*

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**Algorithm 1** CSNP: Combinatorial Sparse Nullspace Pursuit, with Branch-and-Bound as one of its three engines

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- 1: **Input:**  $\Phi \in \mathbb{R}^{N \times M}$ , noise estimate  $\sigma$ ,  $k_{\max}$  (default 6).
  - 2: **SVD** (Golub and Loan, 2013):  $\Phi = U\Sigma V^T$ . Estimate  $r$  via the gap rule of Equation (2) and Remark 3.2 (falling back to the absolute threshold  $\tau = \max(10^{-8}, 10^{-6}\sigma_1)$  when no gap clears  $\rho_{\min}$ ),  $V_{\text{null}} = \text{last } (M - r) \text{ columns of } V$ ,  $d = M - r$ . **If**  $d = 1$ : set  $c \leftarrow$  last column of  $V$  and goto Step 7.
  - 3: **Route by predicted cost:** **if**  $\binom{M}{d-1} \leq 2 \times 10^5$ , enumerate supports directly (cheap rank tests, same objective) and goto Step 6; **else if**  $\sum_{k=1}^{k'} \binom{M}{k} d^2 \leq 2 \times 10^8$  for some  $k' \leq k_{\max}$ , run BB at the largest such support cap  $k'$  (Step 4); **else** goto the anytime fallback (Step 5).
  - 4: **Branch-and-Bound (exact search):** Initialise  $\text{UB} = (\infty, \infty, \infty)$ , priority queue with  $\text{LB} = 0$ . For each partial support  $S_{\text{partial}}$ :
    1. Classify feasibility via  $V_{\text{null}}[\text{not } S]$ .
    2. If feasible, compute  $c = V_{\text{null}}\alpha$ ,  $R(c)$ ,  $k = |S|$ , gap; update  $\text{UB}$  if  $(R, k, \text{gap}) < \text{UB}$ .
    3. If hard infeasible, cache  $S$  for dominance pruning.
    4. Branch by adding one index  $i > \max(S_{\text{partial}})$ .
  - 5: **Anytime fallback** (both exact engines over budget): Parameterise  $c = V_{\text{null}}\alpha$ , fix  $\alpha_1 = 1$ , solve  $\min \|V_{\text{null}}\alpha\|_1$  via LASSO, threshold, reconstruct  $c$ ; this L1 vector seeds the incumbent of a node-capped branch-and-prune over per-coordinate zero/nonzero commitments, each node retrying the two exact engines on its reduced instance. *Note: if the node cap is exhausted the returned incumbent carries no optimality certificate; Proposition 4.1 applies only to the exact engines.*
  - 6: **Post-processing:** Threshold  $|c_i| < \varepsilon$  to zero, with  $\varepsilon = \max(10^{-3}, 3\sigma)$ ; snap-round to rationals (Section 3.8); apply algebraic minimality reduction (Section 3.7); compute reduced Gröbner basis over  $\mathbb{Q}$  (Cox et al., 2015).
  - 7: **Nullspace deflation** for multiple invariants (optional): project out the recovered direction via  $V_{\text{null}} \leftarrow \left(I - \frac{cc^T}{\|c\|^2}\right)V_{\text{null}}$ , set  $d \leftarrow d - 1$ , and repeat from Step 2. No more than 2–3 iterations are recommended to avoid propagating numerical errors into noise-dominated subspaces.
- 

### 3.7 Algebraic Minimality Post-Processing

When the current degree  $D$  exceeds the true degree of the invariant, the recovered polynomial may be a higher-degree multiple of the true generator, for example  $y(x^2 + y^2 - 1)$  in place of  $x^2 + y^2 - 1$ . Degree minimality is enforced through a data-driven factorisation step:

1. Factor the recovered polynomial over  $\mathbb{Q}$  after converting to integer coefficients.
2. For each distinct factor  $f$ , evaluate  $f$  on the dataset. If the maximum absolute residual is below a scale-relative tolerance  $\tau_{\max} = \max(\tau_{\text{floor}}, 3\sigma \cdot g \cdot \sqrt{2 \ln N})$ , where  $g$  is the median gradient norm of  $f$  on the data and  $\tau_{\text{floor}} = \max(10^{-4}, 3\sigma)$ , replace the polynomial with  $f$  and recurse.
3. If no factor passes, attempt division by each variable.

Both tolerances above depart from the aggregate norm check of Section 3.6 for the same underlying reason, an absolute tolerance calibrated for a different statistic reused for a per-point, scale-dependent check; Appendix A.3 works through why that reuse is incorrect and what the corrected tolerances are.

### 3.8 Rational Snap-Rounding

Floating-point SVD coefficients must be converted to exact rationals before Gröbner canonicalisation. The procedure follows Legendre’s theorem on Diophantine approximation (Hardy and Wright, 2008):

1. Normalise  $c$  by its dominant coefficient, and zero out coefficients whose absolute value is below  $\varepsilon = \max(10^{-3}, 3\sigma)$ .

2. For each  $c_i$ , scan all denominators  $q = 1, \dots, Q_{\max}$  with  $Q_{\max} = 16$ , the same denominator ceiling the rationality-cost oracle of Section 3.4 certifies, at every noise level. If some  $p/q$  matches to machine precision ( $|c_i - p/q| < 10^{-4}$ ), take the lowest-bit-cost such exact match: exactness outranks simplicity, so an exact  $3/7$  is never rounded to a nearby cheaper  $2/5$ .
3. Otherwise select from the noise band  $|c_i - p/q| < \tau$  with  $\tau = \max(0.05, 3\sigma)$ , preferring the lowest-bit-cost fraction in the band (the MDL prior). At moderate noise the band can contain both the true fraction and a simpler wrong one; the acceptance gate below, not the band selection, is what adjudicates.
4. Fall back to the best rational approximation with denominator at most  $Q_{\max}$  if no denominator in the scan succeeds.

The snapped candidate is then re-scored on the data: if its relative RMS residual degrades by more than a factor of 1.5 over the unsnapped floating-point candidate, the snap is rejected and retried with nearest-fraction band selection in place of lowest-cost, and if that snap degrades the residual as well, the candidate is discarded. True snaps measured on the benchmarks sit near  $1.0\times$  the float residual while wrong band picks sit near  $1.6\text{--}1.8\times$ , so the gate separates them cleanly, and a coefficient the denominator window cannot represent produces an abstention, not a confidently wrong rational.

### 3.9 Algebra-Guided Pruning

After a candidate invariant is confirmed (via snap-rounding and residual check), we compute a partial reduced Gröbner basis of the ideal generated by all confirmed invariants found so far. We then identify monomials in the active library that are multiples of any leading term of this partial basis and permanently remove them from the search space.

Justifying this removal directly on the undiscovered invariant  $p$ , on the grounds that  $p$  itself always keeps some monomial outside the redundant set, is not valid in general: Section 4.3 gives an explicit  $p$  whose entire support is redundant. The correct object is the normal form of  $p$  under division by the confirmed basis, not  $p$  itself.

The formal statement and proof are given as Lemma 4.3 in Section 4.3: the normal form  $q = \text{NF}_G(p)$  of an undiscovered minimal generator  $p$  is nonzero, lies in the true ideal, has every monomial of its support surviving pruning, has degree at most that of  $p$ , and generates the same extension as  $p$  would.

Pruning therefore never removes the search space needed to recover the next generator: it only means the search recovers  $q$ , not  $p$ , when the two differ, and  $q$  extends  $I_G$  to exactly the same ideal  $p$  would have.

### 3.10 MDL Stopping and Global Degree Selection

The adaptive loop explores degrees from 1 up to a safety bound  $D_{\max}$  (default 10). At each degree, after running CSNP and confirming the best candidate (if any), we record the MDL cost  $\text{Cost}(c)$  of that candidate. We also store the candidate itself. If no valid candidate is found at a degree, we record  $\infty$ .

The full MDL cost is defined as:

$$\text{MDL}(p) = R(c) + \log_2 \binom{M}{k} + \log_2(D_{\max}) + \frac{\log_2 e}{2\sigma^2} \sum_{i=1}^N p(x_i)^2, \quad (7)$$

where  $R(c)$  is the rationality bit-cost,  $\binom{M}{k}$  encodes the number of monomials,  $\log_2(D_{\max})$  encodes the degree, and the final term is the Gaussian negative log-likelihood of the residuals converted to bits by the factor  $\log_2 e$ , so that all four terms are code lengths in the same unit.

After the loop terminates, we select the degree  $D^*$  that achieved the *global minimum* MDL cost over all explored degrees. We then discard all generators found at degrees greater than  $D^*$ , keeping only those found up to  $D^*$ . Global-minimum selection tolerates non-convex cost landscapes better than first-local-minimum selection would, but the criterion itself is a heuristic: Proposition 4.5 (Section 4.5) shows the sweep is exactly correct *when* the true candidate uniquely minimises this cost (that proposition’s assumption (4)), and that assumption is a property of the data and dictionary, proved only for the symplectic sub-case of Section 4.6 and validated empirically elsewhere.

### 3.11 Stopping Criterion

After each degree  $D$ , if  $\text{MDL}(D) - \text{MDL}(D - 1) > \frac{1}{2} \log_2(N)$  (the BIC penalty converted to bits), we stop expanding; this prevents overfitting.

Here  $\text{MDL}(D)$  and  $\text{MDL}(D - 1)$  denote the two most recently recorded entries of  $\mathcal{J}$  (Table 2; the list is maintained by Algorithm 2), not necessarily integer-adjacent degrees: a degree contributes an entry only when a candidate is actually found there, so this comparison silently skips over any empty degree between them and only ever fires between two degrees that both found something. Once the true invariant is found, however, every higher degree’s nullspace also contains redundant multiples of it, so CSNP can keep returning a candidate at every subsequent degree indefinitely. We therefore also track a small consecutive-miss counter, incremented whenever a degree contributes no new information, whether because nothing is found or because the best candidate found is the same invariant, up to a scalar, already accepted at a lower degree, and reset whenever a genuinely different candidate appears. One point matters here: the counter does not advance on an empty degree that precedes the first find: an all-empty prefix of low degrees (a genuine and common situation, since the true invariant’s own degree sets a hard lower bound below which every degree is necessarily empty) must not exhaust  $\mu$  before the sweep has ever had a chance to reach that degree. The sweep stops once this counter reaches a small constant (three, in the experiments reported here). Because the counter only ever resets on a genuinely new candidate, it cannot mask a later, higher-degree competitor: global selection over all explored degrees still sees it if the sweep reaches it.

Algorithm 2 states the full outer loop, calling Algorithm 1 once per degree.

---

**Algorithm 2** Adaptive Degree Sweep (calls Algorithm 1 once per degree)

---

- 1: **Input:** data  $\mathcal{D}$ , variables  $x$ , safety bound  $D_{\max}$  (default 10), consecutive-miss bound  $\mu$  (default 3).
  - 2: **Initialise:**  $D = 1$ , confirmed basis  $G = \emptyset$ ,  $\mathcal{J} = []$ , miss count  $m = 0$ , last accepted invariant  $q_{\text{prev}} = \perp$ .
  - 3: **Prune:** let  $\mathcal{M}_D$  be the monomials of exact degree  $D$ ; remove those divisible by a leading monomial of  $G$  (Lemma 4.3). **If** none of  $\mathcal{M}_D$  survives:  $m \leftarrow m + 1$  *only if*  $\mathcal{J} \neq []$  (the counter never advances before the first candidate is found); **if**  $m \geq \mu$  goto Step 7, else set  $D \leftarrow D + 1$  and repeat this step (terminate at Step 7 if  $D > D_{\max}$ ).
  - 4: **Expand:** extend the active design matrix by the surviving monomials via QR column-insertion, and take a fresh SVD of the resulting  $R$  factor (Section 3.2).
  - 5: **Search:** run Algorithm 1 on this degree- $D$  nullspace basis for a candidate  $p$ . **If** none found:  $m \leftarrow m + 1$  *only if*  $\mathcal{J} \neq []$ ; **if**  $m \geq \mu$  goto Step 7, else set  $D \leftarrow D + 1$  and goto Step 3 (terminate at Step 7 if  $D > D_{\max}$ ).
  - 6: **Accept:** reduce  $p$  to its minimal generator  $q$  (Section 3.7); compute  $\text{MDL}(D)$  (Section 3.10) and append it to  $\mathcal{J}$ ; set  $G \leftarrow G \cup \{q\}$ . **If**  $q$  equals  $q_{\text{prev}}$  up to a scalar,  $m \leftarrow m + 1$ ; otherwise  $m \leftarrow 0$  and  $q_{\text{prev}} \leftarrow q$ . **If**  $m \geq \mu$ , or  $|\mathcal{J}| \geq 2$  and its last two entries differ by more than  $\frac{1}{2} \log_2 N$ : goto Step 7. Otherwise set  $D \leftarrow D + 1$  and goto Step 3 (terminate at Step 7 if  $D > D_{\max}$ ).
  - 7: **Output:**  $D^* = \arg \min$  over  $\mathcal{J}$ ; return  $G$  restricted to invariants found at degree  $\leq D^*$ , canonicalised as a reduced Gröbner basis (Section 3.12).
- 

### 3.12 Gröbner Basis Canonicalisation

The snap-rounded polynomial(s) generate an ideal over  $\mathbb{Q}$ . The **reduced Gröbner basis** of this ideal is computed with respect to graded reverse lexicographic order using SymPy’s implementation of Buchberger’s algorithm with the standard Buchberger criteria (product and chain criteria) for avoiding redundant S-polynomials (Cox et al., 2015). Computation is performed over  $\mathbb{Q}$ , not  $\mathbb{R}$ , because uniqueness requires exact arithmetic. The monomial ordering is grevlex throughout, as stated in Section 3.1. For the polynomial ideals considered in this paper ( $n \leq 8$ ,  $D \leq 4$ ), the Gröbner basis computation is near-instantaneous; Section 6.10 provides detailed timings for larger configurations.

### 3.13 Multiple Invariants via Nullspace Deflation

When  $d > 1$  and multiple independent invariants are present, they can be recovered iteratively by nullspace deflation. After the first invariant  $c_1$  is identified, the nullspace basis is updated by projecting out the

recovered direction:

$$V_{\text{null}} \leftarrow \left( I - \frac{c_1 c_1^T}{\|c_1\|^2} \right) V_{\text{null}}, \quad (8)$$

which reduces  $d$  by 1, after which CSNP is re-applied.

## 4 Theoretical Analysis

### 4.1 Exact Recovery Condition and Stability

The key structural condition for exact recovery is:

**Proposition 4.1** (Exact Recovery, noiseless). *Let  $c^*$  be the true invariant with support  $S^*$ , and let  $V_{\text{null}}$  be a basis for the nullspace of  $\Phi$ . CSNP recovers  $c^*$  uniquely if and only if  $\text{rank}(V_{\text{null}}[\text{not } S^*]) = d - 1$ .*

*Proof.* See Appendix B. This condition is the nullspace analogue of the spark condition in compressed sensing (Donoho and Elad, 2003), characterising uniqueness of the sparse representation within the nullspace.

This is an algebraic characterisation of when a support admits a unique sparse solution; it does not by itself guarantee that any given search terminates having found it. The exact engines this pipeline runs toward that guarantee, direct enumeration and the branch-and-bound of Section 3.6, realise it once they explore the relevant search space fully, and the BB tree is provably finite (Theorem 4.4's proof bounds it by  $\sum_{k=0}^{k_{\max}} \binom{M}{k}$ ), so an exhaustive run always terminates and always returns  $c^*$  under this proposition's hypothesis: nothing about the recovery itself is conditional on a time or compute budget. What the deployed configuration adds is the anytime node budget discussed in Section 4.4, a computational resource chosen for a given problem size, not a hidden extra hypothesis of this proposition; that section also gives a real instance where the finite bound above is large enough to matter, and confirms that the deployed default was never binding on any benchmark in this paper.

### 4.2 Monotonicity of Rank and Dominance Pruning

The dominance pruning in the BB relies on the following monotonicity property:

**Lemma 4.2** (Monotonic Rank). *For any  $S_1 \subseteq S_2 \subseteq \{1, \dots, M\}$ , we have*

$$\text{rank}(V_{\text{null}}[\text{not } S_1]) \geq \text{rank}(V_{\text{null}}[\text{not } S_2]).$$

**Proof.** The matrix  $V_{\text{null}}[\text{not } S_1]$  is obtained from  $V_{\text{null}}[\text{not } S_2]$  by appending the rows corresponding to  $S_2 \setminus S_1$ . Appending rows cannot decrease the rank.  $\square$

**Corollary.** If  $S$  is *Hard Infeasible* ( $\text{rank}(V_{\text{null}}[\text{not } S]) = d$ ), then every  $S' \subseteq S$  is also Hard Infeasible. This justifies the dominance pruning rule: if a partial support is a subset of a cached hard-infeasible set, it can be safely pruned.

### 4.3 Normal-Form Pruning Lemma

We now prove the pruning rule introduced in Section 3.9. The property does not hold directly on  $p$ : it is not true in general that a minimal generator  $p \notin I_G$  keeps some monomial not divisible by any leading monomial of  $G$ , and the Remark following the proof gives an explicit counterexample. What is true, and what pruning actually needs, is the corresponding statement about the normal form of  $p$ .

**Lemma 4.3** (Normal-Form Pruning). *Let  $G$  be a reduced Gröbner basis of an ideal  $I_G \subset \mathbb{Q}[x]$  under a graded monomial order (grevlex, as used throughout this paper, qualifies), and let  $p \in I \setminus I_G$  for some ideal  $I \supseteq I_G$ . Let  $q = \text{NF}_G(p)$  be the remainder of  $p$  on division by  $G$ . Then:*

- (i)  $q \neq 0$ ;
- (ii)  $q \in I$ ;

(iii) every monomial of  $q$  is standard, i.e. divisible by no  $\text{LM}(g)$ ,  $g \in G$ , so every monomial of  $q$  survives pruning;

(iv)  $\deg q \leq \deg p$ ;

(v)  $\langle I_G \cup \{p\} \rangle = \langle I_G \cup \{q\} \rangle$ .

**Proof.** The division algorithm in  $\mathbb{Q}[x]$  writes  $p = \sum_{g \in G} a_g g + q$  with  $q$  a sum of monomials none of which is divisible by any  $\text{LM}(g)$ ; this is (iii). If  $q = 0$  then  $p = \sum_g a_g g \in \langle G \rangle = I_G$ , contradicting  $p \notin I_G$ , which gives (i). Since  $p \in I$  and each  $g \in G$  generates  $I_G \subseteq I$ ,  $q = p - \sum_g a_g g \in I$ , giving (ii). The order is graded, so the division algorithm never produces a remainder monomial of degree exceeding  $\deg p$ , giving (iv). Finally  $q$  is a  $\mathbb{Q}$ -linear combination of  $p$  and elements of  $G$ , and conversely  $p = q + \sum_g a_g g$  is a combination of  $q$  and  $G$ , so adjoining  $p$  to  $I_G$  and adjoining  $q$  to  $I_G$  produce the same ideal, giving (v).  $\square$

**Remark (why the direct claim fails).** Taking  $I_G = \langle x^2 - y \rangle$  and  $p = x^2$ , a minimal generator of  $I = \langle x^2, y \rangle \supset I_G$ , gives a case where the entire support of  $p$  is divisible by  $\text{LM}(x^2 - y) = x^2$ : a minimal generator outside  $I_G$  can be entirely non-standard, so no pruning argument, and no induction over generators such as the one behind Proposition 4.5, can rest on  $p$  keeping a standard monomial. Lemma 4.3 gives the correct statement: the normal form  $q = \text{NF}_G(p) = y$  is nonzero, standard, and extends  $I_G$  to the same ideal  $p$  would have, so pruning still removes  $x^2$  from the active library but the search recovers  $y$  from the surviving columns instead. Appendix F recaps this and additionally bounds how a badly conditioned cofactor could, in principle, amplify noise through this substitution on real data, and why that bound does not affect what the pipeline can be made to output.

#### 4.4 Global Optimality of Branch-and-Bound

**Theorem 4.4** (Global Optimality, Relative to the Feasibility Oracle). *Assume the feasibility classification of Section 3.5 (the  $\rho_{\min}^A$ -gated gap test on  $V_{\text{null}}[\text{not } S]$ 's spectrum, together with that section's  $\kappa\sigma$  noise-scale guard) correctly labels every support  $S$  encountered as Hard Infeasible if and only if  $\text{rank}(V_{\text{null}}[\text{not } S]) = d$ ; we call this the oracle hypothesis. Under it, the Branch-and-Bound with dominance pruning terminates in finite iterations, and upon termination the incumbent support  $S^*$  satisfies:*

$$S^* = \arg \min_{S: \text{rank}(V_{\text{null}}[\text{not } S])=d-1, |S| \leq k_{\max}} (R(c), |S|, \text{gap}(S)) \quad (9)$$

in the lexicographic order.

**Proof Sketch.** The BB tree is finite because we enforce  $|S| \leq k_{\max}$  and canonical branching ( $2^M$  nodes, capped by  $k_{\max}$ ). The algorithm disposes of every node in one of three ways: (i) evaluation, which updates UB; (ii) lower bound pruning, where  $\text{LB} = |S_{\text{partial}}| \geq \text{UB}$ . This is a valid admissible bound because  $R(c) \geq |S|$ : every active monomial contributes at least one bit to the rationality cost under the coding scheme of Section 3.4, so support size can never exceed rationality cost, and a partial support already as large as the incumbent's cost can be pruned without risk of discarding the true optimum, letting sparsity, which is cheap to track incrementally, do the pruning that rationality cost, which is expensive to evaluate per node, would otherwise have to do; (iii) dominance pruning, which is sound by Lemma 4.2 given the classification hypothesis above. Since no node with  $\text{LB} < \text{UB}$  is pruned, and the priority queue ensures exhaustive exploration of all nodes that could improve UB, the incumbent is guaranteed to be the global minimiser conditional on that hypothesis.  $\square$

The certificate also presumes the search ran to exhaustion, which the proof above shows always happens in finitely many iterations, bounded by  $\sum_{k=0}^{k_{\max}} \binom{M}{k}$  node evaluations in the worst case. The deployed configuration additionally imposes an anytime node budget ( $10^5$  nodes evaluated) on the BB sweep, a computational resource choice, not a correctness assumption: a run that hits the budget before exhausting the tree returns its best incumbent *without* the certificate and records that in the search log, and the budget can be raised for a larger instance, up to the full worst-case bound if needed. That bound is why a finite default is kept in place of always running to exhaustion: it genuinely can be very large. An instance with  $M = 45$ ,  $d = 13$ ,  $k_{\max} = 6$  (a monomial-library size well within reach of the certified regime  $n \leq 8$ ,  $D \leq 4$  of Section 6.10) has a raw worst-case node bound  $\sum_{k=0}^6 \binom{45}{k} \approx 9.5 \times 10^6$ , and its predicted-cost value under the routing formula of Section 3.6 ( $d^2$  times that sum,  $\approx 1.6 \times 10^9$  units) exceeds the  $2 \times 10^8$ -unit routing budget



at the full cap, so routing would instead run BB at the largest affordable reduced cap ( $k' = 4$ ,  $\approx 2.8 \times 10^7$  predicted units): the sweep still runs to exhaustion, but its certificate then covers only supports of size at most four, and a denser true invariant would sit beyond it unless the budget were raised. This instance is not one of the benchmarks reported in this paper, but it shows that no single fixed cap could be guaranteed to suffice for an arbitrary future instance. The node-count instrumentation reported at the end of this subsection never observed the deployed  $10^5$  budget being hit on any system in this paper (its largest measured sweep evaluated  $\sim 7 \times 10^4$  nodes, inside but within a factor of 1.4 of the cap); repeating that instrumentation with the cap removed entirely reproduces identical incumbents and identical node counts on every one of those systems, confirming the budget was never the limiting factor on the benchmarks reported here. The certificate is accordingly unconditional given a sufficient budget, and, for every system in this paper, the deployed default already was one.

**The oracle hypothesis can fail.** The feasibility oracle is itself a gap heuristic (Section 3.5), and Remark 3.2 already documents that the analogous top-level rank estimate can fail on flat or gapless spectra. A false Hard Infeasible verdict on some  $S$  is silently propagated by dominance pruning to every subset of  $S$ , which can excise the true optimum from the search without the algorithm ever detecting the loss.  $\rho_{\min}^A$  is set well above the top-level guard’s  $\rho_{\min}$  (Table 20) because this failure mode is worse than an occasional overestimate of  $d$ . The oracle is checked directly against a system whose true generators and true  $d$  are both known in advance, the 2D-oscillator deflation benchmark of Section 5.8 ( $d = 4$ ,  $\sigma = 0.02$ , 30 seeds): does the feasibility oracle classify each of the four true generators’ supports as feasible, or does it wrongly declare one Hard Infeasible? All four generators, the cross term  $xy + v_x v_y$ , angular momentum  $xv_y - yv_x$ , and the two per-axis energy terms  $x^2 + v_x^2$  and  $y^2 + v_y^2$ , are correctly classified feasible in all 30 seeds once the noise-scale guard of Section 3.5 is applied. Without the guard the two energy terms were wrongly declared Hard Infeasible in 6 of 30 seeds (20%) and 1 of 30 (3.3%) respectively, a real, non-trivial false-negative rate, not a merely theoretical one, and asymmetric across algebraically similar generators for reasons not fully understood; a false Hard Infeasible verdict at this stage means the search never considers that generator at all for the affected seed, regardless of what BB does afterward. The guard is an empirical margin, not a numerical-rank proof, and a companion sensitivity sweep across its multiplier and a wider noise range confirms it does not eliminate the failure mode everywhere: at higher noise ( $\sigma = 0.10$ ) on the same diagnostic, the best-attainable rescue rate plateaus around 93%. Theorem 4.4’s certificate should accordingly still be read as relative to the guarded oracle, not as an unconditional guarantee that the pipeline never discards the true answer, and the measured rate at this benchmark’s own noise level, 0/30 for all four generators with the guard, quantifies the remaining gap.

**Search efficiency, measured directly.** The oracle’s correctness just discussed is a separate question from how much work BB actually does to reach its answer. Node counters added to the search record its behaviour without altering it. On the two transition dictionaries examined throughout this section, Kepler angular momentum and the 2D harmonic oscillator, the cost-routed circuit search resolves the invariant through CSNP’s progressive  $d$ -sweep and its direct-enumeration path before general BB is ever invoked, so those counters read zero on both: they record that BB is bypassed on this class of instance instead of measuring its work. The search work is therefore measured where BB genuinely runs, on a high-dimensional sparse case: six variables confined to a rank-two latent subspace, a degree-two library whose nullspace has dimension around twenty-two, and a single sparse rational relation  $x_0 - x_1$  that the pinning enumeration cannot reach within budget. Each system is run at  $\sigma \in \{0, 0.02\}$ , ten seeds,  $N = 5000$ .

Measured directly, dominance pruning removes zero nodes at every noise level, for a structural reason: the exact sweep explores by non-decreasing support size, so by the time any support is proved Hard Infeasible every subset of it has already been evaluated, and a cache populated only by the sweep’s own findings never holds a usable entry. Seeding the cache from the beam warm start’s verdicts would make it fire, but is unsound, since the warm start’s noise-aware rank heuristic is not monotone under the subset relation across different supports; on the fixed- $dt$  rotation flow that variant discarded the true invariant outright (Section 5.8), so the BB search does not keep the warm start’s finds. The admissible lower bound carries what pruning there is. At  $\sigma = 0$ , averaged over the six seeds that recover  $x_0 - x_1$ , a sweep pops and evaluates 50,758 nodes and filters a further 19,656 (about 28% of every node generated) at push time; the other four seeds terminate after a few dozen nodes and abstain, returning no invariant instead of a wrong one, which pulls the all-seed mean (30,466 popped) below these per-recovery figures. At  $\sigma = 0.02$  the noise-aware gap rarely certifies infeasibility at push, so push-time pruning nearly vanishes, the successful sweeps visit 48,000 to 72,000 nodes,

and exact recovery again stands at 6 of 10 seeds.

## 4.5 Conditional Adaptive Exact Recovery

The following proposition extends Proposition 4.1 to the adaptive pipeline. It is deliberately conditional: its assumptions (3) and (4) are substantive, not consequences of the pipeline’s construction. The only part of either that is proved rather than assumed anywhere in this paper is the cost-ordering content of assumption (4), and only for the symplectic sub-case of Section 4.6. What the proposition contributes is not an unconditional guarantee but an exact isolation of what must hold for the adaptive sweep to be correct, so that each condition can be checked, proved, or refuted per setting.

**Proposition 4.5** (Conditional Adaptive Exact Recovery). *Let  $I_{D^*-1}$  be the ideal generated by the invariants already confirmed at degrees below  $D^*$ , with reduced Gröbner basis  $G_{D^*-1}$  (empty if none have been confirmed yet). Assume:*

1. *The true invariant ideal has a minimal generator  $p^*$  of total degree  $D^*$ , with  $p^* \notin I_{D^*-1}$ , and  $q^* = \text{NF}_{G_{D^*-1}}(p^*)$  its normal form (Lemma 4.3).*
2. *The safety bound satisfies  $D_{\max} \geq D^*$ .*
3. *The numerical rank estimation and CSNP thresholds are set such that Proposition 4.1 holds, at degree  $D^*$ , for the fixed design matrix restricted to the monomials that survive pruning against  $G_{D^*-1}$ ; by Lemma 4.3(iii)-(iv) this restricted matrix already contains the full support of  $q^*$ .*
4. *Among all candidates the sweep confirms at any degree in  $1, \dots, D_{\max}$ , the candidate  $q^*$  confirmed at  $D^*$  attains the strictly minimal MDL cost of Equation (7). This excludes spurious minima at lower degrees as well as higher-degree multiples, and it is a condition on the cost landscape, not a consequence of the construction.*

Assumption 3’s appeal to Lemma 4.3 inherits a completeness, not soundness, qualifier on noisy data: part (v) of that lemma is an exact polynomial identity, and on noisy data a badly conditioned cofactor could in principle amplify residual noise enough that the surviving library fails to contain  $q^*$ ’s full support (Appendix F bounds this precisely). Should that happen, this theorem’s hypotheses are simply not met at  $D^*$  and no recovery claim is made, exactly as stated; the pipeline is never made to output an incorrect  $q^*$  by this mechanism, only, in the worst case, to correctly find nothing. Then Algorithm 2 recovers  $q^*$  (up to scale) when the loop reaches  $D = D^*$ , and the final output  $\mathcal{G}$  is a reduced Gröbner basis of the true invariant ideal  $\langle I_{D^*-1} \cup \{p^*\} \rangle = \langle I_{D^*-1} \cup \{q^*\} \rangle$ . Whenever  $p^*$  is already reduced modulo  $G_{D^*-1}$ , in particular whenever it is the first invariant confirmed and  $G_{D^*-1} = \emptyset$ ,  $q^* = p^*$  and the stronger, verbatim recovery statement holds.

**Proof.** The argument is direct and has three steps. *Step 1 (the support survives).* Lemma 4.3 guarantees that pruning against  $G_{D^*-1}$  removes only monomials divisible by one of its leading monomials, and that  $q^*$  is nonzero, lies in the true ideal, and is supported entirely on monomials that survive this pruning, at degree  $\deg q^* \leq D^*$ . By assumption (2) the sweep reaches  $D = D^*$ , and at that iteration the active monomial set contains the full support of  $q^*$ :  $\Phi$  is exactly the fixed design matrix over all monomials up to degree  $D^*$  not redundant against  $G_{D^*-1}$ . *Step 2 (recovery at  $D^*$ ).* Assumption (3) is precisely the hypothesis of Proposition 4.1 for this restricted matrix, so CSNP recovers  $q^*$  (up to scale) and the sweep confirms it at degree  $D^*$ . *Step 3 (selection).* The algorithm returns the confirmed candidate of globally minimal MDL cost over all explored degrees. Assumption (4) states that this minimum is attained, strictly and uniquely, by  $q^*$ , so  $q^*$  is selected. Adjoining  $q^*$  to  $G_{D^*-1}$  and reducing (Section 3.12) yields a reduced Gröbner basis of  $\langle I_{D^*-1} \cup \{q^*\} \rangle$ , which Lemma 4.3(v) identifies with  $\langle I_{D^*-1} \cup \{p^*\} \rangle$ , the true invariant ideal.  $\square$

Assumption (4) is, as stated, a condition imposed on the MDL landscape, not a guarantee derived from it. Section 4.6 proves its cost-ordering content, instead of merely assuming it, for the fixed- $dt$  symplectic-integrator sub-case underlying the transition-invariant benchmarks of Section 5.8; outside that sub-case, and for the rest of the assumption even inside it, the condition remains one to be checked, not a consequence of the pipeline’s construction.

No sample-complexity bound of this kind is available; in its place, Section 5.3’s direct measurement on the unit circle at  $\sigma = 0.01$  finds 100% exact recovery for every  $N \geq 100$  tested, so on this benchmark the proposition’s assumptions are already satisfied at modest sample sizes, not only asymptotically.

## 4.6 Modified-Equation Separation for Fixed-Step Symplectic Sampling

Assumption (4) of Proposition 4.5 is, as stated, a condition on the MDL landscape, not a derived guarantee. One setting where its cost-ordering content can be proved, not merely assumed, is the fixed- $dt$  transition-invariant case of Section 5.8: state pairs  $(x_t, x_{t+1})$  collected at a constant sampling interval from a Hamiltonian system integrated by a fixed-step symplectic method. Backward error analysis for symplectic integrators (Hairer et al., 2006) shows that such a method does not approximately conserve the true Hamiltonian  $H$ ; it *exactly* conserves a nearby modified Hamiltonian  $H_{dt} = H + O(dt^p)$  for a method of order  $p$ . What CSNP recovers from noiseless fixed- $dt$  trajectory data is therefore not a numerical curiosity but the exact invariant of this modified system, and the question of whether the pipeline still recovers  $p^* = H$  itself reduces to how far  $H_{dt}$  is from  $H$  relative to the pipeline’s rationality-cost resolution. The scope of what follows should be fixed before any formula appears: the result is a *pairwise* separation of  $H$  from  $H_{dt}$  within the two-parameter quadratic family, and it resolves exactly that ambiguity. It does not extend to the unrestricted transition dictionary, where Section 5.10 shows an additional exact algebraic tie defeats recovery of  $H$  entirely; that failure is open, and Section 6.5 discusses it as a limitation.

For a quadratic Hamiltonian this separation can be computed in closed form, not only bounded asymptotically, because the one-step map is linear and its exactly conserved quadratic form follows from solving a small linear system. Take the unit harmonic oscillator  $H = \frac{1}{2}(x^2 + v^2)$ . Symplectic Euler,  $v_{t+1} = v_t - dt x_t$ ,  $x_{t+1} = x_t + dt v_{t+1}$ , is a first-order method, and the quadratic form it exactly conserves is

$$Q_{dt}(x, v) = x^2 - dt xv + v^2, \quad (10)$$

so the gap from the true invariant  $x^2 + v^2$  is exactly the single cross term  $dt xv$ . Störmer–Verlet (leapfrog),  $v_{t+1/2} = v_t - \frac{dt}{2}x_t$ ,  $x_{t+1} = x_t + dt v_{t+1/2}$ ,  $v_{t+1} = v_{t+1/2} - \frac{dt}{2}x_{t+1}$ , is symmetric and second order, and its exactly conserved quadratic form is diagonal (Appendix C):

$$Q_{dt}(x, v) = x^2 + \frac{4}{4 - dt^2} v^2 = x^2 + \left(1 + \frac{dt^2}{4} + \frac{dt^4}{16} + \dots\right) v^2, \quad (11)$$

whose series contains only even powers of  $dt$ , exactly as the symmetric-method theory of (Hairer et al., 2006) predicts, with leading separation  $dt^2/4$ . By contrast, non-symplectic one-step methods such as explicit Euler or explicit midpoint RK2 admit no such exact invariant at all: their one-step map has eigenvalues off the unit circle, energy drifts without bound, and there is no fixed quadratic form for the transition-invariant search to find in the first place. The separation result below is therefore specific to structure-preserving integrators, which is also the regime in which fixed- $dt$  mechanical simulation data (the harmonic oscillator, Kepler, and 2D-oscillator benchmarks of Section 5.8) is ordinarily generated.

The proposition below comes in two strengths. Whenever the integrator’s modified invariant gains a monomial that  $H$  does not have, as symplectic Euler does, recovery is unconditional: it holds for every  $dt$ , with no dependence on whether  $dt$  happens to be a small-denominator rational. Whenever the modified invariant instead keeps  $H$ ’s support and only perturbs an existing coefficient, as Störmer–Verlet does, the guarantee is weaker: it holds outside a set of resonant step sizes that is measure-zero only in the zero-tolerance limit  $\varepsilon \rightarrow 0$ , and at the pipeline’s actual finite working tolerance the excluded set has positive, though small and shrinking, measure.

**Proposition 4.6** (Separation of  $H$  from the modified invariant). *Let trajectory data be generated by a fixed-step symplectic integrator of order  $p$  applied to a quadratic Hamiltonian, and let  $Q_{dt}$  be its exactly conserved modified invariant. Suppose CSNP is run over the degree-2 monomial library with rationality tolerance  $\varepsilon$  and maximum denominator  $q_{\max}$  (Section 3.4). If the coefficient-space gap  $\|Q_{dt} - H\|$  is below  $\varepsilon$ , then  $Q_{dt}$  and  $H$  round to the same polynomial under snap-rounding and recovering either recovers the true invariant. If instead the gap exceeds  $\varepsilon$  and, as in the symplectic Euler case (Equation (10)),  $Q_{dt}$  has strictly larger monomial support than  $H$  because the surviving cross term is not present in  $H$ , then  $R(H) = |S_H| < |S_{Q_{dt}}| \leq R(Q_{dt})$  by the admissible lower bound of Section 3.5, so among the supports its feasibility gate admits, BB assigns  $H$  the strictly lower cost, regardless of the numerical value of  $dt$ .*

**Proof sketch.** The first case is immediate: snap-rounding maps any coefficient vector within  $\varepsilon$  of a rational point to that point (Section 3.8), so  $Q_{dt}$  and  $H$  become syntactically identical once resolved. For the second case,  $R(H) = |S_H|$  exactly, since every coefficient of  $x^2 + v^2$  (or its transition-invariant form) is

$\pm 1$  and each attains the admissible bound’s one-bit minimum with no slack;  $Q_{dt}$ , having one more active monomial than  $H$ , cannot cost less than  $|S_H| + 1$  bits under the same admissible bound, independent of what rational value is eventually assigned to the extra coefficient.  $\square$

This proposition compares  $H$  against  $Q_{dt}$  specifically; it does not assert that the global BB optimum over an unrestricted transition dictionary is  $H$ , and the degeneracy documented below shows it need not be.

The statement orders the *cost* of two fixed polynomials, and BB only ever ranks supports its feasibility gate admits. On genuine symplectic-integrator trajectories only  $Q_{dt}$  vanishes exactly;  $H$  itself carries an  $O(dt^p)$  residual and is therefore not a feasible candidate at all, so the proposition does not claim  $H$  is recovered from data that fails to conserve it. Its operative content runs the other way, that the rationality prior never underprices the true  $H$  against the more complex modified invariant  $Q_{dt}$ . Where  $H$  is exactly conserved, the exact-rotation-flow setting of Section 5.8, it is the feasible candidate and is recovered in every seed; where only  $Q_{dt}$  is conserved, recovery of  $H$  is correspondingly 0% (Table 15), and the difference-dictionary restriction of Section 6.5 recovers  $Q_{dt}$ , the invariant the data actually carries.

This closes the cost-ordering part of assumption (4) of Proposition 4.5 for the fixed- $dt$  symplectic-integrator sub-case without any non-resonance condition on  $dt$ , because the argument rests on the number of active monomials, not on whether  $dt$  happens to admit a low-denominator rational approximation. The symmetric (Störmer–Verlet-type) case is weaker in this respect: Equation (11) shows the modified invariant keeps the *same* support as  $H$  and differs only in the value of one coefficient, so recovery instead depends on whether  $4/(4 - dt^2)$  resists a low-denominator rational approximation at the pipeline’s working tolerance  $\varepsilon$ .

The checks in Appendix C work within the two-parameter family  $(B, C)$  of  $Q = x^2 + Bxv + Cv^2$ , the family the proposition is stated in. Handing the full four-variable, degree-2 transition dictionary of Section 5.8 (fifteen monomials) to CSNP’s general search instead turns out to matter. Because  $(x_{t+1}, v_{t+1})$  is an exact linear function of  $(x_t, v_t)$  on noiseless data, the fifteen-monomial dictionary admits several exactly-vanishing degree-2 relations besides  $H$  and  $Q_{dt}$ , some of which tie  $H$  and  $Q_{dt}$  at the same admissible rationality cost. Run over that unrestricted dictionary at  $dt = 0.05$ ,  $\sigma = 0$ , CSNP returns  $(x_t - x_{t+1})^2$ , neither  $H$  nor  $Q_{dt}$ : a symptom of the tie, not a search failure. Whether CSNP’s general search recovers  $H$  over  $Q_{dt}$ , and not some other tied relation, from that wider dictionary is checked directly, at scale, in Section 5.10, which reruns the full search 30 times against orthogonal matching pursuit (OMP) and SINDy-null under both fixed and variable  $dt$ . The answer is no, and for a reason more structural than a noise-floor artifact: the wide dictionary contains a third exact relation, a pure algebraic identity of the symplectic-Euler update rule itself (Equation (12)), that ties  $H$  at the same rationality cost regardless of  $dt$ .

Since the pipeline’s tolerances are hand-set, not derived from  $dt$  and  $p$ , we tested whether a tolerance keyed to the known modified-equation gap instead of to noise could break the tie above; no point in the sweep recovers  $H$ . This confirms the wide-dictionary tie is a genuine degeneracy in exact algebra, so no choice of  $\tau$  can be expected to break it. Appendix E gives the full sweep.

## 4.7 Computational Complexity

The adaptive pipeline avoids building the full degree- $D_{\max}$  library upfront. Instead, monomials are added incrementally, and pruning prevents redundant columns from ever being added, instead of removing them afterward. The pipeline maintains a QR factorization of  $\Phi$ , extended by Householder-based column insertion (cost  $O(NM)$  per new column) (Golub and Loan, 2013), and takes a fresh SVD of the resulting  $R$  factor at each degree, which costs only  $O(M^3)$  since  $R$  is  $M \times M$  regardless of  $N$ . The exact BB search enumerates supports with combinatorial cost  $\sum_{k=1}^{k'} \binom{M_{\text{active}}}{k} d^2$ , run at the largest support cap  $k' \leq k_{\max}$  whose predicted cost fits the  $2 \times 10^8$ -unit budget of Section 3.6; direct enumeration covers the  $\binom{M}{d-1} \leq 2 \times 10^5$  regime, and only when both exact-engine budgets are exceeded does the anytime L1-seeded fallback take over. Table 4 traces this routing on representative problem sizes: the subset and unit counts are analytic (binomial counts under the routing formulas above, at the default  $k_{\max} = 6$ ), and the times are indicative single-core figures for one circuit search at that size.

Table 4: Cost routing at representative problem sizes. “Driver” is the quantity the routing rule of Algorithm 1 (Step 3) compares against its budget:  $\binom{M}{d-1}$  dual subsets for direct enumeration (budget  $2 \times 10^5$ ), predicted units  $\sum_{k \leq k'} \binom{M}{k} d^2$  for BB (budget  $2 \times 10^8$ ). All benchmark problems of Section 5.1 fit the exact-engine budgets and therefore never reach the anytime fallback.

| $M_{\text{active}}$ | $d$   | Driver                                              | Mode                       | Time (typical)               |
|---------------------|-------|-----------------------------------------------------|----------------------------|------------------------------|
| 10                  | 2     | $\binom{10}{1} = 10$                                | Direct enumeration         | $< 0.01$ s                   |
| 21                  | 3     | $\binom{21}{2} = 210$                               | Direct enumeration         | $< 0.1$ s                    |
| 28                  | 12    | $\approx 7.2 \times 10^7$ units at $k' = 6$         | BB, full cap               | $\sim 1$ s                   |
| 45                  | 13    | $\approx 2.8 \times 10^7$ units at $k' = 4$         | BB, reduced cap $k' = 4$   | $\sim 1\text{--}2$ s         |
| $\gg 100$           | large | $M d^2 > 2 \times 10^8$ : even $k' = 1$ over budget | Anytime fallback (L1 seed) | $\sim 1\text{--}5$ s (LASSO) |

## 5 Experiments

**Evaluation metric: exact recovery.** Throughout the experiments, a recovered polynomial (or set of polynomials) counts as an *exact* recovery if, after snap-rounding to  $\mathbb{Q}$ , algebraic minimality reduction (Section 3.7), and computation of its reduced Gröbner basis over  $\mathbb{Q}$  under graded reverse lexicographic order (Section 3.12), some generator of the resulting basis equals the ground-truth invariant up to a nonzero rational scalar. The criterion is applied identically to every method, baselines included, and every method’s output passes through the same snap-rounding and canonicalisation before scoring. The criterion does not by itself penalise extra generators returned alongside the true one, which is why redundancy counts are reported as separate columns in Section 5.1. Where the recovered object is a genuinely multi-generator ideal, the stricter test of exact ideal equality between recovered and true reduced bases is used instead (Table 14).

To confirm that canonicalisation yields the intended algebraic object, we encode the recovered ideal in Z3 (de Moura and Bjørner, 2008) and verify exact ideal equality with the ground truth. On the circle benchmark at  $\sigma = 0$ , Z3 confirms equality on all 30 seeds.

SR-GB+CSNP is evaluated across eight benchmark classes. All experiments were conducted on a standard CPU runtime (Google Colab, single-core CPU). Code and data are available at <https://github.com/TesfayZ/srgb-csnp>.

**Configuration.** Unless a benchmark states otherwise, every problem is sampled at  $N = 5000$  points over 30 random seeds, with independent Gaussian noise at  $\sigma \in \{0.00, 0.02, 0.05\}$  added independently to each observed variable, and recovery rates are reported with 95% Wilson intervals. SR-GB+CSNP runs inside its certified regime ( $n \leq 8$  variables, degree  $D \leq 4$ , support size  $k_{\max} \leq 6$ ) and snaps rational coefficients at a denominator ceiling of 16. Two degree regimes are exercised. When the true degree of the target invariant is known, it can be supplied directly, and the Feynman and AVI comparisons below do exactly that, for every method alike. When the degree is unknown, on the other hand, the adaptive MDL sweep of Section 3.10 discovers it, which is how the transition benchmarks of Section 5.8 are run. The baselines use fixed settings held constant across all benchmarks. KRONIC aggregates its sparse-nullspace estimate over 15 bootstrap resamples drawn at 80% subsampling whenever the nullspace dimension  $d > 1$ ; at  $d = 1$  there is only a single nullspace direction to begin with, nothing for resampling to disambiguate, so KRONIC instead returns that direction directly without entering the bootstrap loop, and several benchmarks below are explicitly  $d = 1$  settings. SINDy-ST runs all-targets implicit STLSQ in the style of SINDy-PI (Kaheman et al., 2020), cycling every library monomial once as the regression target with hard-threshold tolerance  $10^{-2}$  and deduplicating the converged fits up to scalar multiples; OMP is capped at  $\min(4, d)$  nonzero coefficients; and PySR+GB runs 20 search iterations over the operator set  $\{+, -, \times, (\cdot)^2\}$ .

**Baseline Selection.** Nine baselines are run across the eight benchmark classes of this section, none of them on every class:

1. SINDy-null (KRONIC) (Kaiser et al., 2018): nullspace variant with L1 sparsity, no canonicalisation.
2. SINDy-ST (Sequential Thresholding) (Brunton et al., 2016): implicit all-targets STLSQ; every library monomial serves once as the regression target, each fit is sequentially hard-thresholded, and the converged candidates are deduplicated up to scalar multiples.



3. SINDy-AD (automatic differentiation) (Kaheman et al., 2022): uses smoothing splines to estimate derivatives (approximate implementation).
4. SINDy-FD (finite differences): a baseline that computes time derivatives via finite differences (included for comparison).
5. Dense SVD+GB: applies snap-rounding and Gröbner reduction to the leading right singular vector.
6. OMP on nullspace (Tropp and Gilbert, 2007): greedy matching pursuit on the nullspace basis.
7. PySR+GB (Cranmer, 2023): PySR output followed by Gröbner canonicalisation.
8. AVI (Approximate Vanishing Ideal) (Heldt et al., 2009; Kreuzer et al., 2009): the border basis of the approximate vanishing ideal, compared directly in Section 5.7.
9. Row-echelon disambiguation (Oellerich and Emelianenko, 2024): the same SVD nullspace basis reduced to row-echelon form and read off row by row, run in Sections 5.8 and 5.9.

For  $d = 1$ , Dense SVD+GB coincides with SR-GB+CSNP as reported in Table 6, since with a single nullspace direction there is no support set for CSNP or BB to search over: the lexicographic cost tuple of Equation (5) has only one candidate to evaluate, so BB terminates immediately and the two methods are, by construction, the same computation on this class of problems. For  $d > 1$ , the comparison separates the methods.

Dense SVD+GB serves as the canonicalisation-only ablation. Why no “SINDy + GB” hybrid can play that role, and why AVI admits no analogous construction, is argued in Appendix D; Section 5.7 reports a direct empirical comparison against AVI itself.

**Benchmark roles.** The benchmarks below serve four distinct purposes, and Table 5 assigns each one its role. *Breadth checks* are single-invariant ( $d = 1$ ) by construction, so they establish correctness without stressing disambiguation at all: with one candidate to select between, none of them can separate rationality-based selection from a sparsity-only rule. *Necessity demonstrations* show a baseline’s shortcut failing and identify the mechanism that closes the gap. The *discriminator* is narrower than either: the sparsity-first-vs-rationality-first ablation (Table 11) holds every other variable fixed and varies only the selection key, which is what makes it a clean test of SR-GB+CSNP’s central claim rather than of the pipeline in general; the fixed-versus-variable- $dt$  table (Table 15) applies the same isolation to the data-generating regime at 30-seed scale. Finally, several results are *failure modes*, catalogued in Section 6.



Table 5: What each experiment tests. “Breadth” benchmarks are single-invariant ( $d = 1$ ) and establish correctness or competitiveness without stressing disambiguation. “Necessity” benchmarks show a specific SR-GB+CSNP mechanism succeeding where a specific baseline shortcut fails. The “discriminator” isolates SR-GB+CSNP’s central claim, rationality over sparsity, by holding every other variable fixed; the “regime test” applies the same isolation to the data-generating process rather than to the selection key. “Failure mode” entries are settings where the method does not recover the target.

| Benchmark                                                   | Role                 | What it isolates                                                                              |
|-------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------|
| Synthetic circle/sphere/cubic & holonomic constraints (5.1) | Breadth              | $d = 1$ parity with baselines, incl. a robotics-motivated case; no disambiguation to test     |
| Feynman polynomial subset (5.5)                             | Breadth              | noise sensitivity across 26 single-invariant polynomial equations                             |
| PySR-GB, loop invariants & DIG (5.4)                        | Breadth              | explicit-vs-implicit and exact-trace scope boundaries, not head-to-head tests                 |
| Degree-lifting over-lift (5.2)                              | Necessity            | algebraic minimality vs. SINDy-null’s 3.3%                                                    |
| Noise sensitivity, snap-rounding (5.3)                      | Necessity            | noise-tolerant rounding vs. naïve rounding                                                    |
| Kepler angular momentum (5.8)                               | Necessity            | CSNP recovers a $d = 2$ invariant baselines cannot isolate                                    |
| 2D-oscillator deflation (5.8)                               | Necessity            | full 4-generator recovery vs. SINDy-null/FD/AD returning none                                 |
| OMP on nullspace (5.6)                                      | Necessity            | greedy pursuit succeeds only when nullspace geometry is incidentally favourable               |
| AVI baseline (5.7)                                          | Necessity            | border-basis cardinality inflates with noise; CSNP reduces it back                            |
| Sparsity-vs-rationality (5.8, Table 11)                     | <b>Discriminator</b> | SR-GB+CSNP’s one claimed mechanism, isolated with every other variable fixed                  |
| Fixed-vs-variable- $dt$ (5.10, Table 15)                    | <b>Regime test</b>   | same isolation applied to the data-generating regime, at 30-seed scale against real baselines |
| Fixed- $dt$ vs. variable- $dt$ , wide dictionary (5.10)     | Failure mode         | update-rule identity ties $H$ ; 0% for every method, ours included                            |
| Point vortices, $n_v \geq 4$ (5.9)                          | Failure mode         | dense higher-degree generator lost as pooled dimension grows                                  |
| Unit-invariance stress test (6.4)                           | Failure mode         | rationality prior is not scale-invariant; 3 of 4 benchmarks collapse                          |
| Noise ceiling, degree-3 (6.1)                               | Failure mode         | spectral-gap collapse bounds the usable degree under noise                                    |

## 5.1 Synthetic Benchmarks: Redundancy Elimination and Holonomic Constraints

**Setup.** Three algebraic varieties are considered: the circle ( $x^2 + y^2 = 1$ ), the sphere ( $x^2 + y^2 + z^2 = 1$ ), and a cubic ( $x^3 + 2xy + y^2 = 0$ ). All three have  $d = 1$ . SINDy-null, the L1-sparsity nullspace variant of KRONIC described above, recovers a single generator in every trial, matching Dense SVD+GB and SR-GB+CSNP. SINDy-ST, however, still returns multiple redundant polynomials (Table 6), confirming that sequential thresholding alone does not eliminate algebraic equivalence.

Table 6: Redundancy elimination results over 30 seeds ( $\sigma = 0.0$ ). Notations: Red. = Redundancy; Rec. = Reconstruction; FDR = False Discovery Rate. SINDy-null (KRONIC) with L1 sparsity recovers a single generator for  $d = 1$  like ours; SINDy-ST still produces redundancy. SINDy-FD and SINDy-AD do not appear: both require a time dimension for finite-difference or automatic-differentiation derivative estimation, which this static point-cloud benchmark does not have. They are evaluated where trajectory data exists, in Table 12.

| Method                | Red. Circle   | Red. Sphere   | Red. Cubic    | Rec. error | FDR      |
|-----------------------|---------------|---------------|---------------|------------|----------|
| SINDy-null (KRONIC)   | $1.0 \pm 0.0$ | $1.0 \pm 0.0$ | $1.0 \pm 0.0$ | 0.00       | 0.00     |
| SINDy-ST              | $4.0 \pm 0.0$ | $7.0 \pm 0.0$ | $8.0 \pm 0.0$ | 0.16       | 0.83     |
| Dense SVD+GB ablation | $1.0 \pm 0.0$ | $1.0 \pm 0.0$ | $1.0 \pm 0.0$ | 0.00       | 0.00     |
| <b>SR-GB+CSNP</b>     | <b>1</b>      | <b>1</b>      | <b>1</b>      | <b>0</b>   | <b>0</b> |

**Holonomic equality constraints.** The same  $d = 1$  breadth check extends to two synthetic holonomic

equality constraints motivated by robotic kinematic settings: a **linear** constraint ( $q_1 - q_2 = 0$ ) and a **nonlinear** distance-equality constraint ( $(x_1^2 + y_1^2) - (x_2^2 + y_2^2) = 0$ ). SR-GB+CSNP recovers both constraints exactly in every trial at every noise level tested ( $\sigma \in \{0, 0.02, 0.05\}$ , 30 seeds each: 100%, Wilson 95% CI [0.89, 1.00]), and SINDy-null (KRONIC) with L1 sparsity again reaches 100% recovery at every noise level, the same  $d = 1$  parity seen above; the all-targets SINDy-ST ensemble again finds the constraint among its candidates at every noise level but returns it inside a redundant set rather than a single canonical generator, a mean of 2 candidates on the linear constraint and 12 on the nonlinear one.

## 5.2 Degree Lifting Robustness

When the lifting degree  $D$  is chosen larger than the true degree of the invariant, the nullspace of  $\Phi$  contains both the true generator and its higher-degree multiples. For the unit circle lifted to degree 3, the true generator  $p = x^2 + y^2 - 1$  (3 terms) and the multiple  $q = yp = x^2y + y^3 - y$  (also 3 terms) are both present in the nullspace. SINDy-null, which relies on L1 sparsity, may return either candidate, leading to a low success rate. Dense SVD+GB, in contrast, extracts the dominant null vector and then applies snap-rounding and reduced Gröbner basis computation with factorisation, which reduces  $q$  to  $p$  and returns the canonical generator.

Table 7 shows the exact recovery rates on the circle with  $D = 3$  over 30 seeds. SINDy-null recovers the true invariant in a single trial (3.3%), even when every candidate it returns is scored; in the rest it comes back with a higher-degree multiple instead, whereas Dense SVD+GB succeeds in 100%: the algebraic minimality step is what matters when the true degree is unknown.

Table 7: Exact recovery on the unit circle with over-lifting ( $D = 3$ ,  $N = 5000$ , 30 seeds).

| Method              | Exact recovery | 95% Wilson CI |
|---------------------|----------------|---------------|
| SINDy-null (KRONIC) | 3.3%           | [1%, 17%]     |
| Dense SVD+GB        | 100%           | [89%, 100%]   |

## 5.3 Noise Sensitivity

Snap-rounding achieves 100% exact recovery on the circle ( $N = 5000$ ) up to  $\sigma = 0.14$ , with 96.7% at  $\sigma = 0.16$ , 60% at  $\sigma = 0.18$ , and 3.3% at  $\sigma = 0.20$ ; beyond that recovery drops to 0%. This extends the noise tolerance substantially compared to naïve rounding, which already fails at  $\sigma = 0.04$ , as shown in Table 8.

**Correlated noise.** Replacing i.i.d. noise with a stationary AR(1) process along the sampling order, at a correlation strength ( $\phi = 0.9$ ) that shrinks the effective sample size by roughly  $19\times$ , leaves recovery essentially unaffected: unchanged at  $\sigma \in \{0.02, 0.05\}$  for every  $\phi$  tested, and at  $\sigma = 0.10$ , 100% for  $\phi = 0$ , 100% for  $\phi = 0.5$ , and 96.7% for  $\phi = 0.9$  (Table 8(b)). This isolates the SVD’s reliance on noise averaging, as distinct from noise magnitude; Appendix A.7 gives the construction in full.

Table 8: Circle noise sensitivity,  $N = 5000$ , 30 seeds, 95% Wilson CI. (a) Exact recovery vs. noise level  $\sigma$  under i.i.d. noise, naïve rounding vs. snap-rounding. (b) Exact recovery under AR(1)-correlated noise at three correlation strengths  $\phi$ , isolating whether the SVD’s noise-averaging is disrupted by correlation, not just magnitude.

| (a)       |                   |                           | (b)                                                                                                                             |          |            |            |
|-----------|-------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|------------|------------|
| $\sigma$  | Naïve             | Snap-round.               | $\sigma$                                                                                                                        | $\phi=0$ | $\phi=0.5$ | $\phi=0.9$ |
| 0.00      | 100% [0.89, 1.00] | <b>100%</b> [0.89, 1.00]  | 0.02                                                                                                                            | 100%     | 100%       | 100%       |
| 0.02      | 60% [0.42, 0.75]  | <b>100%</b> [0.89, 1.00]  | 0.05                                                                                                                            | 100%     | 100%       | 100%       |
| 0.04–0.14 | 0% [0.00, 0.11]   | <b>100%</b> [0.89, 1.00]  | 0.10                                                                                                                            | 100%     | 100%       | 96.7%      |
| 0.16      | 0% [0.00, 0.11]   | <b>96.7%</b> [0.83, 0.99] | 95% Wilson CI matches panel (a) at equal point estimates:<br>100% $\rightarrow$ [0.89, 1.00], 96.7% $\rightarrow$ [0.83, 0.99]. |          |            |            |
| 0.18      | 0% [0.00, 0.11]   | <b>60.0%</b> [0.42, 0.75] |                                                                                                                                 |          |            |            |
| 0.20      | 0% [0.00, 0.11]   | <b>3.3%</b> [0.01, 0.17]  |                                                                                                                                 |          |            |            |
| 0.22–0.30 | 0% [0.00, 0.11]   | 0% [0.00, 0.11]           |                                                                                                                                 |          |            |            |

**Sample size and monomial degree.** For the unit circle at  $\sigma = 0.01$ , SR-GB+CSNP reaches 100% exact recovery for all  $N \geq 100$  and for every lifting degree  $D \in \{2, 3, 4\}$  (30/30 seeds each, 95% Wilson

CI [88.6%, 100%]; degree-1 lifting fails), with the algebraic minimality step (Section 3.7) reducing every over-lifted degree’s higher-degree multiples back to the true generator.

## 5.4 Scope Boundaries: PySR-GB, Loop Invariants, and DIG

Applying Gröbner canonicalisation (Cox et al., 2015) to the output of PySR (Cranmer, 2023) fails on the unit circle (multi-valued variety, 0% recovery over 30 seeds) but succeeds on  $F = ma$  (single-valued, 100% over 30 seeds). PySR addresses explicit regression  $y = f(x)$ ; implicit constraints  $p(x) = 0$ , SR-GB+CSNP’s target, fall outside its design scope.

We also evaluate SR-GB+CSNP on classic loop invariants (sum, ps2, ps3, egcd, hash) and against DIG (Nguyen et al., 2014), the closest prior work in spirit on exact trace data. Like the PySR-GB comparison above, this is a scope boundary, not a head-to-head benchmark; detailed results are in Appendix H.

## 5.5 Feynman Polynomial Benchmark

This section evaluates SR-GB+CSNP on the polynomial subset of the Feynman symbolic-regression database, the implicit relations  $p(x) = 0$  underlying 26 of its equations, not the explicit  $y = f(x)$  form the database is usually posed in. As Section 2 explains, AI Feynman and PySR address that complementary explicit-regression problem and are not run here as head-to-head baselines. The baselines reported below, KRONIC and SINDy-ST, are instead the methods that target implicit, possibly-multi-invariant recovery directly, the same problem SR-GB+CSNP is built for.

Evaluation on 26 polynomial equations (14 degree-2, 12 degree-3) shows that SR-GB+CSNP achieves 100% exact recovery at zero noise. Degree-2 equations hold **100%** exact recovery through  $\sigma = 0.02$  and give way only at  $\sigma = 0.05$ , where they average **96.7%**, while degree-3 equations are markedly more noise-sensitive: at  $\sigma = 0.02$  five of the twelve (`algebraic_cubic_toy`, `charge_force_toy`, `I.13.4`, `kinetic_energy_scaled`, and `kinematics_position`) recover in every seed, and by  $\sigma = 0.05$  only `I.13.4` and `kinetic_energy_scaled` persist, both at 100%. This higher-degree sensitivity is traced to the spectral gap in Section 6.1 (Table 9).

The degree-2 drop at  $\sigma = 0.05$  is likewise concentrated, not uniform: `momentum_conservation_1d` ( $m_1v_1 + m_2v_2 = P$ ) and `I.39.22` ( $pV = nT$ ) are partial at 66.7% and 86.7%, respectively, while the other twelve degree-2 equations stay at 100% (Table 26). Direct inspection of the failing runs identifies the mechanism, and it is not the exact multi-relation tie seen for Kepler angular momentum in Section 5.7: in every failing seed, a sparse spurious relation beats the true generator under the noise-widened feasibility tolerance: confined to a single narrowly-ranged variable, it stays uniformly small (*Chebyshev-small*, in the minimax sense that a low-degree polynomial can be made small everywhere over a short interval) across that variable’s whole sampled range, so it slips under the tolerance despite being spurious, a bias rather than a variance effect that raising  $N$  does not fix. Appendix I works the mechanism through in full, including the same trap’s role in `algebraic_cubic_toy`’s loss at  $\sigma = 0.05$ .

This benchmark compares against KRONIC (sparse-nullspace SINDy-null) and SINDy-ST as static-data baselines, since the Feynman equations are sampled as independent variable tuples, not trajectories; SINDy-FD and SINDy-AD, which require a time dimension, are instead evaluated on the transition-invariant benchmarks of Section 5.8. The SINDy-ST baseline runs as the all-targets implicit STLSQ ensemble of Section 5, cycling every library monomial once as the regression target instead of fixing a single arbitrary one, so that its reported performance reflects the method itself and not an avoidable implementation bias. Both baselines run at the ensemble sizes fixed there, which credit each method at its strongest configuration. Table 9 reports both baselines at the same scale as SR-GB+CSNP (all 26 equations, 30 seeds), with the same errors-in-variables column correction applied to both baselines as to SR-GB+CSNP itself. All three methods agree at  $\sigma = 0$  (100% each). Under noise they separate: KRONIC falls to 28.2% at  $\sigma = 0.02$  and 20.4% at  $\sigma = 0.05$ ; SINDy-ST, for its part, holds up better at 42.2% and 29.7%; and SR-GB+CSNP’s rationality-cost tie-break keeps it at **73.1%** and **59.7%** over the same range. A gap of roughly 30 points therefore remains under noise, and inspecting the failing cells shows it is a matter of search reach as much as of tie-breaking. On `I.13.4` at  $\sigma = 0.02$ , where SR-GB+CSNP recovers the two-term generator in every seed, the ensemble returns twenty candidates of six to ten terms each: hard thresholding walks a fixed path through coefficient space and, on a noisy degree-3 dictionary, that path never reaches the true sparse support at all, whereas BB enumerates small supports directly and its feasibility oracle still certifies the correct two-term circuit.

Where both methods do reach comparable candidates, the rationality tie-break then does the remaining work. Neither baseline has either mechanism, which CSNP +BB specifically targets. The rate gap is not just numerically large but statistically decisive: a McNemar exact test on the same paired per-trial outcomes rejects equal error rates against both baselines at every noise level tested ( $p < 10^{-70}$  against KRONIC and against SINDy-ST at both  $\sigma = 0.02$  and  $\sigma = 0.05$ ), and on every trial where the two methods disagree it is SR-GB+CSNP that succeeds, never the baseline.

Recovery is only half of this comparison; the compute bill is the other half. Timed cell by cell across all 26 equations and three noise levels (30 seeds per cell), the all-targets SINDy-ST configuration consumes about  $0.75\times$  the wall time of the entire SR-GB+CSNP pipeline in aggregate, though with wide per-cell variance (its per-cell median is about  $1.13\times$  and its worst cell, `gravitational_pe_orbit` at  $\sigma = 0$ , costs  $24.8\times$  SR-GB+CSNP), while the bootstrapped KRONIC estimate finishes in about  $0.24\times$  SR-GB+CSNP’s time in aggregate. Per-trial variance is wide for all three methods, not just SINDy-ST: over every individual (equation,  $\sigma$ , seed) trial, mean wall time is 0.56s for SR-GB+CSNP (std 0.97s, IQR [0.02, 0.94]s), 0.42s for SINDy-ST (std 0.69s, IQR [0.03, 0.56]s), and 0.13s for KRONIC (std 0.35s, IQR [0.00, 0.07]s), reflecting the same per-equation cost heterogeneity the worst-cell figure already hints at. The Branch-and-Bound search, whatever its combinatorial worst case, is in practice priced within a small constant factor of the sparse-regression baselines it is compared against.

Table 9: Feynman polynomial benchmark, all 26 polynomial equations of the subset described in Section 5.5, 30 seeds each. **(a)** Exact recovery rates by degree subset. **(b)** SR-GB+CSNP’s All-26 row from (a) against KRONIC (SINDy-null) and the all-targets SINDy-ST ensemble, same  $\sigma$  grid.

| (a)      |               |                   |                   |    | (b)      |                   |                   |  |
|----------|---------------|-------------------|-------------------|----|----------|-------------------|-------------------|--|
| Eq. type | $\sigma=0.00$ | $\sigma=0.02$     | $\sigma=0.05$     | #  | $\sigma$ | KRONIC            | SINDy-ST          |  |
| All 26   | <b>100%</b>   | 73.1% [69.9,76.1] | 59.7% [56.3,63.1] | 26 | 0.00     | 100% [99.5,100]   | 100% [99.5,100]   |  |
| Degree-2 | <b>100%</b>   | 100% [99.1,100.0] | 96.7% [94.5,98.0] | 14 | 0.02     | 28.2% [25.2,31.5] | 42.2% [38.8,45.7] |  |
| Degree-3 | <b>100%</b>   | 41.7% [36.7,46.8] | 16.7% [13.2,20.9] | 12 | 0.05     | 20.4% [17.7,23.4] | 29.7% [26.6,33.0] |  |

SR-GB+CSNP at these  $\sigma$ : 100%, 73.1%, 59.7% (panel (a)), All 26).

## 5.6 Comparison with OMP on Nullspace

The OMP (Tropp and Gilbert, 2007) baseline was evaluated on both transition invariant benchmarks (Table 10).

Table 10: OMP vs CSNP exact recovery (30 seeds, 95% Wilson CI).

| Problem                                        | CSNP (ours)       | OMP baseline      |
|------------------------------------------------|-------------------|-------------------|
| Harmonic oscillator ( $d = 2$ , $M = 10$ )     | 100% [0.89, 1.00] | 0% [0.00, 0.11]   |
| Kepler angular momentum ( $d = 2$ , $M = 21$ ) | 100% [0.89, 1.00] | 100% [0.89, 1.00] |

CSNP achieves 100% on both problems; OMP succeeds only when the nullspace geometry is incidentally favourable (Kepler), and fails when it is balanced (harmonic oscillator), where it returns dense combinations of the two nullspace basis vectors whose coefficients do not simplify to integers under snap-rounding, so the resulting Gröbner basis never matches the ground-truth ideal.

## 5.7 AVI Baseline

AVI computes the full border basis of the vanishing ideal from noisy data using a numerically adapted Buchberger–Möller procedure. Its  $\varepsilon$ -truncated membership test needs a noise-calibrated tolerance just as every threshold in this pipeline does; throughout this comparison it is set to the smallest multiple of  $\hat{\sigma}$  at which AVI returns a numerically full-rank border basis, not an under-merged, spuriously empty one, on every system in Table 24, a criterion evaluated on AVI’s systems independent of how it compares to SR-GB+CSNP. Unlike SR-GB+CSNP, AVI lacks a rationality tie-break mechanism, and its border-basis cardinality inflates with noise even when the true ideal has not changed (e.g., Kepler angular momentum: mean cardinality 3.2 at  $\sigma = 0$  to 20.3 at  $\sigma = 0.02$ ). We provide a detailed empirical comparison, including per-system cardinality

breakdowns, in Appendix G. The main conclusion is that SR-GB+CSNP’s rationality cost and BB search reduce the inflated border basis back to the minimal canonical generator, while AVT’s  $\varepsilon$ -truncated membership test has no such mechanism.

**Fixed- $dt$  versus variable- $dt$ .** Having compared against both alternative-paradigm and nullspace competitors on single- and multi-invariant systems, Sections 5.8 and 5.10 turn to the regime identified above (Section 5) as carrying the discriminator and the regime test: recovery under fixed-step symplectic sampling, where a modified invariant  $H_{dt}$  competes with the true  $H$ , along the fixed/variable- $dt$  axis.

## 5.8 Transition Invariants

**Fixed sampling interval (constant  $dt$ ).** Data here is generated by the exact analytic rotation flow of the harmonic oscillator,  $x_{t+1} = \cos(dt) x_t + \sin(dt) v_t$ ,  $v_{t+1} = -\sin(dt) x_t + \cos(dt) v_t$ , at  $dt = 0.1$ , not by a numerical integrator; a rotation preserves the Euclidean norm exactly, so  $x^2 + v^2$  is conserved with no  $O(dt^p)$  error term at all. Using the full BB criterion, CSNP recovers the 4-term energy invariant in **100%** of trials (30 seeds). A sparsity-only ablation fails at 0% by selecting a  $dt$ -specific 3-term invariant with irrational coefficients; the greedy sparsity-first OMP selector run on the same exact-rotation dictionary in Section 5.10 reproduces that 0%. The rationality component is therefore essential for this setting. A wider sweep confirms this is not an artifact of the single  $dt = 0.1$  point: Table 11 repeats the sparsity-first vs. rationality-first comparison, both selectors searching the identical candidate pool on the identical transition-pair dictionary, across eleven  $dt$  values from 0.05 to 0.30, and every one shows the same split (pooled over  $n = 330$  trials: sparsity-first 0.0% [0.0%, 1.2%] vs. rationality-first **100.0%** [99%, 100%]).

Table 11: Sparsity-first vs. rationality-first selection on the fixed- $dt$  harmonic oscillator’s transition-pair dictionary ( $N = 5000$ ,  $\sigma = 0$ , 30 seeds per  $dt$ ). Both selectors search the identical candidate pool; only the final selection key differs.

| $dt$                 | Sparsity-first  | Rationality-first                         |
|----------------------|-----------------|-------------------------------------------|
| 0.05                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.08                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.10                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.12                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.15                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.18                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.20                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.22                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.25                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.28                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| 0.30                 | 0% [0.00, 0.11] | <b>100%</b> [ <b>0.89</b> , <b>1.00</b> ] |
| Pooled ( $n = 330$ ) | 0% [0.00, 0.01] | <b>100%</b> [ <b>0.99</b> , <b>1.00</b> ] |

This 100% result is not in tension with the wide-dictionary tie documented for symplectic Euler in Section 4.6: that tie arises specifically because a numerical integrator’s modified invariant  $Q_{dt}$  differs from  $H$  by a genuine  $O(dt^p)$  term, giving BB a real competing low-cost candidate to choose incorrectly. The exact rotation flow used here has no such competitor, since there is no integration error to generate one, and we confirmed directly that swapping in symplectic Euler’s data for this identical search code reproduces the Section 4.6 tie instead of the 100% result above.

**Variable sampling interval (random  $dt$ ).** When the sampling interval varies, the energy is the unique invariant and CSNP recovers it in 100% of trials automatically.

**Kepler angular momentum.** CSNP recovers the 4-term invariant in 100% of trials ( $d = 2$ ,  $M = 21$ ; resolved by the direct-enumeration engine of Section 3.5, as the node instrumentation of Section 4.4 records).

**2D harmonic oscillator, multi-invariant deflation.** As a direct test of the nullspace-deflation procedure of Section 3.13, a 2D harmonic oscillator with independent, equal-frequency  $x$ - and  $y$ -axes is sampled at fixed  $dt$ . The dictionary is built as a degree-2 monomial library over the 4-variable state  $(x, y, v_x, v_y)$ , with the difference  $\Phi_{\text{old}} - \Phi_{\text{new}}$  forming the nullity-4 target of deflation. CSNP with deflation recovers all four minimal quadratic generators of the conserved-quantity ideal in **100%** of 30 seeds:  $x^2 + v_x^2$ ,



$y^2 + v_y^2$ ,  $xy + v_x v_y$ , and the angular momentum  $xv_y - yv_x$ . Run on the identical difference dictionary, SINDy-null fails to recover the full set of generators in any seed; it returns non-canonical linear mixtures, consistent with the observation of Kaiser et al. (Kaiser et al., 2018) that, once the nullspace is multi-dimensional, individual invariants are not generally recoverable as raw SVD basis vectors. SINDy-FD and SINDy-AD, run on their smaller, restricted 4-variable state dictionary  $(x, y, v_x, v_y)$  rather than the difference dictionary, fare no better: SINDy-FD’s nullspace-dimension estimate comes out at  $d = 10$  against a true  $d = 4$ , since its finite-difference derivative is only a first-order approximation even at  $\sigma = 0$ , and the single candidate it returns before its sparsity gate discards it is a dense, physically meaningless 8–9 term mixture, the same arbitrary-mixture failure mode Kaiser et al. (Kaiser et al., 2018) describe, surviving even in a dictionary much smaller than the one that produces it for CSNP and SINDy-null. Unlike SINDy-null, however, SINDy-FD and SINDy-AD return only a single candidate per seed, so the full four-generator set is structurally unreachable for either regardless of accuracy; per-invariant scoring against their single returned candidate is reported in Table 13 instead.

A fifth baseline, run on the identical difference dictionary, is row-echelon disambiguation (Oellerich and Emelianenko, 2024): reduce the same SVD nullspace basis CSNP consumes to reduced row-echelon form (RREF), with no sparsity or rationality criterion involved, and read the resulting rows off directly as the recovered generators. Here it ties CSNP at 100% on all four. The tie is not surprising once the monomial supports are laid out: the four generators  $x^2 + v_x^2$ ,  $y^2 + v_y^2$ ,  $xy + v_x v_y$ , and  $xv_y - yv_x$  each touch only two of the fourteen library monomials, and no two generators share a monomial, so the nullspace is already triangular under essentially any fixed pivot order and there is nothing for row-echelon reduction to tangle. This system alone would suggest the open question raised by Kaiser et al. (Kaiser et al., 2018) is resolved by row reduction after all; the point-vortex system of Section 5.9, where support genuinely overlaps, shows otherwise.

Results are summarised in Table 12: CSNP solves all transition-invariant tasks, recovering the correct invariant(s) in 100% of trials in each case. For the SINDy variants one distinction matters and is made explicit in the table. SINDy-null and SINDy-ST operate on the same transition-pair dictionary as CSNP and recover the transition invariant in no trial. SINDy-FD is a different kind of method: it consumes the time-ordered trajectory, not transition pairs, and fits a *state* polynomial  $p(x_t, v_t)$ , so it has two defensible scoring targets, the state energy  $x^2 + v^2$  directly, and the induced transition form  $p(x_t, v_t) - p(x_{t+1}, v_{t+1})$ . Under both scorings it recovers the noiseless fixed- $dt$  oscillator’s energy in every trial.

Table 12: Transition invariant recovery rates.

| System                           | Mode                              | Exact recovery    | Redundancy |
|----------------------------------|-----------------------------------|-------------------|------------|
| Harmonic oscillator, fixed $dt$  | CSNP                              | 100% [0.89, 1.00] | 1.0        |
| Harmonic oscillator, random $dt$ | CSNP                              | 100% [0.89, 1.00] | 1.0        |
| Kepler angular momentum          | CSNP                              | 100% [0.89, 1.00] | 1.0        |
| Harmonic oscillator, fixed $dt$  | sparsity-only                     | 0% [0.00, 0.11]   | —          |
| All transition tasks             | SINDy-null / SINDy-ST             | 0% [0.00, 0.11]   | —          |
| Harmonic oscillator, fixed $dt$  | SINDy-FD, state-energy target     | 100% [0.89, 1.00] | —          |
| Harmonic oscillator, fixed $dt$  | SINDy-FD, induced transition form | 100% [0.89, 1.00] | —          |

Table 13: Multi-invariant recovery via nullspace deflation (2D harmonic oscillator, 4 simultaneous generators, 30 seeds). SINDy-null, RREF, and CSNP search the identical difference dictionary; SINDy-FD and SINDy-AD search a smaller, restricted state-only dictionary and each return a single candidate per seed, so the full 4-generator set is structurally unreachable for either (marked —). SINDy-AD differs from SINDy-FD only in its derivative estimator (a smoothing spline in place of a finite difference); it is a lightweight proxy for the joint denoising/noise-model method of Kaheman et al. (Kaheman et al., 2022), not a reimplement of it, which is a fair comparison at this table’s  $\sigma = 0$  since there is no noise for the joint noise model to characterize. RREF (Oellerich and Emelianenko, 2024) rows are read directly off the row-echelon-reduced nullspace basis, with no further Gröbner reduction, matching what that method actually delivers as output.

| Invariant            | CSNP             | SINDy-null   | RREF             | SINDy-FD     | SINDy-AD     |
|----------------------|------------------|--------------|------------------|--------------|--------------|
| Energy (per-axis)    | 100% [89%, 100%] | 0% [0%, 11%] | 100% [89%, 100%] | 0% [0%, 11%] | 0% [0%, 11%] |
| Cross term           | 100% [89%, 100%] | 3% [1%, 17%] | 100% [89%, 100%] | 0% [0%, 11%] | 0% [0%, 11%] |
| Angular momentum     | 100% [89%, 100%] | 0% [0%, 11%] | 100% [89%, 100%] | 0% [0%, 11%] | 0% [0%, 11%] |
| Full 4-generator set | 100% [89%, 100%] | 0% [0%, 11%] | 100% [89%, 100%] | —            | —            |



## 5.9 Point Vortices: Row-Echelon’s Nonlinear Limit

The 2D-oscillator deflation above is synthetic and low-dimensional. A harder, genuinely physical test is a system of  $n_v$  identical unit-circulation point vortices in the plane, evolving under the Kirchhoff equations ( $n_v$  is the vortex count; the state has  $n = 2n_v$  coordinates, so the symbol stays distinct from the variable count  $n$  used throughout). Its energy is logarithmic and so not a target, but translational and rotational symmetry give three polynomial first integrals: the linear impulses  $P = \sum_j x_j$  and  $Q = \sum_j y_j$  (both degree 1) and the angular impulse  $I = \sum_j (x_j^2 + y_j^2) - I_0$  (degree 2). Each initial condition is projected onto the shared level set  $P = Q = 0, I = I_0$  before integration, so pooling trajectories from different initial conditions leaves exactly the ideal  $\langle P, Q, I \rangle$  vanishing on the whole pool; pooling independent-value trajectories would cancel the invariants instead. The pooled state is lifted to a degree-2 library over its  $2n_v$  coordinates and handed to the full-nullspace deflation path. Because the Gröbner basis rewrites  $I$  into its normal form modulo  $P$  and  $Q$ , a per-generator scalar-multiple check against the raw  $I$  spuriously fails; recovery is scored instead by ideal equality against  $\langle P, Q, I \rangle$ .

The result, in Table 14, is a clean map of where the method holds and where it fails. For  $n_v = 3$  the recovered ideal equals  $\langle P, Q, I \rangle$  exactly, in every seed. At  $n_v = 4$  the two linear impulses are still recovered in every seed, but the degree-2 angular impulse is not: in the reduced basis its support grows quadratically with  $n_v$  (it is a  $(2n_v+1)$ -term sum of squares before reduction, denser still after elimination against  $P$  and  $Q$ ), and the sparse circuit search no longer reaches it within budget. Raising the support cap does not help, since the target is genuinely dense, not merely capped. By  $n_v = 5$  the over-lifted library’s singular spectrum has no usable gap and rank estimation returns nothing at all. This is the same deflation-numerics and combinatorial-scalability boundary discussed in Section 6.8, made concrete on a physical system: the low-degree, sparse invariants come out reliably, while a higher-degree invariant that is dense in the monomial basis is the first thing lost as the pooled dimension grows.

This system is also where row-echelon disambiguation (Oellerich and Emelianenko, 2024) stops tying CSNP.  $P$  and  $Q$  are linear and share no monomial with  $I$  or with each other, so RREF isolates both as individual rows exactly, at every  $n_v$  tested, reproducing the same success it had on the 2D-oscillator generators above and on the worked example of Oellerich and Emelianenko.  $I$  is different: it is the dense quadratic generator, and RREF’s raw output never contains a row that is a scalar multiple of it, at any  $n_v$ , including  $n_v = 3$  where CSNP recovers the full ideal  $\langle P, Q, I \rangle$  in every one of 30 seeds. Because RREF performs no further reduction, the fair test here is a direct match against each raw generator, not the ideal-equality criterion used for CSNP’s output above: any complete basis of a numerically correct nullspace spans the same ideal once run through a Gröbner computation, RREF’s dense, uncanonicalized rows included, so scoring RREF’s output that way would credit it with the very canonicalisation step it does not perform. Table 14 reports the direct per-generator match. This is the sharper instance the 2D-oscillator tie above could not settle: with genuine overlapping support, row reduction disentangles the linear part of the ideal but not the nonlinear part, exactly as Section 2 anticipates.

Table 14: Point-vortex multi-invariant recovery via full-nullspace deflation, degree-2 lift over the pooled state, mean over the seed grid ( $\sigma = 0$ ).  $P = \sum_j x_j$  and  $Q = \sum_j y_j$  are the degree-1 linear impulses;  $I = \sum_j (x_j^2 + y_j^2) - I_0$  is the degree-2 angular impulse. CSNP’s  $P/Q/I$  columns and “Ideal match” test ideal membership of the true generator against CSNP’s canonicalized (reduced Gröbner basis) output. The RREF (Oellerich and Emelianenko, 2024) columns instead test a direct scalar-multiple match against each raw generator, with no further reduction, since that is what row-echelon disambiguation actually delivers; an ideal-equality reading of RREF’s raw, non-minimal nullspace basis is close to tautological once the nullspace itself is numerically correct (Section 5.8) and is omitted here for that reason. 30 seeds, Wilson 95% CI on Ideal match.

| System     | $P$  | $Q$  | $I$  | Ideal match      | RREF $P$ | RREF $Q$ | RREF $I$  |
|------------|------|------|------|------------------|----------|----------|-----------|
| 3 vortices | 100% | 100% | 100% | 100% [89%, 100%] | 100%     | 100%     | <b>0%</b> |
| 4 vortices | 100% | 100% | 0%   | 0% [0%, 11%]     | 100%     | 100%     | <b>0%</b> |
| 5 vortices | 0%   | 0%   | 0%   | 0% [0%, 11%]     | 100%     | 100%     | <b>0%</b> |

## 5.10 Fixed- $dt$ versus Variable- $dt$ : The Regime Test

**The tie, confirmed at scale.** The full fifteen-monomial search is rerun 30 times at  $\sigma = 0$ ,  $dt = 0.1$ , against OMP-on-nullspace and SINDy-null as competitors, under three data-generating regimes: the exact rotation flow of Section 5.8, fixed- $dt$  symplectic Euler (the setting just analysed), and symplectic Euler with  $dt$  redrawn independently per step from  $[0.05, 0.15]$ . Table 15 reports exact recovery of  $H = x_t^2 + v_t^2 - x_{\text{next}}^2 - v_{\text{next}}^2$  in each cell. On the exact flow, where  $H$  is the only exactly-vanishing quadratic relation, CSNP recovers it in **100%** of seeds while OMP and SINDy-null recover 0%, reproducing Section 5.8’s result under a second, independently written harness. On fixed- $dt$  symplectic Euler, CSNP falls to **0%**: inspecting the returned candidates shows it is not recovering  $Q_{dt}$  either, but a third relation,

$$x_t^2 - x_t x_{\text{next}} + v_t v_{\text{next}} - v_{\text{next}}^2 = 0, \quad (12)$$

which we verified is a pure algebraic identity of the symplectic-Euler update rule itself, true for every  $dt$  by direct substitution of  $v_{\text{next}} = v_t - dt x_t$  and  $x_{\text{next}} = x_t + dt v_{\text{next}}$  (the two  $dt$ -dependent cross terms this substitution generates cancel exactly, leaving no  $dt$  anywhere in the result). Its coefficients are all  $\pm 1$ , so it costs exactly as many rationality bits as  $H$  itself, and BB’s tie-break has no structural reason to prefer one over the other. Because Equation (12) holds for every  $dt$  and not one specific value, it survives unchanged when  $dt$  is redrawn per step: the variable- $dt$  column also recovers  $H$  in **0%** of seeds, with CSNP returning Equation (12) again in every seed inspected. This is a different mechanism from the Section 5.8 variable- $dt$  ablation, where varying  $dt$  under the exact rotation flow *destroys* the fixed- $dt$  irrational-coefficient artifact and leaves  $H$  standing alone; here, varying  $dt$  leaves the artifact standing and  $H$  never becomes recoverable in the first place, because  $H$  is not even exactly conserved by symplectic Euler at any single  $dt$ , fixed or otherwise. OMP and SINDy-null fare no better on either symplectic-Euler column, but not for a shared reason: at fixed  $dt$ , OMP returns a candidate with no special structure, an approximate fit with maximum residual  $\approx 0.04$ , not an exact relation, while SINDy-null abstains outright in every seed inspected; at variable  $dt$ , both OMP and SINDy-null instead converge on Equation (12) exactly, the same artifact CSNP finds, once the wider dictionary’s other fixed- $dt$ -specific ties have been eliminated by the varying step size. The qualitative observations above (OMP’s  $\approx 0.04$  residual, SINDy-null’s abstention, which relation each method converged on) come from direct inspection of individual runs, not from an automated summary statistic.

**SINDy-FD and SINDy-AD: a tie specific to this system.** Both baselines were run on the same three-regime harness, each scored two ways, against the state energy  $x_t^2 + v_t^2$  directly and against the induced transition form  $p(x_t, v_t) - p(x_{\text{next}}, v_{\text{next}})$ . On the exact flow they tie CSNP at **100%** under both scorings (Table 15). This is not a competing resolution of the same ambiguity: SINDy-FD and SINDy-AD search a much smaller, restricted state-only dictionary  $(x_t, v_t)$  (six monomials) rather than the full fifteen-monomial transition-pair dictionary every other method in this table searches, and in that smaller dictionary  $H$  is the only exactly-vanishing quadratic relation, a consequence of the rotational  $\text{SO}(2)$  symmetry of the oscillator, with no competing candidate at all. They win by having no ambiguity to resolve, not by resolving it. On fixed- $dt$  symplectic Euler, where a real trajectory (not the exact rotation flow) is used for the first time, both baselines drop to **0%** under both scorings, returning the modified invariant  $Q_{dt} = x^2 - dt xv + v^2$  in every seed instead of  $H$ , the same failure Section 4.6 analyses for CSNP itself; variable  $dt$  has no single fixed step for either baseline’s derivative estimator and is not run. The scope of this table’s principal result is therefore narrower than the harmonic oscillator alone: it is a clean test against sparsity-first selection (Table 11) and against OMP and SINDy-null on the shared wide dictionary, but not against SINDy-FD or SINDy-AD’s shortcut of searching a smaller, hand-restricted dictionary with no built-in competitor. Table 13’s multi-invariant deflation benchmark is where that shortcut stops working: even in SINDy-FD and SINDy-AD’s smaller state-only dictionary, the 2D oscillator has four simultaneous invariants, real ambiguity survives, and both baselines return nothing usable there.

Table 15: Fixed- $dt$  vs. variable- $dt$  regime test: exact recovery over 30 seeds ( $N = 5000$ ,  $\sigma = 0$ ). SR-GB+CSNP, OMP, and SINDy-null search the full fifteen-monomial transition-pair dictionary and are scored against  $H$  directly. SINDy-FD and SINDy-AD search a different, restricted state-only dictionary  $(x_t, v_t)$ , have no entry on variable  $dt$  (no single fixed step for their derivative estimators), and are each scored two ways: against the state energy directly, and against  $H$  via the induced form  $p(x_t, v_t) - p(x_{\text{next}}, v_{\text{next}})$ . Every reported cell is 0% or 100%; the corresponding 95% Wilson intervals over 30 seeds are  $[0.00, 0.11]$  and  $[0.89, 1.00]$ ; the fixed and variable  $dt$  are for symplectic Euler; FD and AD stands for SINDy-FD and SINDy-AD.

| Regime               | SR-GB+CSNP  | OMP | SINDy-null | FD (state, transition) | AD (state, transition) |
|----------------------|-------------|-----|------------|------------------------|------------------------|
| Exact rotation flow  | <b>100%</b> | 0%  | 0%         | <b>(100%, 100%)</b>    | <b>(100%, 100%)</b>    |
| Euler, fixed $dt$    | 0%          | 0%  | 0%         | (0%, 0%)               | (0%, 0%)               |
| Euler, variable $dt$ | 0%          | 0%  | 0%         | —                      | —                      |

Table 15’s two Euler rows falsify nothing that Proposition 4.6 claims: the pairwise statement about  $H$  against  $Q_{dt}$  stands, and what fails to extend is recovery over the unrestricted dictionary, where the update-rule identity ties  $H$  in every seed tested, at 30-seed scale and not only at the single  $dt = 0.05$  probe of Section 4.6, and independently of  $\sigma$  or the particular  $dt$ .

## 6 Limitations and Future Work

The noise ceiling, the denominator cap, the unit-dependence of the rationality prior, and the Gröbner stage’s scaling are each measured rather than asserted, with a table in this section or its appendix.

### 6.1 Noise Ceiling for Higher-Degree Invariants

Degree-3 invariants degrade sharply from **100%** at  $\sigma = 0$  to **41.7%** at  $\sigma = 0.02$  and **16.7%** at  $\sigma = 0.05$ , far faster than degree-2 invariants, which hold **100%** and then **96.7%** over the same range (Table 9). This is the most consequential limitation in practice, since it bounds the degree at which the noisy benchmarks of Section 5.5 succeed. Generic noise amplification does not explain the gap’s sharpness by itself, since degree-2 monomials are also amplified by noise, just far more slowly.

**Mechanism.** CSNP’s starting point is the right singular vector of  $\Phi$  associated with its smallest singular value  $\sigma_M$  (Section 3.3); at  $d = 1$  this vector *is* the coefficient candidate before any support search or snap-rounding runs. A Davis–Kahan/Wedin-type perturbation bound (Davis and Kahan, 1970; Wedin, 1972; Golub and Loan, 2013) states that the angle between the noiseless null direction and the one recovered from a perturbed  $\Phi + E$  is controlled by  $\|E\|/\text{gap}$ , where  $\text{gap} = \sigma_{M-1} - \sigma_M$  is the absolute separation between the smallest singular value and its neighbour. Because  $\Phi$ ’s monomial columns span every degree from 0 to  $D$  at once (Section 3.3’s library construction never restricts to a single degree), a degree- $D$  dictionary has a column-norm spread that widens with  $D$ , and gap shrinks correspondingly even before any noise enters  $E$ . Remark 3.2 already documents one symptom of exactly this: on the Feynman benchmark’s degree-3 equations the largest consecutive ratio  $\rho_i$  is only  $\approx 1.5$ , a spectral elbow so shallow that the rank estimate needs its  $\rho_{\min}$  guard to avoid misfiring. A shallow elbow at  $\sigma = 0$  means gap starts small; the perturbation bound then predicts that a given absolute noise level  $\|E\| \propto D\sigma$  produces a far larger angular, and hence coefficient-space, deviation for degree 3 than for degree 2, not because the numerator  $D\sigma$  alone differs by much between  $D = 2$  and  $D = 3$ , but because the denominator gap is already an order of magnitude smaller for degree 3 at every noise level tested.

**Empirical check.** This is tested directly by comparing one degree-2 equation (`circle_locus`,  $x^2 + y^2 = r^2$ ) and one degree-3 equation (I.12.2,  $Fr^2 = kq_1q_2$ ) from the same benchmark suite, at  $N = 5000$  and  $\sigma \in \{0, 0.005, 0.01, 0.02, 0.05\}$ , 10 seeds each. For every seed it takes the raw smallest-singular-vector coefficient candidate, normalises it against the true polynomial’s leading coefficient, and reports its  $L_2$  distance from the true coefficient vector, alongside the consecutive-ratio gap  $\rho_{M-1}$  of Equation (1). Table 16 reports the result: at  $\sigma = 0.005$ , the smallest noise level tested beyond zero, degree-2’s coefficient error is already three orders of magnitude smaller than degree-3’s ( $4.1 \times 10^{-3}$  versus  $\sim 8 \times 10^3$ ), and degree-2 does not reach a comparably large error until  $\sigma = 0.05$  (mean 14.9).  $\rho_{M-1}$  tells the same story: degree-2 starts at  $\rho_{M-1} \approx 7.6$  and only approaches the degenerate  $\rho_{M-1} \approx 1$  regime by  $\sigma = 0.05$ , while degree-3 sits at  $\rho_{M-1} \approx 4.8$  already at  $\sigma = 0.005$  and never recovers a wide margin over the tested range. The degree-3

errors also carry far larger seed-to-seed variance (standard deviation exceeding the mean at several noise levels), consistent with a near-degenerate smallest singular vector being essentially unidentifiable, not a small, well-behaved perturbation of the truth.

Table 16: Raw nullspace-coefficient  $L_2$  error and mean gap ratio  $\rho_{M-1}$ , degree-2 vs. degree-3 (10 seeds,  $N = 5000$ ).

| Equation                           | $\sigma$    |                      |                      |                      |                   |
|------------------------------------|-------------|----------------------|----------------------|----------------------|-------------------|
|                                    | 0.000       | 0.005                | 0.010                | 0.020                | 0.050             |
| circle_locus (deg 2), $L_2$ error  | $\approx 0$ | $4.1 \times 10^{-3}$ | $1.5 \times 10^{-2}$ | $8.0 \times 10^{-2}$ | 14.9              |
| circle_locus (deg 2), $\rho_{M-1}$ | $\gg 1$     | 7.6                  | 3.8                  | 1.9                  | 1.3               |
| I.12.2 (deg 3), $L_2$ error        | $\approx 0$ | $8.2 \times 10^3$    | $6.2 \times 10^3$    | $1.1 \times 10^4$    | $3.2 \times 10^4$ |
| I.12.2 (deg 3), $\rho_{M-1}$       | $\gg 1$     | 4.8                  | 4.8                  | 4.7                  | 4.3               |

This connects directly to the Denominator Ceiling of Section 6.2: the snap-rounding tolerance  $\tau$  never exceeds  $\max(0.05, 3\sigma)$ , so once the raw coefficient error reaches order  $10^0$  or larger, as degree-3 already does at every tested  $\sigma > 0$ , no denominator search or MDL tie-break downstream can rescue the candidate, the estimate handed to BB is simply too far from the true rational for any  $q \leq Q_{\max}$  to fall within  $\tau$ . Degree-3’s collapse is therefore not a downstream selection failure but an upstream one: the nullspace estimate itself is already unusable by the time CSNP begins its combinatorial search.

This is a first-order, order-of-magnitude account grounded in a two-equation diagnostic, not a sharp closed-form noise ceiling as a function of  $(\sigma, D, N)$ ; a non-asymptotic bound tying gap to the specific monomial degree spread of an arbitrary dictionary, sharp enough to predict the exact  $\sigma$  at which any given equation collapses, instead of only explaining an observed collapse after the fact, is left as an open problem.

**A second, degree-2 mechanism: spurious sparse relations on narrow-range variables.** The degree-2 losses at  $\sigma = 0.05$  (momentum\_conservation\_1d and I.39.22 in Table 26) come from a different mechanism than the spectral-gap collapse above, diagnosed directly in Section 5.5: when one variable is confined to a short interval, a Chebyshev-small univariate quadratic in that variable passes the noise-widened feasibility tolerance and, being sparser than the true generator, wins the lexicographic objective. Unlike the degree-3 ceiling this is a bias, and more data does not remove it. It is also the one failure mode in this paper where the sparsity preference itself is the culprit, which is why a fit-quality-only baseline (SINDy-ST) survives those specific cells. No counter to this failure mode is implemented here.

## 6.2 Denominator Ceiling

Snap-rounding and the rationality-cost oracle both scan denominators up to a single fixed constant  $Q_{\max}$  (default 16, Table 20), the same constant at every noise level (Section 3.8). The natural alternative, deriving the ceiling from the snap tolerance itself as  $Q_{\max} = \max(1, \lfloor 1/(2\tau) \rfloor)$  with  $\tau = \max(0.05, 3\sigma)$ , shrinks as  $\sigma$  grows and is pinned at  $Q_{\max} = 10$  at the noise floor. That design has two failure modes: an exact, in-range coefficient whose reduced denominator sits just above the shrunk bound abstains outright even at  $\sigma = 0$ , and under noise a representable coefficient can fall outside the narrower band and get misrounded to a wrong, simpler fraction instead. The pipeline therefore fixes  $Q_{\max}$  at 16 regardless of  $\tau$ , and leaves the acceptance gate, not the window width, to reject a bad snap.

A fixed  $Q_{\max}$  still leaves a real ceiling, just not the  $\tau$ -derived one: any true invariant whose coefficients need a reduced denominator larger than  $Q_{\max}$  is unrecoverable outright, since no  $q \leq Q_{\max}$  can ever match it. Section E.2 measures this ceiling directly and confirms it is this clean: recovery is 0% whenever the invariant’s minimum representable denominator  $q^*$  exceeds  $Q_{\max}$  and 100% the moment  $Q_{\max} \geq q^*$ , with no additional margin needed. Here  $q^*$  must be computed under snap-rounding’s own normalisation, dividing the coefficient vector by whichever entry has the largest magnitude, since that is the entry snap-rounding actually divides by; a normalisation chosen instead to minimise the resulting denominator can understate the true floor substantially. This is distinct from and additional to the coefficient-irrationality restriction of Section 6.9.

### 6.3 Denominator-Cap and Tolerance Sensitivity

Two constants govern the rationality machinery throughout, the denominator cap  $Q_{\max}$  and the snap tolerance  $\varepsilon$  used for the exact-match track (Section 3.8), and neither has been varied elsewhere in this paper. A three-arm sensitivity sweep (Appendix E.2, 30 seeds per arm) measures exact recovery against  $Q_{\max}$  alone on six conics of increasing minimum representable denominator  $q^*$ , checks that widening  $Q_{\max}$  never manufactures a false positive on two negative controls, and sweeps the rationality-window tolerance  $\varepsilon$  across four orders of magnitude. The principal finding: recovery is a clean step function of  $q^*$  (computed under snap-rounding’s own normalisation, Section 6.2), 0% below it and 100% at or above it with no observed margin, so a coefficient needing a denominator up to the low twenties can already exceed the default  $Q_{\max} = 16$  and abstain outright; raising  $Q_{\max}$  costs nothing on the negative controls tested, so the default is a reasonable, conservative choice, not one the method is fragile around, and  $\varepsilon$  needs no per-benchmark tuning at all.

### 6.4 Unit-Dependence of the Rationality Prior

Every benchmark elsewhere in this paper is generated in units chosen so the true invariant already has small-integer coefficients, the unit circle, the unit-mass oscillator, and so on, so the rationality prior has never been stress-tested against an arbitrary choice of units. It is not scale-invariant: relabelling a variable  $v$  as  $cv$  for some constant  $c$  turns a coefficient that was 1 into  $1/c^2$ , which need not be a simple rational under the Denominator Ceiling of Section 6.2 at all, and the mechanism there is exactly what is at stake here, an arbitrary  $c$  has no reason to leave  $1/c^2$  within tolerance of any  $p/q$  with  $q \leq Q_{\max}$ .

This is made concrete on four benchmarks, the circle, the fixed- $dt$  harmonic oscillator, Kepler angular momentum, and Feynman `momentum_conservation_1d` ( $m_1v_1 + m_2v_2 - P$ , chosen because its five variables span three physically distinct quantities, mass, velocity, and momentum), each rescaled by an independent random factor per physical quantity (log-uniform in  $[0.15, 8]$ ) and rerun under two conditions: no preprocessing, and each column divided by its own empirical standard deviation before the monomial library is built. Table 17 reports exact recovery of the true invariant, expressed in whatever coordinate system the pipeline actually saw, over 30 seeds.

Table 17: Unit-invariance stress test: exact recovery under random per-quantity rescaling (30 seeds each). “Baseline” is the unmodified benchmark already reported elsewhere in this paper.

| Benchmark                                     | Baseline | Rescaled    | Standardized |
|-----------------------------------------------|----------|-------------|--------------|
| Circle                                        | 100%     | 0%          | 0%           |
| Oscillator, fixed $dt$                        | 100%     | 0%          | 0%           |
| Kepler angular momentum                       | 100%     | <b>100%</b> | 0%           |
| Feynman <code>momentum_conservation_1d</code> | 100%     | 0%          | 0%           |

The damage is total on three of the four benchmarks, exactly as the denominator-ceiling argument predicts, and naive per-column standardization does not rescue any of them: dividing by the empirical standard deviation removes the synthetic factor  $c$  from each column almost exactly, in expectation, but replaces it with  $1/\text{std}(x)$ , the reciprocal of that variable’s own natural spread in whatever units it happened to be generated in, which is no more likely to be a simple rational than  $c$  was. Standardizing to unit variance is not the same as standardizing to simple-rational invariant coefficients, and nothing in the pipeline currently bridges that gap.

Kepler angular momentum is the instructive exception, and the reason is structural, not a failure of the argument above. Every term of  $L = x_tvy_t - y_tv_x - x_{\text{next}}vy_{\text{next}} + y_{\text{next}}vx_{\text{next}}$  is a product of exactly one position variable and one velocity variable, so rescaling all four position variables by one factor  $c_{\text{pos}}$  and all four velocity variables by another,  $c_{\text{vel}}$ , multiplies every term by the same  $c_{\text{pos}}c_{\text{vel}}$ : a single global scalar, not a change in the coefficients’ relative ratios, and the rationality cost depends only on the latter after lead-coefficient normalisation (Section 3.4). This robustness is a property of  $L$ ’s bilinear, single-position-times-single-velocity monomial structure, not a general immunity of physically meaningful invariants to unit changes, and it is fragile: per-column standardization assigns each of the eight variables its *own independent* empirical scale, not one shared scale per physical quantity, which breaks exactly the group-level coherence that made the raw rescaling harmless. Recovery on Kepler accordingly drops from **100%** rescaled to **0%**



standardized, the one case in Table 17 where the mitigation actively destroys a robustness the raw data already had, worse than doing nothing.

The practical conclusion is narrower than “standardize your data”: the rationality prior assumes coordinates already close to the units the true invariant is simple in, an assumption every other benchmark in this paper satisfies by construction and this one does not, and no preprocessing tested here restores that assumption once it is violated.

## 6.5 Modified-Equation Separation: What Remains Open

Proposition 4.6 proves recovery once the coefficient-space gap  $\|Q_{dt} - H\|$  falls below the rationality tolerance  $\varepsilon$ , but it does not go the other direction and derive  $\varepsilon$  (or the snap tolerance  $\tau$ ) from  $dt$  and the integrator order  $p$ . The direct version of that fix was tried and fails for a structural reason: the tolerance-sweep experiment of Section E shows the wide-dictionary failure is a set of exactly-vanishing relations tied at the same integer rationality cost, so no choice of  $\tau$  changes which one the tie-break lands on, and Section 5.10 identifies the specific competitor at 30-seed scale, the update-rule identity of Equation (12), which holds for every  $dt$  and so survives per-step redraws. What remains open is therefore a tie-break that uses the integrator’s structure to discount relations implied by the update rule alone; the difference-dictionary restriction below is a first concrete step. The tie itself, on the unrestricted dictionary, has only been probed at  $\sigma = 0$ , so whether representative sampling noise changes which candidate wins it is untested; the restriction’s own behaviour under noise is measured below.

One structural mitigation does already work. Restricting the transition dictionary to difference features  $p(x_t, v_t) - p(x_{t+1}, v_{t+1})$ , one per state monomial  $p$ , excludes Equation (12) by construction: the identity contains the mixed monomials  $x_t x_{t+1}$  and  $v_t v_{t+1}$ , and no difference feature contains a mixed monomial. Run on the same symplectic-Euler data ( $dt = 0.1, \sigma = 0$ ), the search over this difference dictionary returns exactly  $x^2 - \frac{1}{10}xv + v^2$ , that is,  $Q_{dt}$  of Equation (10), in every seed probed. This is the best case achievable here:  $H$  itself is not conserved by the integrator, so no exact method can recover it from this data, and the modified invariant is the physically meaningful object backward error analysis says the data actually carries. Dictionary structuring therefore converts the failure from returning a meaningless update-rule artefact into returning the correct modified invariant; recovering  $H$  itself would additionally require the  $Q_{dt} \rightarrow H$  step, for instance snapping the  $dt$ -dependent coefficient to zero when  $dt$  is known, which is again integrator-aware post-processing, not search.

Two remarks bound the scope of this mitigation. The restriction can never discard a true conserved quantity: for any polynomial invariant  $Q$  of the map,  $Q(x_t, v_t) - Q(x_{t+1}, v_{t+1})$  expands term by term into difference features, so the true target always remains representable. Nor is it specific to this oscillator. The multi-invariant deflation benchmark of Section 5.8 already runs end to end on exactly this dictionary and recovers all four generators, and the linear update relations of that benchmark’s fixed- $dt$  rotation map are excluded for an analogous reason: a relation of the form  $x_{t+1} - cx_t - sv_t$  lies in the span of the difference features only when the map is the identity. What is not established is the converse guarantee: nothing proven here rules out an integrator whose update rule implies an exactly-vanishing relation that itself lies in the difference span, so the exclusion argument remains evidence of generality, not a proof of it.

The empirical scope, however, is wider than a single zero-noise cell. The restriction is rerun along the two axes the paragraph above leaves open, a second integrator and representative sampling noise, deriving the exactly conserved quadratic  $Q_{dt}$  of each (integrator,  $dt$ ) cell in closed form as ground truth: symplectic Euler at  $dt \in \{1/10, 1/2\}$  and Störmer–Verlet at  $dt \in \{1/2, 1\}$ , whose conserved form  $x^2 + \frac{4}{4-dt^2}v^2$  has no cross term at all, each at  $\sigma \in \{0, 0.01, 0.02, 0.05\}$  over 30 seeds (Table 18). At  $\sigma = 0$  the modified invariant is recovered in every seed of every cell, so the exclusion carries unchanged to an integrator it was not designed around. Under noise the restriction inherits the pipeline’s usual coefficient-resolution limit instead of failing in any new way. Recovery stays at or near 100% wherever the  $dt$ -dependent coefficients sit well above the noise-widened threshold; Störmer–Verlet at  $dt = 1/2$  falls back at  $\sigma = 0.05$  to the true energy  $x^2 + v^2$  in 96.7% of seeds, which is the interpretable failure expected once  $3\sigma$  exceeds its coefficient gap of  $1/15$  between  $Q_{dt}$  and  $H$ ; and symplectic Euler at  $dt = 1/10$  abstains outright at every  $\sigma > 0$  tested, returning no invariant at all, not a wrong one.



Table 18: Difference-dictionary generality probe: recovery of the integrator’s exactly conserved quadratic  $Q_{dt}$  ( $N = 5000$  transitions, 30 seeds, 95% Wilson CI). The  $H$  column counts seeds returning the true energy  $x^2 + v^2$  instead of  $Q_{dt}$ , the fallback mode expected once  $3\sigma$  exceeds the coefficient gap between  $Q_{dt}$  and  $H$ .

| Integrator       | $dt$ | $\sigma$  | $Q_{dt}$ recovery  | $H$ instead |
|------------------|------|-----------|--------------------|-------------|
| Symplectic Euler | 1/10 | 0.00      | 100% [0.89, 1.00]  | 0%          |
| Symplectic Euler | 1/10 | 0.01–0.05 | 0% [0.00, 0.11]    | 0%          |
| Symplectic Euler | 1/2  | 0.00–0.05 | 100% [0.89, 1.00]  | 0%          |
| Störmer–Verlet   | 1/2  | 0.00–0.01 | 100% [0.89, 1.00]  | 0%          |
| Störmer–Verlet   | 1/2  | 0.02      | 93.3% [0.79, 0.98] | 0%          |
| Störmer–Verlet   | 1/2  | 0.05      | 0% [0.00, 0.11]    | 96.7%       |
| Störmer–Verlet   | 1    | 0.00–0.05 | 100% [0.89, 1.00]  | 0%          |

## 6.6 Combinatorial and $L_1$ -Fallback Scalability

The combinatorial search becomes infeasible for  $M > 100$ . Past both exact-engine budgets the pipeline enters the anytime fallback of Section 3.6, whose incumbent is seeded by the  $L_1$  relaxation; the fallback can only improve on that seed under the lexicographic objective, but once its node cap is exhausted it carries no optimality certificate. On the 4-sphere stress test, the  $L_1$  seed achieves only 13.3%, 10.0%, and 0% recovery at  $\sigma = 0.01, 0.02$ , and  $0.05$ , respectively. For practical use with large dictionaries, this fallback path should be considered a heuristic, not a guaranteed recovery method.

## 6.7 Feasibility Oracle: A Guarded, Not Proven, Rank Test

Theorem 4.4’s global-optimality certificate is conditional on the feasibility classification of Section 3.5 labelling every support correctly, not an unconditional guarantee that BB never discards the true generator; Section 4.4 works through why (a false Hard Infeasible verdict is silently propagated by dominance pruning to every subset of the support) and measures the gap directly on the 2D-oscillator diagnostic, where the  $\kappa\sigma$  guard closes it at that benchmark’s own noise level but only narrows it at higher noise, plateauing around a 93% rescue rate at  $\sigma = 0.10$ . The certificate should accordingly be read as relative to this guarded oracle, with the residual failure rate that section quantifies, not as a proof that the pipeline always finds the minimal generator once BB is invoked.

## 6.8 Deflation Numerics

Step 7 of Algorithm 1 projects the recovered direction  $c$  out of  $V_{\text{null}}$  before repeating the search (Section 3.13). In the implementation,  $c$  at this point is the *snap-rounded*, rational coefficient vector, not the raw pre-rounding SVD estimate. Snap-rounding is what restores exactness on a genuine invariant, but it also means  $c$  is no longer guaranteed to lie exactly in  $\text{span}(V_{\text{null}})$ : the rounding perturbation, small as it is, is generic, not aligned with the nullspace, so the projection  $V_{\text{null}} \leftarrow (I - cc^T/\|c\|^2)V_{\text{null}}$  removes a direction close to, but not exactly equal to, the true recovered generator. The practical effect is a small residual leakage of the just-found invariant back into the deflated nullspace, not a clean rank drop of exactly one. This leakage is the reason Section 3.6 recommends no more than 2–3 deflation iterations: each iteration’s leakage is small on its own, but repeated projections against an already-imperfect basis compound, and the numerical distance between the deflated subspace and the true remaining invariants’ span grows correspondingly with iteration count.

A second limitation is structural rather than numerical. The projection in Equation (8) removes  $\text{span}(c)$ , a single dimension, whereas the ideal  $\langle c \rangle$  contributes an entire graded component to the nullspace at any search degree  $D$  above  $\deg c$ : every product  $m \cdot c$  with  $\deg m \leq D - \deg c$ , of dimension  $\binom{n+D-\deg c}{D-\deg c}$ . At  $D = \deg c$ , the case for every generator reported in Sections 5.8 and 5.9 that is recovered inside a library set to its own degree, the two coincide and the projection is exact. Where a found generator’s degree sits below the search degree instead, as with the point-vortex system’s linear impulses  $P, Q$  recovered inside a degree-2 library (Section 5.9), the deflated basis genuinely retains trivial multiples that one projection cannot remove. This does not corrupt the reported generator sets: the deflation collection loop separately tests every subsequent candidate for ideal membership against the generators already accepted, by reducing it modulo

their Gröbner basis, and rejects it if it is redundant rather than reporting it. The cost is search effort, not correctness, since each rejected trivial multiple still consumes one of the collection loop’s iterations before a genuinely new generator is reached.

## 6.9 Polynomial-Only Restriction and Coefficient Irrationality

SR-GB+CSNP is designed for low-description-complexity *rational* invariants; both restrictions below follow from that design choice, not from an incidental implementation gap, and algebraic-number or non-polynomial extensions are future work, not a present capability.

The most fundamental restriction is that SR-GB+CSNP recovers only polynomial invariants of bounded degree. This is not merely a scoping choice but follows directly from the paper’s central theoretical guarantee: the canonical, minimal generating set that SR-GB+CSNP certifies is the reduced Gröbner basis of a polynomial ideal, and reduced Gröbner bases, along with their uniqueness for a fixed monomial ordering (Cox et al., 2015), are defined only over polynomial rings. Trigonometric, exponential, or other transcendental relations do not form a commutative polynomial ring in the same sense, so there is no directly analogous canonical-form theory to invoke; extending canonicalisation to such relations would need a different algebraic foundation (differential-algebraic or Pfaffian methods, for instance), not a straightforward generalisation of the Gröbner basis machinery this paper builds on. Many physical laws,  $E = mc^2$ , exponential decay, trigonometric identities, are not polynomial. Two small MuJoCo (Todorov et al., 2012) pendulum benchmarks included in the repository make both sides of this restriction concrete with real rigid-body simulation, not synthetic trajectories. A linear spring-mass oscillator, whose discrete RK4 dynamics conserves the polynomial energy  $\dot{q}^2 + q^2$  exactly, is recovered by SR-GB+CSNP in all 30 trials. Replacing the linear restoring force with a genuine gravity pendulum instead gives a conserved energy  $0.5\dot{\theta}^2 - (g/L)\cos\theta$  that is transcendental in  $\theta$ ; there, the pipeline abstains in all 30 trials instead of snap-rounding some spurious near-fit, the correct behaviour given that no polynomial invariant of bounded degree exists in the raw coordinates  $(\theta, \dot{\theta})$ . Recovery would require  $(\cos\theta, \sin\theta)$  to be supplied as input coordinates, whereas PySR could in principle discover the trigonometric transformation itself given the right operators. Extending the method to non-polynomial terms via dictionary expansion (including  $\sin$ ,  $\cos$ ,  $\exp$ ) is future work.

A second, narrower restriction concerns coefficients, not monomials, and splits into cases the pipeline handles differently. Under constant-interval sampling, the nullspace contains both the physical invariant and a sampling-induced artefact with irrational coefficients; here the adaptive global MDL criterion (Proposition 4.5) resolves the ambiguity, since the true rational invariant has strictly lower rationality cost and sparsity than the irrational artefact (validated in Section 5.8: 100% recovery for the full pipeline against 0% for a sparsity-only ablation). What remains genuinely unsolved is a true invariant with inherently irrational coefficients: since snap-rounding and Gröbner canonicalisation over  $\mathbb{Q}$  require exact rational arithmetic, a true invariant such as  $x^2 + \sqrt{2}y^2 - 1 = 0$  gets its  $\sqrt{2}$  snapped to the nearest rational  $p/q$  within tolerance, losing exact algebraic structure. Extending the method to algebraic field extensions (e.g.  $\mathbb{Q}(\sqrt{2})$ ) or to irrational coefficients while preserving canonicalisation would need a generalisation of the Gröbner basis framework to polynomial rings over algebraic number fields, or a lattice-reduction approach such as LLL (Lenstra et al., 1982). A related edge case is also open: if the sampling interval  $dt$  happens to be exactly a rational multiple of  $\pi$ , both the physical invariant and the sampling artefact have rational coefficients, and the rationality cost alone cannot resolve the ambiguity; the method instead applies a guided mode or issues a warning in that rare case, independently of the adaptive degree discovery.

## 6.10 Gröbner Basis Scalability

Computing reduced Gröbner bases scales poorly with both the number of variables and the polynomial degree (Cox et al., 2015). Table 19 provides measured timings for larger configurations, on random sparse ideals of the shape snap-rounding actually hands to the Gröbner stage (3–6 term generators with small rational coefficients, forced support overlap). Median times stay below a fifth of a second through  $n = 12$ ,  $D \leq 3$ , but the distribution is heavy tailed: individual instances at  $n = 12$ ,  $D = 2$  can exhaust a 120-second per-instance budget outright, which is the practical failure mode (occasional blowup, not smooth degradation). The certified regime  $n \leq 8$ ,  $D \leq 4$  quoted throughout is chosen to stay clear of that tail.

Table 19: Gröbner basis computation timings (SymPy, grevlex over  $\mathbb{Q}$ ) for random sparse polynomial ideals with the stated number of generators. Median over 3 seeded instances per row, with a 120 s per-instance cap; the timeout column counts instances hitting the cap (excluded from the median).

| $n$ | $D$ | $M$ | Generators | Median time | Timeouts / 3 |
|-----|-----|-----|------------|-------------|--------------|
| 4   | 2   | 15  | 1          | 0.0009 s    | 0            |
| 6   | 2   | 28  | 2          | 0.003 s     | 0            |
| 8   | 2   | 45  | 3          | 0.07 s      | 0            |
| 10  | 2   | 66  | 4          | 0.01 s      | 0            |
| 12  | 2   | 91  | 5          | 0.09 s      | 2            |
| 8   | 3   | 165 | 2          | 0.002 s     | 0            |
| 10  | 3   | 286 | 3          | 0.11 s      | 0            |

A step-by-step operational protocol distilled from these limitations (choosing  $N$ , reading an abstention, reading an L1-fallback answer) is given in Appendix A.1.

## 6.11 Future Work

The most direct next test of the method is state-transition data from trained reinforcement-learning agents, which was its original motivation. Unlike the closed-form mechanical systems and program traces covered here, whether such trajectories carry any recoverable polynomial invariant is itself unvalidated: actuated, dissipative control tasks have no general reason to conserve a fixed quantity across a transition, so the more plausible target is an action-independent kinematic constraint among redundant state coordinates (link lengths, angle-embedding identities, forward-kinematic relations) rather than a dynamical conservation law. Establishing which control benchmarks admit such constraints, and which admit none, is a prerequisite for any downstream use, and a method that reports an invariant where none exists would be worse than one that abstains.

Several further extensions remain open, independent of that direction. Section 6.8 identifies a structural gap in the current deflation rule; replacing projection with normal-form reduction would close it and let the adaptive loop lift degree without leakage. Relatedly, since physical constraints typically involve few coordinates at a time, grading the combinatorial search by the number of distinct variables a candidate support touches would decompose one large search into many small ones, each inside the certified regime of Proposition 4.1. Richardson extrapolation across step sizes could in principle recover  $H$  to higher accuracy than any single fixed  $dt$  permits, by running the same system at several step sizes and extrapolating  $Q_{dt} \rightarrow H$  as  $dt \rightarrow 0$ , turning coarse- $dt$  trajectory data that is otherwise stuck below the modified-equation floor into usable input; this would require its own sampling and error-propagation analysis. The closed-form separation of Section 4.6 also depends on the one-step map being linear, which only holds for quadratic Hamiltonians; for a general Hamiltonian, backward error analysis still guarantees a modified invariant  $H_{dt} = H + O(dt^p)$ , but only as an asymptotic series without a closed form or explicit remainder bound, so the monomial-support argument behind Proposition 4.6 does not obviously carry over. A systematic comparison against Koopman-eigenfunction invariant discovery (Section 2) remains to be run. The parallel comparison against row-echelon disambiguation (Oellerich and Emelianenko, 2024) does not: Sections 5.8 and 5.9 report it directly on our nonlinear multi-invariant benchmarks, the 2D-oscillator deflation and the point-vortex system. Hybrid numerical-symbolic methods, including learned Gröbner basis computation (Kera et al., 2024), could extend the pipeline to noisier, higher-dimensional settings. Uncertified L1-fallback candidates (Appendix A.1) are a natural input to an exact checker such as DIG (Nguyen et al., 2014): DIG’s guess-and-check loop expects exactly this kind of candidate relation to verify or refute against further data, a hybrid neither system currently implements.

## 7 Conclusion

This paper introduced SR-GB+CSNP, a pipeline that recovers minimal, canonical polynomial invariants from noisy data by combining Combinatorial Sparse Nullspace Pursuit with an information-theoretic rationality cost, lexicographic Branch-and-Bound optimisation, noise-tolerant snap-rounding, and reduced Gröbner

basis canonicalisation. Its contributions are the rationality cost as the primary selection criterion in place of sparsity, the certified BB search that minimises it, the adaptive monomial expansion that discovers the required polynomial degree instead of requiring it as input, and the canonicalisation stage that makes the output unique. We proved three theoretical results: (i) the exact recovery condition for CSNP, (ii) the global optimality of the BB search with a certificate, and (iii) a Normal-Form Pruning Lemma together with a conditional adaptive exact-recovery proposition that extend these guarantees, at the level of the recovered ideal, to the full adaptive pipeline. On the Feynman polynomial benchmark, SR-GB+CSNP achieves 100% exact recovery at zero noise, and degree-2 equations retain 100% at  $\sigma = 0.02$  and 96.7% at  $\sigma = 0.05$ .

Comparing against **SINDy-null (KRONIC)** shows that sparsity-based nullspace methods are sufficient for identifying a single invariant ( $d = 1$ ) but fundamentally fail to disentangle multiple coexisting invariants ( $d > 1$ ). On the 2D-oscillator deflation benchmark, SR-GB+CSNP recovers all four entangled generators in every seed, whereas SINDy-null on the identical difference dictionary never returns the full set. That same disentanglement gap reappears on physical dynamics: the point-vortex system at  $n_v = 3$  recovers an ideal exactly equal to  $\langle P, Q, I \rangle$  in every seed. Degree lifting tells a related story: algebraic minimality turns out to be essential when the true degree is unknown, since SINDy-null achieves only 3.3% recovery on the over-lifted circle, while Dense SVD+GB reaches 100% through factorisation. SR-GB+CSNP addresses these gaps through combinatorial search, algebraic minimality, adaptive degree discovery, and global optimality certification. This makes canonical forms the default output of the discovery pipeline, which is what downstream consumers that need decidable equality between relations, SOS solvers and SMT verifiers among them, actually require.

Three failure modes bound these results (Table 5). The rationality prior is not scale-invariant: Section 6.4 shows three of four benchmarks collapse under an arbitrary but physically reasonable choice of units, and naive standardization does not reliably rescue them. The wide-dictionary tie of Section 6.5 is a genuine algebraic degeneracy, not a miscalibrated tolerance, so no choice of  $\tau$  resolves it; only restricting the dictionary’s structure does. The point-vortex system of Section 5.9 shows the same combinatorial search losing a dense higher-degree generator once the pooled dimension grows past  $n_v = 3$ , the specific regime where row-echelon disambiguation also fails. Unit-robust recovery and the wide-dictionary tie remain the two open problems whose resolution would most extend the certified regime.

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## A Implementation Notes

### A.1 A Practitioner’s Protocol

This subsection condenses the limitations above into operational guidance; per-benchmark commands, outputs, and further usage detail are documented in the accompanying repository, <https://github.com/TesfayZ/srgb-csnp>.

**Choosing  $N$ .** For single-invariant ( $d = 1$ ) problems at moderate noise,  $N \geq 100$  already achieves 100% exact recovery on the unit circle at  $\sigma = 0.01$  (Section 5.3) and sets the right order of magnitude. Transition-invariant and multi-dimensional-nullspace problems need substantially more; every such benchmark in this paper uses  $N = 5000$ .

**Reading an abstention.** A run returning no invariant has at least three causes, which the pipeline does not distinguish automatically. (i) No polynomial invariant exists at the searched degrees: the MDL history (Section 3.10) is then flat or monotonically increasing, with no interior minimum. (ii) The data are in the wrong units (Section 6.4): rescaling each variable by its empirical standard deviation and rerunning diagnoses this, though Table 17 shows the rescaling does not reliably repair it. (iii) Noise exceeds the degree-dependent ceiling (Section 6.1): a candidate is found but rejected at the final residual gate, visible in the search log.

**Reading an L1-fallback answer.** Past the exact engines’ budgets (Section 3.5), the returned candidate carries no optimality certificate (Section A.5) and should be treated as a hypothesis for external verification, not as equivalent to a certified BB output.

### A.2 Environment and Reproducibility

All results in this paper were generated with the repository’s pinned dependency versions (Python 3.11.9, NumPy, SciPy, SymPy, scikit-learn, pandas); every table is backed by its own script and output file in the repository, cross-referenced against this paper in the repository’s README. Every benchmark reporting “30 seeds” uses seeds  $0, 1, \dots, 29$  unless a table caption states otherwise. The one exception to the script-per-table rule is the DIG comparison, produced by driving the external `dynaroars/dig` tool at one run per system and noise level ( $9 \times 3 = 27$  runs); its commit and comparison protocol are given in Section H.

### A.3 Scale-Relative Residual Acceptance

The norm-scaled tolerance  $\tau_{\text{resid}} = \max(10^{-4}, 3\sigma) \cdot \sqrt{N} \cdot \Phi_{\text{rms}}$  of Section 3.6 is calibrated for the aggregate norm check  $\|\Phi c\|$ , whose scale grows with  $\sqrt{N}$  under i.i.d. noise; reusing it directly for a *per-point maximum* residual (Section 3.7) is incorrect on two counts. First, the maximum of  $N$  noise samples grows like  $\sqrt{2 \ln N}$ , not  $\sqrt{N}$ , which is roughly  $8\times$  too loose at  $N = 1000$ . This is why the per-factor check of Section 3.7 instead uses the gradient-scaled, extreme-value-corrected tolerance  $3\sigma \cdot g \cdot \sqrt{2 \ln N}$ . Second, an *absolute* threshold is not scale-invariant: the same relation  $p(x) = 0$  evaluated on data ranging over  $[0, 5]$  with squared terms produces residuals about  $25\times$  larger than on  $[0, 1]$  data at identical noise, so an absolute cutoff wrongly rejects genuine invariants on large-magnitude data while accepting spurious fits on small-magnitude data. This second problem is addressed at the final acceptance gate applied after the recursion in Section 3.7 (immediately before Gröbner canonicalisation, Section 3.12): the maximum residual there is gated *relative to* the candidate polynomial’s own characteristic term magnitude on the data (the RMS value of its non-constant monomial terms), which restores the same scale-invariance the normalised  $\|\Phi c\|$  check already enjoys via the normalisation of  $c$ , again combined with the  $\sqrt{2 \ln N}$  extreme-value scaling, not  $\sqrt{N}$ . Both fixes address the same underlying error, an aggregate, absolute tolerance reused for a per-point, scale-dependent check, at the two different points in the pipeline where a maximum residual is evaluated.

$\tau_{\text{resid}}$  itself carries a related correction, the  $\Phi_{\text{rms}}$  factor above: the RMS entry magnitude of the active monomial library. Without it, an absolute floor tuned for unit-scale data either rejects an exact invariant outright on a large-dynamic-range library (the ps3-type case, where even the true relation’s float residual routinely exceeds a bare  $10^{-4}$ ) or goes vacuous on tiny-scale data, the same absolute-threshold failure mode the per-factor and final-acceptance gates above guard against. A further detail is easy to miss from the formula alone: the  $\sqrt{N}$  factor multiplies the whole  $\max(10^{-4}, 3\sigma)$  term, not just the  $3\sigma$  branch inside it, so at very small  $\sigma$  (where the  $10^{-4}$  floor governs) the tolerance still grows with  $\sqrt{N}$  through  $\Phi_{\text{rms}}$ , not sitting flat at  $10^{-4}$ . This is a deliberate consequence of keeping the aggregate norm check’s  $\sqrt{N}$  scaling, which is a

property of the check itself (summing  $N$  residual contributions), not of the noise estimate, and should apply regardless of which branch of the noise-dependent max is active.

## A.4 Numerical Thresholds

When  $\sigma$  is unknown, we estimate it as  $\hat{\sigma} = \text{median}(\text{tail singular values})/\sqrt{N}$  following the Marchenko–Pastur law (Marchenko and Pastur, 1967; Golub and Loan, 2013). Rank tests on candidate-support submatrices fall back to the absolute tolerance  $\max(10^{-8}, 10^{-6}\sigma_1)$  when no spectral gap clears the ratio threshold (Remark 3.2).

Table 20 collects the hand-set constants used throughout Sections 3 and 4 in one place; each is a tuning choice, not a value a proof depends on holding exactly (the noise-adaptive tolerances such as  $\varepsilon = \max(10^{-3}, 3\sigma)$  and  $\tau_{\text{resid}}$  are not repeated here since their functional *form* is what the recovery guarantees rely on, not a fixed number, and each is derived where it is used).

Table 20: Hand-set implementation constants, none fit to a particular benchmark. Defaults are as stated in Section 5.

| Constant                 | Default         | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Q_{\max}$               | 16              | Denominator ceiling for snap-rounding (Section 3.8) and the rationality-cost oracle (Section 3.4); sensitivity measured directly in Section E.2.                                                                                                                                                                                                                                                                                                                     |
| $\rho_{\min}$            | 2.0             | Minimum spectral gap ratio trusted for top-level rank estimation (Remark 3.2).                                                                                                                                                                                                                                                                                                                                                                                       |
| $\rho_{\min}^A$          | 20              | Minimum gap ratio trusted for per-support feasibility classification (Section 3.5); higher than $\rho_{\min}$ because a spurious gap is more likely in a small submatrix.                                                                                                                                                                                                                                                                                            |
| $\kappa$                 | 3.5             | Multiplier for the noisy-rank guard’s absolute near-kernel tolerance ( $\kappa\sigma$ , Section 3.5), which admits a per-support classification the $\rho_{\min}^A$ gap test alone would wrongly call Hard Infeasible. A rank-classification margin, not a residual or coefficient tolerance, so it is not held to the same default as those; its sensitivity, and the cost of leaving it out entirely, are both measured directly on the diagnostic in Section 3.5. |
| Direct-enum. bud-<br>get | $2 \times 10^5$ | Subset count below which CSNP enumerates supports exhaustively instead of routing to BB (Algorithm 1, Step 3).                                                                                                                                                                                                                                                                                                                                                       |
| BB budget                | $2 \times 10^8$ | Predicted-unit ceiling below which the exact BB search runs, at the largest affordable support cap (Algorithm 1, Step 3).                                                                                                                                                                                                                                                                                                                                            |
| Beam width $w$           | 20              | Warm-start beam search width used only to seed BB’s initial upper bound, not its dominance cache (Section 3.5).                                                                                                                                                                                                                                                                                                                                                      |
| $D_{\max}$               | 10              | Safety bound on the adaptive degree sweep (Section 3.10).                                                                                                                                                                                                                                                                                                                                                                                                            |
| $\mu$                    | 3               | Consecutive-miss bound that stops the degree sweep (Section 3.10).                                                                                                                                                                                                                                                                                                                                                                                                   |
| Deflation iters.         | 2–3             | Recommended cap on nullspace-deflation steps before numerical error propagates into noise-dominated subspaces (Algorithm 1, Step 7).                                                                                                                                                                                                                                                                                                                                 |

## A.5 L1 Fallback Details

The parameterisation  $c = V_{\text{null}}\alpha$  is used with  $V_{\text{null}} \in \mathbb{R}^{M \times d}$ . Fixing  $\alpha_1 = 1$ , we solve  $\min \|V_{\text{null}}\alpha\|_1$  via LASSO, implemented using scikit-learn. The L1 fallback’s raw candidate is passed through the same post-processing tail as every other candidate (Section 3.7), so its coefficients are thresholded at the same noise-adaptive  $\varepsilon = \max(10^{-3}, 3\sigma)$  of Section 3.4, not a flat  $10^{-3}$ , before the result is snap-rounded. The L1 vector does not stand alone as the fallback’s answer: it seeds the incumbent of the anytime fallback of Section 3.6, which retries the exact engines on reduced instances and can only improve on the seed under the lexicographic objective. Section 6.6 reports the seed’s exact-recovery rate on the 4-sphere stress test.

## A.6 Incremental QR Updates and the Nullspace Basis

To avoid recomputing a full SVD after each monomial addition, the pipeline maintains a QR factorization  $\Phi = QR$ , extended by Householder-based column insertion (Golub and Loan, 2013) as each new batch of monomials arrives, at cost  $O(NM)$  per column, significantly cheaper than a full SVD recomputation  $O(NM^2)$

for typical  $M$ . Pruning (Section 3.9) only ever prevents a column from being added in the first place; it does not remove columns already present, so no downdate is needed. Since  $Q$  has orthonormal columns,  $R$ 's singular values and right singular vectors are exactly  $\Phi$ 's, so a fresh SVD of  $R$  alone, taken at every degree instead of chained across degrees, recovers the nullspace basis to machine precision at cost  $O(M^3)$ , independent of  $N$ .

## A.7 Construction of the Correlated-Noise (AR(1)) Probe

The correlated-noise probe of Section 5.3 reruns the circle benchmark with the points ordered as a sweep around the circle and the additive noise following a stationary AR(1) process over that acquisition order,  $e_i = \phi e_{i-1} + \sqrt{1-\phi^2} \sigma w_i$ , whose marginal standard deviation is exactly  $\sigma$  for every correlation  $\phi$ ; the pipeline itself receives only  $\sigma$ , never  $\phi$ . Correlation at  $\phi = 0.9$  shrinks the effective sample size by roughly  $19\times$  without changing any pointwise statistic, which is what isolates the SVD's reliance on noise averaging (as opposed to noise magnitude, already covered by the  $\sigma$  sweep of Table 8(a)) as the specific thing this probe stresses.

## B Full Proof of Proposition 4.1

**Proof.** Let  $V \in \mathbb{R}^{M \times d}$  be the nullspace basis, and let  $S$  denote the support of the true invariant  $c^*$ . A vector  $c = V\alpha$  has support contained in  $S$  if and only if, for all  $i \notin S$ ,  $(V\alpha)_i = 0$ , i.e.,  $V_{[M] \setminus S} \alpha = 0$ . The set of all nullspace vectors supported on  $S$  is therefore  $V \cdot \ker(V_{[M] \setminus S})$ , with dimension  $\dim \ker(V_{[M] \setminus S})$ . By hypothesis,  $c^*$  spans a one-dimensional subspace, so  $\dim \ker(V_{[M] \setminus S}) = 1$ . By the rank-nullity theorem,  $d - \text{rank}(V_{[M] \setminus S}) = 1$ , whence  $\text{rank}(V_{[M] \setminus S}) = d - 1$ . Conversely, rank  $d - 1$  implies a unique (up to scale) null direction  $\alpha$ , and  $c = V\alpha$  is then the unique sparse invariant supported on  $S$ . This is the nullspace analogue of the spark condition in sparse recovery (Donoho and Elad, 2003).  $\square$

## C Modified-Equation Closed Forms: Derivation and $dt$ -Scaling Checks

Section 4.6 states  $Q_{dt}(x, v) = x^2 + \frac{4}{4-dt^2} v^2$  as the exactly conserved quadratic form of leapfrog Störmer–Verlet without deriving it; the two-line version is: writing the one step  $v_{1/2} = v - \frac{dt}{2}x$ ,  $x_1 = x + dt v_{1/2}$ ,  $v_1 = v_{1/2} - \frac{dt}{2}x_1$  as the linear map  $x_1 = (1 - \frac{dt^2}{2})x + dt v$ ,  $v_1 = -dt(1 - \frac{dt^2}{4})x + (1 - \frac{dt^2}{2})v$ , substitute into  $Q(x, v) = x^2 + c v^2$  and collect the  $xv$  cross term of  $Q(x_1, v_1) - Q(x, v)$ : it vanishes exactly when  $c(1 - \frac{dt^2}{4}) = 1$ , i.e.  $c = 4/(4 - dt^2)$ . Substituting this  $c$  back, the  $x^2$  and  $v^2$  coefficients of  $Q(x_1, v_1) - Q(x, v)$  both vanish identically as well (an over-determined system that is consistent here because the map is symplectic and must conserve some quadratic form), confirming  $Q_{dt}$  is exactly invariant, not only invariant to leading order in  $dt$ .

**$dt$ -scaling of the gap, from the closed forms.** Fitting a log-log slope to Equation (10) over  $dt \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$  recovers 1.0000 against symplectic Euler's formal order, since the gap there is a pure linear term in  $dt$  with no higher-order remainder at all. Störmer–Verlet is second order in the same closed-form sense, but Equation (11) is an asymptotic series, not a pure power law, so the same fit over that full range returns 2.0029: at  $dt = 0.2$  the  $dt^4/16$  term is already large enough to bend the fit away from the leading  $dt^2$  behaviour. Restricting the fit to the smaller range  $dt \in \{0.001, 0.002, 0.005, 0.01, 0.02\}$ , where the leading term dominates and the series has not yet started to bend, recovers 2.0000 to four decimal places.

**The same law from sampled trajectories.** The two slopes above come from a symbolic substitution, not sampled data, so we checked separately that the same law holds once actual trajectories enter the pipeline. Solving by ordinary least squares for the coefficients  $B, C$  of  $Q = x^2 + B x v + C v^2$  directly from the transition-difference monomials  $x_t^2 - x_{t+1}^2$ ,  $x_t v_t - x_{t+1} v_{t+1}$ , and  $v_t^2 - v_{t+1}^2$ , evaluated on finite- $N$  sampled trajectories, reproduces the same slopes from data: 1.0000 for symplectic Euler and 2.0013 for Störmer–Verlet over  $dt \in \{0.001, \dots, 0.2\}$ . This check targets the two-parameter family  $(B, C)$  the proposition is stated in directly; what happens when the unrestricted fifteen-monomial dictionary is searched instead is treated in Sections 4.6 and 5.10.

## D Baseline Construction Rationale

Dense SVD+GB, and not a “SINDy + GB” baseline for any of the SINDy variants of Section 5, is used to isolate the effect of canonicalisation, because appending Gröbner reduction directly to a SINDy variant’s output is neither a clean ablation nor numerically well-posed. It is not clean because SINDy-null, SINDy-ST, and SINDy-FD all use Lasso-based or thresholded sparse regression, not SVD nullspace estimation, so any difference between “SINDy+GB” and SR-GB+CSNP could no longer be attributed to canonicalisation alone, it would also reflect the choice of estimator. It is not well-posed because Gröbner basis computation requires exact coefficients over  $\mathbb{Q}$ , while the redundant polynomials these SINDy variants return are, by construction, only *approximately* proportional to one another; that inexactness is why they are returned as syntactically distinct polynomials in the first place. Passing several such inexact floating-point polynomials to an exact symbolic Gröbner computation does not gracefully reduce to the single true generator, depending on the numerical noise in each input, the ideal they jointly generate over  $\mathbb{Q}$  can degenerate outright instead of reducing to the geometrically correct relation. Dense SVD+GB avoids this failure mode because it snap-rounds a *single* coefficient vector to exact rationals *before* canonicalisation, the same ordering SR-GB+CSNP itself follows (Sections 3.8–3.12). The SINDy variants are reported without this post-processing because their role in this paper is to represent the redundancy and nullspace-ambiguity failure modes as they are actually encountered in practice: these methods are not, in standard usage, followed by algebraic-geometry canonicalisation, and adding it here would misrepresent the baselines as well as remove the very phenomena Tables 6 and 12 are constructed to illustrate.

The argument above is about well-posedness, not about how badly the hybrid fails, so it was also checked empirically. The naive hybrid, SINDy-null’s candidate list (each candidate already snap-rounded, per Section 3.8) passed jointly to a single Gröbner basis computation, was run on the over-lifted circle of Table 7 ( $D = 3$  against a true degree-2 relation, 30 seeds), the same ambiguous-nullspace scenario Dense SVD+GB is built to handle. At  $\sigma = 0$ , where SINDy-null does return multiple candidates (mean 1.2 per seed), the joint Gröbner merge recovers the true relation in **20%** of seeds (6/30), never crashing and never collapsing to the trivial unit ideal, just landing on a non-matching multi-generator basis in the remaining 80%. That is worse than any individual candidate needs to be: SINDy-null’s raw per-candidate match rate on this scenario is 3.3%, so the merge recovers more often than a single un-merged candidate does, yet still far short of Dense SVD+GB’s 100% on the identical seeds. At  $\sigma = 0.05$  the hybrid fails completely (0/30), but for a different reason than the merge itself: SINDy-null’s L1 bootstrap fallback returns zero valid sparse candidates in every seed at this noise level on this dictionary (confirmed directly, the nullspace dimension is still correctly estimated at  $d = 3$ ; the sparse solver simply never lands inside its sparsity cap), so there is nothing for the Gröbner step to act on. The two failure modes are distinct: at  $\sigma = 0$  the ill-posedness argued for above is the one actually observed; at  $\sigma = 0.05$  the estimator-choice confound already noted is the limiting factor.

One further choice deserves explicit statement. Although the SINDy variants’ outputs are not post-processed into canonical form, they are *scored* through the same snap-rounding and Gröbner-based exact-recovery check as SR-GB+CSNP (Section 5): each candidate a baseline returns is snap-rounded and canonicalised before the proportionality test, and a baseline is credited if any of its candidates passes. This is generous to the baselines: a floating-point candidate close enough to the true invariant is credited as exact even though the method itself never produces a rational form. Because the check runs on our snap-rounding and canonicalisation machinery, the convention is stated explicitly here.

AVI’s canonicalisation is, for a related but distinct reason discussed at length in Appendix G, intrinsic to the algorithm’s Buchberger–Möller construction, not a post-processing step that could be added or withheld, so there is no well-posed “AVI + snap-rounding + GB” hybrid to run in the way there is for a raw SVD vector. That does not preclude running AVI itself, however, and Section 5.7 reports a direct empirical comparison between AVI’s border basis and SR-GB+CSNP’s minimal generator on ten individual systems drawn from this paper’s benchmark suite (Table 24).

## E Threshold and Tolerance Sensitivity

### E.1 Modified-Equation Tolerance Sweep

A natural fix for the wide-dictionary tie of Section 4.6 is a tolerance keyed to the known modified-equation gap instead of to noise. We tested this directly on the degeneracy probe of Section 4.6 by supplying CSNP with a synthetic  $\sigma_{\text{est}} = dt^p/3$  (so that  $\tau = \max(0.05, 3\sigma_{\text{est}})$  tracks the closed-form gap, not an assumed noise level) and, more broadly, by sweeping  $\sigma_{\text{est}}$  from 0 to 0.1 at  $dt = 0.05$ , which covers every value either  $p = 1$  or  $p = 2$  scaling would plausibly suggest. Neither the derived value nor any point on the sweep recovers  $H$ : the returned relation changes with  $\sigma_{\text{est}}$ , from the original quadratic tie ( $-v_{t+1}^2 + v_{t+1}v_t - x_{t+1}x_t + x_t^2$  at  $\sigma_{\text{est}} = 0$ ) to degenerate linear artefacts such as  $v_t - v_{t+1}$  once  $\sigma_{\text{est}} \gtrsim 0.01$ , but never to  $H$  or  $Q_{dt}$  themselves. This confirms that the wide-dictionary tie is a genuine degeneracy in exact algebra, not a numerically miscalibrated tolerance, so no choice of  $\tau$ , hand-set or  $dt$ -derived, can be expected to break it.

### E.2 Denominator Cap and Rationality-Window Sensitivity

Two constants govern the rationality machinery throughout, the denominator cap  $Q_{\text{max}}$  and the snap tolerance  $\varepsilon$  used for the exact-match track (Section 3.8), and neither has been varied elsewhere in this paper. Three arms isolate their effect, each at 30 seeds.

Arm A measures exact recovery against  $Q_{\text{max}}$  alone, holding  $\sigma = 0$  so that representability, not noise, is the only thing that varies: six conics with coprime integer coefficients and increasing minimum representable denominator  $q^*$ , from the unit circle ( $q^* = 1$ ) up to  $13x^2 + 17y^2 - 19$  ( $q^* = 19$ ) and  $17x^2 + 19y^2 - 23$  ( $q^* = 23$ ). Here  $q^*$  is computed the same way snap-rounding itself normalises a candidate vector, dividing by whichever coefficient has the largest magnitude (Section 6.2), not by whichever coefficient happens to give the smallest denominator; for every system in this arm the constant term is the largest-magnitude coefficient, so it is the one snap-rounding actually divides by. Table 21 reports the result. Recovery is a clean step function of  $q^*$ , exactly as the necessary-floor argument of Section 6.2 predicts: 0% at every  $Q_{\text{max}}$  below  $q^*$  and 100% at every  $Q_{\text{max}}$  at or above it, with no observed margin or lag anywhere in the grid. The  $q^* = 19$  invariant, for instance, sits at 0% through the default  $Q_{\text{max}} = 16$  and reaches 100% immediately at  $Q_{\text{max}} = 32$ , the very next grid point past its own floor. Runtime grows mildly with  $Q_{\text{max}}$ , from 0.006s at  $Q_{\text{max}} \leq 8$  to 0.038s at  $Q_{\text{max}} = 64$ , a wider window costs more snap-rounding candidates to score but stays cheap in absolute terms.

Table 21: Exact recovery vs the rational-denominator cap  $Q_{\text{max}}$  on conics with coprime integer coefficients, sampled exactly on the variety ( $\sigma = 0$ , 30 seeds). Each invariant’s minimum representable denominator  $q^*$  is listed: the smallest  $Q_{\text{max}}$  at which the invariant is representable under snap-rounding’s own normalisation (divide by whichever coefficient has the largest magnitude, not whichever normalisation happens to be cheapest). Recovery is a clean step function of  $q^*$ : 0% at every  $Q_{\text{max}} < q^*$  and 100% at every  $Q_{\text{max}} \geq q^*$ , with no observed margin, so e.g. the  $q^* = 19$  row stays at 0% through the default  $Q_{\text{max}} = 16$  and turns on immediately at  $Q_{\text{max}} = 32$ .

| Invariant                 | $q^*$ | $Q_{\text{max}}=2$ | $Q_{\text{max}}=4$ | $Q_{\text{max}}=8$ | $Q_{\text{max}}=16$ | $Q_{\text{max}}=32$ | $Q_{\text{max}}=64$ |
|---------------------------|-------|--------------------|--------------------|--------------------|---------------------|---------------------|---------------------|
| $x^2 + y^2 - 1$           | 1     | 100%               | 100%               | 100%               | 100%                | 100%                | 100%                |
| $3x^2 + 5y^2 - 7$         | 7     | 0%                 | 0%                 | 100%               | 100%                | 100%                | 100%                |
| $5x^2 + 7y^2 - 11$        | 11    | 0%                 | 0%                 | 0%                 | 100%                | 100%                | 100%                |
| $7x^2 + 11y^2 - 13$       | 13    | 0%                 | 0%                 | 0%                 | 100%                | 100%                | 100%                |
| $13x^2 + 17y^2 - 19$      | 19    | 0%                 | 0%                 | 0%                 | 0%                  | 100%                | 100%                |
| $17x^2 + 19y^2 - 23$      | 23    | 0%                 | 0%                 | 0%                 | 0%                  | 100%                | 100%                |
| <i>median runtime (s)</i> |       | 0.006              | 0.006              | 0.006              | 0.021               | 0.036               | 0.038               |

Arm B checks the failure mode the tie-break argument above raises, that widening  $Q_{\text{max}}$  might manufacture a false positive by giving a non-invariant more low-cost fractions to snap to. Two negative systems that admit no low-denominator rational invariant at all, a generic point cloud with no polynomial relation and a conic with an irrational coefficient ( $x^2 + \pi y^2 - 1$ ), are run across the same  $Q_{\text{max}}$  grid. Table 22 shows the false-positive rate stays at 0% throughout, including at  $Q_{\text{max}} = 64$ , four steps past the default. Raising  $Q_{\text{max}}$  costs recoverability headroom on the positive side of Table 21, not spurious answers on the negative side.



Table 22: False-positive rate (a nonzero generator returned when the method should abstain) vs the denominator cap  $Q_{\max}$ , 30 seeds. Neither a generic point cloud nor a variety with an irrational coefficient admits a low-denominator rational invariant; raising  $Q_{\max}$  well past the default 16 does not manufacture one.

| Negative system                  | $Q_{\max}=2$ | $Q_{\max}=4$ | $Q_{\max}=8$ | $Q_{\max}=16$ | $Q_{\max}=32$ | $Q_{\max}=64$ |
|----------------------------------|--------------|--------------|--------------|---------------|---------------|---------------|
| Generic cloud (no relation)      | 0%           | 0%           | 0%           | 0%            | 0%            | 0%            |
| $x^2 + \pi y^2 - 1$ (irrational) | 0%           | 0%           | 0%           | 0%            | 0%            | 0%            |

Arm C sweeps the rationality-window tolerance  $\varepsilon$  itself across four orders of magnitude,  $10^{-2}$  down to  $10^{-5}$ , around the pipeline’s default  $10^{-4}$ , on the noisy circle ( $\sigma = 0.01$ ) for recovery and the irrational conic for abstention. Table 23 shows both figures flat at 100% and 0% respectively across the whole range. Part of this flatness is structural, not a coincidence of these two systems:  $\varepsilon$  only ever acts through  $\max(\varepsilon, 3\sigma)$  (Section 3.4), so at  $\sigma = 0.01$  every tested value at or below  $10^{-2}$  is dominated by the  $3\sigma = 0.03$  noise floor on the recovery side, and the sweep mainly demonstrates that the floor, not a hand-tuned  $\varepsilon$ , is what is actually doing the work under noise. The abstention side has no such floor at  $\sigma = 0$  and stays at 0% regardless, so the irrational conic is rejected at every  $\varepsilon$  tested down to  $10^{-5}$ . Either way,  $\varepsilon$  is not a knob that needs re-tuning per benchmark.

Table 23: Sensitivity to the snap / rational-window tolerance  $\varepsilon$ , 30 seeds. Recovery is measured on the unit circle at  $\sigma = 0.01$ ; the false-positive column is abstention on the irrational conic  $x^2 + \pi y^2 - 1$ . Recovery is stable across two orders of magnitude of  $\varepsilon$  around the default  $10^{-4}$ .

| $\varepsilon$ | Circle recovery | False positive (irrational) |
|---------------|-----------------|-----------------------------|
| $10^{-2}$     | 100%            | 0%                          |
| $10^{-3}$     | 100%            | 0%                          |
| $10^{-4}$     | 100%            | 0%                          |
| $10^{-5}$     | 100%            | 0%                          |

Taken together, the three arms show the default  $Q_{\max} = 16$  is not fitted to any benchmark, every true invariant in this paper has small integer coefficients well inside it, but it is also a real, exact ceiling: recovery is 0% for any system whose minimum representable denominator  $q^*$  (computed under snap-rounding’s own normalisation, Section 6.2) exceeds  $Q_{\max} = 16$ , and 100% the moment  $Q_{\max} \geq q^*$ , with no extra margin needed once that floor is met. Raising  $Q_{\max}$  past 16 costs nothing on the negative controls tested here, so the default is a reasonable, conservative choice, not a value the method is fragile around, and  $\varepsilon$  needs no per-benchmark tuning at all.

## F Normal-Form Pruning: Counterexample and Cofactor Amplification

**The counterexample.** Section 4.3 works the  $I_G = \langle x^2 - y \rangle$ ,  $p = x^2$  counterexample and Lemma 4.3’s resolution through in full; in brief, dividing  $p$  by  $G = \{x^2 - y\}$  leaves the standard remainder  $q = \text{NF}_G(p) = y$ , so pruning removes the column  $x^2$  exactly as before while the search recovers  $y$  from the surviving columns instead.

**Noisy-data cofactor amplification.** Part (v) of Lemma 4.3 is an exact identity on the polynomials themselves; on noisy data it does not automatically carry over to an identity on residuals. Writing  $q = p - \sum_g a_g g$ , the residual of  $q$  on the dataset satisfies  $\text{resid}(q) \leq \text{resid}(p) + \sum_g \|a_g\| \text{resid}(g)$ , so a badly conditioned cofactor  $a_g$  could in principle amplify noise already present in a confirmed generator  $g$  into the target  $q$ . This does not leak into the pipeline’s accepted output, for two reasons that hold by construction, not by assumption. First, pruning only ever deletes monomial columns from the active library (Section 3.9); it never forms a new column as an explicit combination  $p - \sum_g a_g g$ , so no correlated-noise column of this kind is ever constructed for CSNP to search over. Second, every candidate CSNP proposes on the surviving, standard monomial columns is validated against the raw dataset directly, through the same scale-relative residual gate used everywhere else in the pipeline (Section 3.4), not against a reconstructed cofactor combination. The cofactor bound above therefore limits only what a badly conditioned  $G$  might cause the search to miss near



the noise floor; it has no bearing on what the pipeline can be made to output.

## G Comparison with Approximate Vanishing Ideals

The Approximate Vanishing Ideal (AVI) framework (Heldt et al., 2009; Kreuzer et al., 2009) computes the full border basis of the vanishing ideal from noisy data using a numerically adapted Buchberger–Möller procedure: it grows an order ideal of “standard” monomials degree by degree, and at each candidate monomial performs a rank test (structurally similar to the SVD-based feasibility test of Section 3.5) to decide whether that monomial’s evaluation vector is numerically dependent on those already accepted, using an  $\varepsilon$ -truncated ideal membership test. Every relation AVI returns is fixed by this fixed-order, fixed-tolerance construction, exactly as the reduced Gröbner basis is fixed by a monomial ordering; AVI’s canonicalisation is therefore intrinsic to the algorithm, not a separable post-processing step, unlike the raw SVD vector that Dense SVD+GB post-processes with snap-rounding and Gröbner reduction. Beyond the object-of-interest and selection-mechanism distinctions already drawn in Section 2, the two also differ in cost: AVI computes a complete border basis with complexity  $O(N \cdot \text{poly}(M))$ , while SR-GB+CSNP’s combinatorial search is exponential in  $k$  but typically much faster for small sparsity  $k$  and  $d$ .

For  $d = 1$ , the two approaches coincide in outcome, though not in mechanism: the border basis reduces to a single generator, and the first relation AVI’s construction discovers at the target degree is exactly the generator that Dense SVD+GB also recovers, which is why Dense SVD+GB is reported in Table 6 as a practical stand-in for AVI at  $d = 1$ , not a substitute for running AVI’s code directly. For  $d > 1$ , AVI’s border basis is the full set of relations needed to reconstruct the standard-monomial staircase of the entire finite sample, not the one geometrically meaningful invariant that SR-GB+CSNP recovers; isolating that invariant is the capability CSNP and BB add. A direct entry for AVI in Table 6 or Table 12 at  $d > 1$  would therefore report a quantity of a different kind (border-basis cardinality of the sampled point set), not the redundancy count reported for the other methods, so it is reported separately, as border-basis cardinality against SR-GB+CSNP’s exact-recovery rate, in Table 24 of Section 5.7.

### G.1 Per-System Exceptions in the Empirical Comparison

Five systems in Table 24 depart from the clean  $d = 1$ , cardinality-1 pattern; the sphere matches the circle at cardinality 1 for every noise level tested and is grouped into its row, not listed separately. Under an under-calibrated  $\varepsilon$ , the sphere is where AVI’s documented under-merging failure mode instead appears: its  $\varepsilon$ -truncated membership test can, for an unlucky noise realization, judge every remaining monomial independent of the order ideal built so far, so the order ideal never stops growing and the border basis  $G$  simply stays empty, never returning a wrong or partial generator, dragging the seed average below one true generator even though no seed returns a partial invariant. This reflects a threshold miscalibration, not a sphere with fewer than one true invariant, and the calibration criterion of Section 5.7 avoids it directly; SR-GB+CSNP is unaffected either way and recovers the true generator in 100% of seeds at every noise level tested here.

**Kepler angular momentum** has border-basis cardinality 3.2 on average even at zero noise, since the ambient degree-2 dictionary over its 8 transition variables admits several exact algebraic relations beyond angular-momentum conservation; CSNP isolates the single geometrically meaningful generator from that genuinely multi-dimensional nullspace, while AVI returns several of them. Under noise its cardinality grows sharply, to a mean of 21.0 at  $\sigma = 0.01$  and 20.3 at  $\sigma = 0.02$ ; SR-GB+CSNP’s exact recovery on the same noisy runs falls to 0%, the same multi-dimensional-nullspace ambiguity seen from AVI’s side as the inflated cardinality.

**The harmonic oscillator transition invariant** and **Feynman I.12.2** are recovered exactly by SR-GB+CSNP at  $\sigma = 0$  but in 0% of seeds once noise is added, at the same  $N = 5000$  sample size used for the main results on these same systems elsewhere in this paper (Section 5.8, Table 9). The recovered polynomial is simply not the true generator, a genuine noise-ceiling effect at this sample size, not an artifact of a smaller comparison harness. **Feynman kinematics\_position** is the instructive contrast. AVI’s cardinality still climbs under noise, through an intermediate value of 5.6 at  $\sigma = 0.01$  to 9.0 at  $\sigma = 0.02$ , yet SR-GB+CSNP recovers the true generator in every seed at both noise levels (100% at  $\sigma = 0.01$  and  $\sigma = 0.02$ ). Here the

rationality cost does what AVI’s  $\varepsilon$ -truncated membership test cannot: it reduces the noise-inflated border basis back to the single canonical generator.

Table 24: AVI border-basis cardinality vs SR-GB+CSNP, mean over 30 seeds ( $N = 5000$ , matching the sample size used elsewhere in this paper). A single true generator underlies every system; AVI’s cardinality is the size of the border basis it returns.

| System                                                               | AVI cardinality |               |               | SR-GB+CSNP exact |               |               |
|----------------------------------------------------------------------|-----------------|---------------|---------------|------------------|---------------|---------------|
|                                                                      | $\sigma=0$      | $\sigma=0.01$ | $\sigma=0.02$ | $\sigma=0$       | $\sigma=0.01$ | $\sigma=0.02$ |
| Circle, Sphere                                                       | 1.0             | 1.0           | 1.0           | 100%             | 100%          | 100%          |
| Cubic (algebraic toy)                                                | 1.0             | 3.0           | 3.0           | 100%             | 100%          | 100%          |
| Kepler angular momentum                                              | 3.2             | 21.0          | 20.3          | 100%             | 0%            | 0%            |
| Harmonic oscillator (transition)                                     | 7.0             | 6.0           | 6.0           | 100%             | 0%            | 0%            |
| Feynman <code>circle_locus</code> , <code>angular_momentum_2d</code> | 1.0             | 1.0           | 1.0           | 100%             | 100%          | 100%          |
| Feynman <code>newtons_second_law</code>                              | 1.0             | 1.0           | 2.0           | 100%             | 100%          | 100%          |
| Feynman I.12.2                                                       | 1.0             | 10.0          | 13.1          | 100%             | 0%            | 0%            |
| Feynman <code>kinematics_position</code>                             | 1.0             | 5.6           | 9.0           | 100%             | 100%          | 100%          |

## H Extended Results: Loop Invariants and DIG

**Loop invariants.** SR-GB+CSNP is run on five classic loop invariants (`sum`, `ps2`, `ps3`, `egcd`, `hash`) (Sharma et al., 2013; Nguyen et al., 2014; Sankaranarayanan et al., 2004; Rodríguez-Carbonell and Kapur, 2007). The three nullspace-reading baselines, SINDy-null, OMP, and Dense SVD+GB, share an identical exact integer nullspace; SR-GB+CSNP itself runs its full floating-point pipeline on the raw traces. SINDy-ST is included as a contrast that does not read the nullspace: it runs the same all-targets implicit STLSQ ensemble as every other benchmark.

Table 25: Loop invariant recovery rates (30 seeds, 95% Wilson CI). The three exact-arithmetic baselines (SINDy-null, OMP, Dense SVD+GB) read the same exact integer nullspace, while SR-GB+CSNP runs its full floating-point pipeline on the raw traces; all four return identical results on every benchmark, so one column reports all four. SINDy-ST is reported separately.

| Benchmark | Nullspace methods | SINDy-ST         |
|-----------|-------------------|------------------|
| sum       | 100% [89%, 100%]  | 100% [89%, 100%] |
| ps2       | 100% [89%, 100%]  | 100% [89%, 100%] |
| ps3       | 100% [89%, 100%]  | 0% [0%, 11%]     |
| egcd      | 100% [89%, 100%]  | 100% [89%, 100%] |
| hash      | 0% [0%, 11%]      | 0% [0%, 11%]     |

**DIG.** DIG (Nguyen et al., 2014) is run directly from the DIG authors’ `dynaroars/dig` repository, at that repository’s own commit `d081604`, pinned there for reproducibility, against nine systems: the circle and sphere of Section 5.1, a cubic  $x^3 + y^3 = 1$ , Kepler angular momentum, the 2D harmonic oscillator transition invariant, and four Feynman equations of Section 5.5, at  $\sigma \in \{0.00, 0.02, 0.05\}$ , 300 traces and one run per cell. It recovers **zero equality invariants in all 27 runs**, including at nominal  $\sigma = 0.0$ , because DIG’s equation engine requires every candidate to evaluate to *exactly* zero in rational arithmetic on every trace, with no tolerance anywhere in that check, and this paper’s trigonometric or square-root data generators leave even “noiseless” traces satisfying their invariant only to within roundoff ( $\sim 10^{-16}$ ). A positive control rules out a harness bug: on a circle trace built from an exact rational parametrisation, DIG immediately recovers it. SR-GB+CSNP, by contrast, recovers all nine systems at 100% over 30 seeds.

## I Feynman Polynomial Benchmark: Full Equation List

Table 26 lists the 26 polynomial equations used in Section 5.5, together with their per-equation exact recovery rates; all 26 achieve 100% at  $\sigma = 0$ . An entry carries an official Feynman-database ID (`I.x.y`) only where the equation was verified against the database’s published equation list; the remaining entries

carry descriptive labels, typeset as code. Two of these, `algebraic_cubic_toy` and `charge_force_toy`, are synthetic polynomial-form stress cases with no physical source.

**The  $\sigma = 0.05$  spurious-relation mechanism, in full.** Two degree-2 equations are partial rather than 100% at  $\sigma = 0.05$ : `momentum_conservation_1d` ( $m_1v_1 + m_2v_2 = P$ ) at 66.7% and I.39.22 ( $pV = nT$ ) at 86.7%. In every failing seed inspected, the pipeline returns a sparse spurious relation in a single narrowly ranged variable: one in  $m_2 \in [0.5, 2]$  for `momentum_conservation_1d` (for instance  $m_2^2 - 3m_2 + \frac{9}{5}$ ), and one in temperature  $T$  for I.39.22 (for instance  $T^2 - \frac{8}{3}T + \frac{5}{3}$ ). A low-degree polynomial can be uniformly small over a variable confined to a short interval, in the Chebyshev sense, and at  $\sigma = 0.05$  the noise-widened feasibility tolerance becomes loose enough to accept such a relation as vanishing. Once accepted, its three-term support with small rational coefficients beats the true generator, five monomials in `momentum_conservation_1d`'s expanded form and four in I.39.22, outright under the lexicographic rationality-and-sparsity objective. This failure is a bias, not a variance, effect: raising  $N$  from 5000 to 20000 changes nothing, because the spurious relation genuinely is small over the sampled marginal. The same trap, no longer confined to a single variable, accounts for `algebraic_cubic_toy`'s loss at  $\sigma = 0.05$ : its failing seeds return the quadratic  $x^2 - xy - \frac{8}{15}y^2$ , which is uniformly small over the sampled patch of the cubic curve.

Table 26: Benchmark suite: the 26 polynomial equations in the implicit polynomial form actually used (not the original, sometimes non-polynomial, physics formula each is derived from; Vars and Deg. are read from the generating source), with per-equation exact recovery rates over 30 seeds per condition. IDs of the form I.x.y are verified official Feynman database equations; typewriter-styled entries are descriptive non-Feynman labels.

| ID                                     | Invariant form ( $p(x) = 0$ )         | Vars | Deg. | $\sigma = 0.00$ | $\sigma = 0.02$ | $\sigma = 0.05$ |
|----------------------------------------|---------------------------------------|------|------|-----------------|-----------------|-----------------|
| I.8.14                                 | $(x_2 - x_1)^2 + (y_2 - y_1)^2 - d^2$ | 5    | 2    | 1.00            | 1.00            | 1.00            |
| <code>circle_locus</code>              | $x^2 + y^2 - r^2$                     | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>newtons_second_law</code>        | $F - ma$                              | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>momentum_conservation_1d</code>  | $m_1v_1 + m_2v_2 - P$                 | 5    | 2    | 1.00            | 1.00            | 0.67            |
| <code>work_energy</code>               | $W - Fd$                              | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>momentum_definition</code>       | $p - mv$                              | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>distance_3d</code>               | $x^2 + y^2 + z^2 - d^2$               | 4    | 2    | 1.00            | 1.00            | 1.00            |
| <code>angular_momentum_2d</code>       | $xv_y - yv_x - L$                     | 5    | 2    | 1.00            | 1.00            | 1.00            |
| <code>power_definition</code>          | $P - Fv$                              | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>frequency_shift_linear</code>    | $f - f_0 - vf$                        | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>wavenumber_dispersion</code>     | $\omega - \omega_0 - vk$              | 4    | 2    | 1.00            | 1.00            | 1.00            |
| I.34.27                                | $E - h\nu$                            | 3    | 2    | 1.00            | 1.00            | 1.00            |
| <code>planck_relation_duplicate</code> | $h\nu - E$                            | 3    | 2    | 1.00            | 1.00            | 1.00            |
| I.39.22                                | $pV - nT$                             | 4    | 2    | 1.00            | 1.00            | 0.87            |
| <code>algebraic_cubic_toy</code>       | $x^3 + 2xy + y^2$                     | 2    | 3    | 1.00            | 1.00            | 0.00            |
| I.12.2                                 | $Fr^2 - kq_1q_2$                      | 5    | 3    | 1.00            | 0.00            | 0.00            |
| <code>centripetal_force</code>         | $rF - mv^2$                           | 4    | 3    | 1.00            | 0.00            | 0.00            |
| <code>charge_force_toy</code>          | $F - q_1q_2r$                         | 4    | 3    | 1.00            | 1.00            | 0.00            |
| I.13.4                                 | $KE - \frac{1}{2}mv^2$                | 3    | 3    | 1.00            | 1.00            | 1.00            |
| <code>kinetic_energy_scaled</code>     | $2KE - mv^2$                          | 3    | 3    | 1.00            | 1.00            | 1.00            |
| I.14.3                                 | $U - mgh$                             | 4    | 3    | 1.00            | 0.00            | 0.00            |
| <code>kinematics_position</code>       | $x - ut - \frac{1}{2}at^2$            | 4    | 3    | 1.00            | 1.00            | 0.00            |
| <code>parallel_axis_theorem</code>     | $I - I_c - md^2$                      | 4    | 3    | 1.00            | 0.00            | 0.00            |
| <code>gravitational_pe_orbit</code>    | $Ur + Gm_1m_2$                        | 5    | 3    | 1.00            | 0.00            | 0.00            |
| I.43.31                                | $D - kT\mu$                           | 4    | 3    | 1.00            | 0.00            | 0.00            |
| <code>polarization_correction</code>   | $n kT - n_0 kT - n_0 pE_f$            | 5    | 3    | 1.00            | 0.00            | 0.00            |